FYPHub - Complete Technical Defence Guide

Prepared for project defence and viva preparation. Inspection date: 11 November 2026, Asia/Karachi. Repository: FYPHub, workspace F:/p1/fyphub-p1.

This guide explains the implementation actually found in the repository and running PostgreSQL catalog. It distinguishes visible workflows, supporting APIs, alternative source files, and historical database objects. Do not present a file's existence as proof that its feature is exposed in the current portal. The generated appendices provide exact tables, columns, foreign keys, registered endpoints, source-file locations, migration revisions, and test names.

Use the PDF to study and print; use the Word document to add your own speaking notes; use the Markdown version to search exact filenames. The companion technical-inventory.json supports detailed lookups. No account records, passwords, session tokens, private environment values, or uploaded student documents are included.

Reading map

Part	What you will learn
Chapters 1-3	Defence introduction, architecture, technologies and repository navigation
Chapters 4-5	Feature-to-file-to-table map and database relationships
Chapters 6-9	Authentication, academic configuration, supervision and real document uploads
Chapters 10-13	Proposal checks/AI, Excel archives/search, progress and evaluation algorithms
Chapters 14-17	Storage/deployment, security, tests and known gaps
Chapters 18-20	Demonstration sequence, read-only SQL, viva questions and revision checklist
Appendix A	Inspection scope, counts, migration state and schema differences
Appendix B	All installed database tables, columns, constraints, indexes and foreign keys
Appendix C	Registered APIs with exact backend handler locations and contract references
Appendix D	Every inspected source file, symbols, frontend imports and schema fields
Appendix E	Alembic migration revision/parent graph
Appendix F	Rebuilding the evidence and generated documents

1. How to study and explain the project

1.1 A two-minute introduction

FYPHub is a group-based final-year-project management application. Administrators configure academic cycles, group policies, user access, supervisor capacity, templates and document requirements. Students register a shared group account, submit proposals and required documents, request a supervisor, track project work and search previous projects. Supervisors review requests and inspect their assigned groups. Coordinator and evaluator interfaces expose selected review and evaluation workflows. The backend enforces roles, ownership and workflow state; PostgreSQL stores the academic records and the application stores uploaded files through feature-specific storage mechanisms.

The application is a modular monolith: one React browser application talks to one FastAPI application containing multiple feature routers. PostgreSQL is the common relational database. Docker Compose runs the application, database and local AI service in separate containers. Separate containers do not make each academic feature an independently deployed microservice.

1.2 Your universal answer pattern

For every feature, explain five things in this order:

1. Who uses it, and what academic problem it solves.

2. Which React screen collects input or displays results.

3. Which API endpoint handles the request and which backend function owns it.

4. Which table stores the parent record, which tables store versions/history, and which foreign keys connect them.

5. Which authorization, validation, transaction and test checks protect the workflow.

Example: for document submission, name StudentRequiredDocumentsDashboard.tsx, POST /api/v1/student/group/requirements/{id}/upload, document_requirements.upload, document_submissions plus document_submission_versions, the document storage volume, and ownership/deadline/type/size/signature checks.

1.3 Suggested preparation order

- First session: architecture, technology stack, role routing, authentication and database relationships.

- Second session: registration, supervisor selection, assignment/capacity, document uploads and discussions.

- Third session: archive import, proposal extraction/compliance, progress and evaluation algorithms.

- Fourth session: deployment, migrations, limitations, demonstration rehearsal and viva questions.

- During rehearsal, navigate to the exact source symbol listed in the appendices instead of memorizing every line.

2. Architecture and technology choices

2.1 Request flow

```
Browser: React + TypeScript
    |
    | HTTP fetch, JSON or multipart file, session cookie
    v
Nginx on localhost:5173
    |-- static HTML/CSS/JavaScript from /usr/share/nginx/html
    |-- /api/* and /health/* reverse proxy
    v
FastAPI / Uvicorn, backend:8000 (host localhost:8010)
    |-- Pydantic request validation
    |-- cookie authentication and role/ownership checks
    |-- feature router and workflow functions
    |-- SQLAlchemy request-scoped Session
    v
PostgreSQL 17 + pgvector (db:5432; host localhost:55732)
    |-- relational academic records, JSONB metadata, histories
    |-- some feature document bytes in BYTEA

FastAPI also accesses:
    |-- uploaded-files volume via DOCUMENT_STORAGE_ROOT
    |-- Ollama HTTP service for supported advisory AI workflows
    |-- optional configured Groq provider and SMTP delivery
```

2.2 Actual stack and its responsibilities

Layer	Technology	Why it is used here	Authoritative file
User interface	React 19, React DOM	Components, state, effects and role-specific screens	frontend/src/main.tsx, App.tsx
Type checking	TypeScript 6	Compile-time data contracts and component properties	frontend/tsconfig.json
Frontend toolchain	Vite 8	Local development, proxy and production asset build	frontend/vite.config.ts
Styling	Tailwind CSS 4 plus CSS files	Utility styles, layouts, shared theme and feature-specific presentation	frontend/src/index.css, design-system.css
API	FastAPI	HTTP routing, dependencies, input validation and OpenAPI documentation	backend/app/main.py
Application server	Uvicorn	Runs app.main:app	backend/Dockerfile
Validation	Pydantic / EmailStr	Request schemas, enums, field restrictions and validation	backend/app/schemas/
ORM	SQLAlchemy 2.1	Explicit SQL queries, transactions and model definitions	backend/app/database.py, models/
Database driver	psycopg 3	PostgreSQL communication	backend/requirements.txt
Persistent database	PostgreSQL 17	Constraints, transactional records, JSONB and foreign keys	docker-compose.yml
Vector extension	pgvector 0.8.6	Installed database extension; availability is part of readiness	database/init/001_extensions.sql
Schema evolution	Alembic	Ordered migration files and installed revision tracking	backend/alembic/
Password hashing	Argon2id	Salted, computationally expensive password verification	backend/app/security.py
Excel reading	openpyxl, xlrd	XLSX and legacy XLS import	backend/app/archived_projects.py
Documents	pypdf, PyMuPDF, python-docx	Text/format extraction and document handling	backend/app/pdf_extraction.py
Report generation	ReportLab	PDF report output	backend/app/evaluations.py, proposal compliance router
Advisory AI	Ollama; configurable Groq	Schema-validated feedback/planning with explicit fallback	backend/app/ai_provider.py
Deployment	Docker Compose and Nginx	Container startup, networking, persistent volumes and serving	docker-compose.yml, frontend/nginx.conf

Package versions above are taken from this checkout. They are not recommendations about current public package releases. Consult backend/requirements.txt and frontend/package-lock.json when reproducing the build.

2.3 Why PostgreSQL instead of a database per feature

All main academic features share one PostgreSQL database, normally named fyphub_p1. A feature's database is its related tables within this database. There is not one physical database file for each React screen. Tables connect through identifiers such as team_id, academic_cycle_id, supervisor_id, requirement_id and submission_id. This permits integrity checks and atomic updates spanning a request, assignment and audit event.

PostgreSQL manages the physical files inside its Docker data directory. Python model files describe the intended database structure; they do not contain live student data. Alembic migrations change the structure. The SQL backup file at repository root is a historical export, not the live database and not necessarily current.

3. Repository map and application entry points

Location	What to look for	How to explain it
frontend/src/main.tsx	React root and global CSS imports	Browser application entry point
frontend/src/App.tsx	Auth pages, signup steps, login, session restoration and portal selection	Top-level application controller
frontend/src/AdminPortal.tsx	Active administrator feature navigation	Which admin screens are actually mounted
frontend/src/portals/student/StudentPortal.tsx	Student feature selection and shared group identity	Student workspace shell
frontend/src/portals/supervisor/SupervisorPortal.tsx	Requests, assignments, capacity and history	Supervisor workspace shell
frontend/src/portals/coordinator/CoordinatorPortal.tsx	Request monitoring and proposal reviewer workspace	Current coordinator screen composition
frontend/src/portals/student/modules/StudentEvaluatorPortal.tsx	Active evaluator entry imported by App.tsx	Evaluator role routing
frontend/src/portals/shared/	Sidebar and shared portal pieces	Reusable presentation and navigation
backend/app/main.py	FastAPI app, middleware, auth, admin configuration and router inclusion	Backend application entry and substantial governance logic
backend/app/models/	SQLAlchemy table definitions	Where to find each table and relationship
backend/app/schemas/	Request/response contracts	Validation and accepted payload shape
backend/app/.py	Feature routers and helpers	Business logic and security checks
backend/app/modules/module_03_proposal/	Compliance rules, template extraction, reports	Proposal checking implementation
backend/alembic/versions/	Revision graph and DDL changes	History of schema changes
database/init/001_extensions.sql	CREATE EXTENSION vector	Initial extension setup, not all feature tables
scripts/	Bootstrap and seed tools	Initial admin and demonstration data setup
backend/tests/, frontend/tests/	Verification source	Evidence of covered scenarios; test existence is not a fresh passing result
docs/	Prior feature documentation and this guide	Useful context; code and live schema take priority when they differ
sample-data/, outputs/, tmp/	Local examples, exports and temporary artifacts	Not application entry points or authoritative databases

3.1 Important naming distinctions

- UI words group and team generally refer to the same teams table. A request may call its identifier group_id while an ORM field is team_id.

- PANEL_MEMBER is the stored role value for evaluator users. Evaluator is a UI label.

- Proposal is not one universal table in this application. Supervisor-request proposal data, proposal compliance submissions, proposal references, archived projects, and foundation proposals/proposal_versions are separate workflows.

- Project information in the supervisor workspace is assembled from the proposal request, group assignment and recorded progress. There is no general projects table in the current ORM. Do not invent a projects.project_id relationship during the defence.

- Files with names Complete, Final, Clean, AllFormats or V2 are not automatically the active version. Imports decide what is rendered.

3.2 Frontend routing and state

The application uses React useState/useEffect and manual browser history handling. It does not declare React Router as a dependency. App.tsx restores a session through /api/v1/auth/me and selects a portal from the persisted role. AdminPortal.tsx recognizes certain /admin/... paths. SupervisorAssignedTeams.tsx supports /supervisor/groups/{group_id}. Some screens remain state-driven and do not have complete bookmark/back-button behavior.

The same-origin API base and Nginx/Vite proxies allow the browser to call /api rather than a hardcoded backend container hostname. Fetch requests generally use credentials: include for cookie authentication. Mutations submit JSON or multipart FormData. Responses update local component state; selected supervisor views poll every ten seconds, while group discussions poll every five seconds. This is polling, not WebSocket push.

4. Feature-to-file and database map

Feature	Active frontend or UI owner	Backend implementation	Main tables / storage
Sign in / sign out / session restoration	App.tsx	main.py, auth.py, security.py	users, auth_sessions
Shared group signup	App.tsx member and credentials steps	main.py register_student_group / register_team; schemas/team_registration.py	users, teams, team_members, academic_cycles, academic_group_policies
Password recovery	App.tsx recovery views	password_reset.py	password_reset_tokens, auth_sessions, users, admin_audit_events
Users, roles and account activation	AdminUsersRoles.tsx	main.py admin user handlers	users, auth_sessions, admin_audit_events
Student group administration	AdminStudentGroups.tsx	main.py student_group_detail and admin group endpoints	teams, team_members, users, supervisor_assignments, supervisor_requests
Academic cycles and group policy	AdminCycleManager.tsx, AcademicCycleDashboard.tsx	main.py, academic_cycle_overview.py	academic_cycles, academic_group_policies
Supervisor capacity	AdminSupervisorCapacity.tsx, SupervisorCapacity.tsx	main.py, coordinator_operations.py, supervisor_requests.py	supervisor_capacity_policies, supervisor_assignments
Document requirements and submission list	AdminDocumentRequirementManagerClean.tsx	document_requirements.py	document_upload_requirements, document_submissions, document_submission_versions; file volume
Student required-document upload/history	StudentRequiredDocumentsDashboard.tsx	document_requirements.py	Same requirement/submission/version tables
Proposal template administration	modules/module-3/AdminProposalTemplates.tsx	modules/module_03_proposal/router.py, template_extractor.py	template_references, template_reference_versions, academic_rule_versions; file volume
Student proposal compliance	modules/module-3/ProposalComplianceWorkspace.tsx	modules/module_03_proposal/router.py, compliance_engine.py, pdf_extraction.py	proposal_compliance_submissions, proposal_compliance_reports
Supervisor selection and request	StudentSupervisorSelection.tsx re-export / StudentSupervisorSelectionPdf.tsx	supervisor_requests.py, pdf_extraction.py	supervisor_requests, supervisor_request_documents, status history, submission-note versions
Supervisor request review	SupervisorRequestsPdf.tsx, SupervisorGroupDetails.tsx	supervisor_workspace.py, supervisor_requests.py	Requests joined with group, assignment, progress and document records
Assigned group details	SupervisorAssignedTeams.tsx, SupervisorGroupDetails.tsx	supervisor_workspace.py	No new snapshot table; joins existing feature tables
Request conversations	RequestConversation.tsx (shared re-exports)	supervisor_request_conversations.py	supervisor_request_messages
Assignment-scoped group discussion	SupervisorGroupDetails.tsx, shared/CollaborationChannels.tsx where mounted	collaboration_channels.py	collaboration_channels, collaboration_participants, collaboration_messages
Earlier group channels / moderation	group-channel clients and module helpers	group_channels.py, module2_collaboration.py	group_messages, module2_channel_reads, module2_notifications
Project progress and meetings	StudentProgressDashboard.tsx, supervisor project tab	project_workflow.py, supervisor_workspace.py	project_milestones, project_tasks, supervision_meetings
Historic Excel project import	OldProjectArchiveImport.tsx, AdminArchiveProjects.tsx	archived_projects.py	archive_import_batches, archived_projects JSONB source data
Archive search and project detail	student/modules/knowledge-search/KnowledgeSearch.tsx and detail	archived_projects.py	archived_projects, archived_project_documents
Proposal reference search	ProposalReferenceSearch variants if mounted	proposal_reference.py	proposal_references, proposal_reference_documents
Academic deadlines	AdminDeadlines.tsx when mounted; dashboard deadline components	main.py deadline handlers, module2_collaboration.py	academic_deadlines, academic_deadline_versions
Reminder configuration	AdminReminderRules.tsx when mounted	main.py reminder handlers	reminder_rules, reminder_rule_versions
Dashboard requirements	components/DashboardDeadlinesRequirements.tsx where mounted	dashboard_requirements.py	dashboard_requirements
Defense schedules and panel scoring	EvaluatorWorkflow.tsx, PanelScheduling.tsx if mounted	project_workflow.py	defense_schedules, defense_panel_members, panel_evaluations, panel_evaluation_scores
Formal marks and publication	EvaluationMarksWorkspace.tsx where mounted; API	evaluations.py, schemas/evaluations.py	evaluation_records, evaluation_audit_events, evaluation_corrections, evaluation_aggregation_configs
Audit history	API / governance screens	audit_history.py, main.py add_admin_audit_event	admin_audit_events; separate evaluation audit table
Group AI helper	Supporting backend/UI if mounted	group_ai.py	module2_chatbot_sessions
Advisory proposal AI review	Supporting API; not current supervisor compliance screen	ai_provider.py, supervisor_requests.py	proposal_ai_review_runs
Agentic supervision foundation	Supervisor navigation removed; backend still registered	agent_supervision.py	supervision_agent_runs, then milestones/tasks only after permitted execution

5. Database design in depth

5.1 Central identity and ownership graph

```
users (shared student login or staff account)
  |-- auth_sessions.user_id
```

```
  |-- password_reset_tokens.user_id
  |-- teams.shared_user_id  [one shared login per group]
  |       |-- team_members.team_id
  |       |-- supervisor_requests.team_id
  |       |-- supervisor_assignments.team_id
  |       |-- project_milestones.team_id
  |       |-- project_tasks.team_id
  |       |-- supervision_meetings.team_id
  |       |-- document_submissions.group_id
  |       |-- proposal_compliance_submissions.team_id
  |       |-- defense_schedules.team_id
  |
academic_cycles
  |-- academic_group_policies.academic_cycle_id
  |-- teams.academic_cycle_id
  |-- document_upload_requirements.academic_cycle_id
  |-- supervisor_requests.academic_cycle_id
  |-- supervisor_assignments.academic_cycle_id

supervisor_requests
  |-- supervisor_request_documents.request_id
  |-- supervisor_request_status_history.request_id
  |-- supervisor_request_messages.request_id
  |-- supervisor_request_submission_note_versions.request_id
  |-- assignment_id -> supervisor_assignments.id

supervisor_assignments
  |-- collaboration_channels.assignment_id
          |-- collaboration_participants.channel_id
          |-- collaboration_messages.channel_id

document_upload_requirements
  |-- document_submissions.requirement_id
          |-- document_submission_versions.submission_id
```

5.2 Normalization, JSONB and version tables

Relational parent tables provide stable identities and foreign keys. Version tables retain historical content or configuration without silently overwriting the parent identity. For example, a document requirement has a submission parent for a group and multiple document_submission_versions. Template references and rubric definitions likewise have version tables.

JSONB is used for variable data such as imported Excel columns/cells, member arrays in archived records, AI payloads and compliance report structures. This is practical because historical workbooks do not all share one fixed set of columns. Stable fields such as title, year, session and source batch are still stored as explicit columns for search and indexing.

Tradeoff: JSONB permits flexibility but weakens column-level constraints on individual embedded values. The application must validate payload structure and preserve provenance. Editing a JSONB object requires assigning/merging the new value in a way SQLAlchemy detects; do not assume an arbitrary in-place nested mutation is automatically persisted.

5.3 Important integrity controls

- users has normalized email/registration checks and uniqueness. Student-role records require a registration value at the database level even though login uses the shared group email.

- teams links to an academic cycle and a shared user. Team status is constrained to forming, complete, locked or dissolved.

- team_members has a composite group/cycle relationship, active-cycle uniqueness checks for registration/email, and a partial uniqueness rule for the creator/leader.

- supervisor_assignments has a partial unique index allowing only one active assignment for a team where ended_at is null.

- supervisor_requests constrains allowed statuses, positive revision numbers, and active request uniqueness. assignment_id provides a reverse link to the created assignment.

- document_submission_versions has a unique submission/version pair. The submission parent is also constrained by requirement/student; group-level selection is additionally performed in code.

- project tasks and milestones constrain completion percentages to 0-100.

- archived_projects has active normalized title/year/group uniqueness; archive_import_batches stores import provenance.

- Many foreign keys use RESTRICT to preserve academic records. Selected child histories or infrastructure use CASCADE/SET NULL. Read each exact delete policy in the generated database appendix.

5.4 Model, migration, installed schema and backup are different

Artifact	Purpose	Where it exists
Model	Intended ORM mapping, fields and constraints	backend/app/models/.py
Pydantic schema	Accepted/returned API data; not a database table	backend/app/schemas/.py

Artifact	Purpose	Where it exists
Migration	A specific schema transition and dependency on earlier revisions	backend/alembic/versions/.py
Live table	Actual installed records and schema	PostgreSQL named volume
alembic_version	Installed migration revision tracking	Table inside the same PostgreSQL database
SQL dump	Point-in-time backup/export	Root .sql export or a chosen backup location
Uploaded file	Original bytes for relevant features	File volume, or BYTEA for some document tables

6. Authentication, authorization and group signup

6.1 Authentication sequence

1. App.tsx submits an email and password to POST /api/v1/auth/login.

2. main.login calls validate_mutating_origin and finds the user by normalized email. Registration-number login is deliberately not accepted by this handler.

3. security.verify_password verifies the Argon2id hash. A dummy hash is used for unknown users to avoid an obviously different hashing path.

4. The account must be active. Student users must have an active shared-group link.

5. A random opaque token is generated. Only its SHA-256 hash is stored in auth_sessions; the browser receives the raw token through an HttpOnly cookie named fyphub_session.

6. Subsequent requests include the cookie. auth.get_current_user joins auth_sessions to users, checks revocation/expiration/activity and verifies the student group link.

7. require_roles checks the stored role. Feature handlers then check ownership of the requested group, document, request or evaluation.

8. GET /api/v1/auth/me supplies safe account and group context, allowing the browser to restore the appropriate portal.

This is server-side session authentication, not JWT authentication. The server can revoke sessions through the database. Passwords are not encrypted and later decrypted; Argon2id is a one-way hash. SHA-256 is appropriate here for high-entropy random tokens, while human passwords need the slower Argon2 password hash.

6.2 Cookie, CORS and origin checks

The cookie is HttpOnly, SameSite=Lax, path=/ and Secure when the backend request scheme is HTTPS. The default session lifetime is eight hours. Authentication helpers explicitly permit the local browser origins http://localhost:5173 and http://127.0.0.1:5173. Cross-origin credentials do not use a wildcard origin.

Nginx forwards /api calls to the backend. This avoids many cookie/CORS problems for the local frontend. However, validate_mutating_origin still checks a supplied Origin against a fixed local allowlist. A new LAN hostname or production domain needs appropriate configuration/code review; the same-origin proxy alone does not guarantee that every new Origin is allowed. Trusting forwarded HTTPS headers must also match the reverse-proxy deployment.

Do not describe the current origin check as a complete standalone CSRF-token mechanism. It is an explicit origin policy used with SameSite cookies. Public routes, no-Origin callers and endpoint-specific permission checks have distinct behavior.

6.3 Signup and members

The signup form has academic-cycle selection, group members, and shared credentials steps. Each member now supplies full_name, email and registration_number. StudentGroupMemberInput requires a full text name and uses validate_full_name. The backend normalizes emails and registration identifiers and rejects invalid input. The first member or explicitly designated leader becomes the group creator.

register_student_group translates the public group payload into the established register_team transaction. register_team validates registration mode, cycle/policy availability, group limits and duplicate accounts/members. It creates the shared User, Team and TeamMember records, refreshes group status and writes an audit event. A failure rolls back the transaction.

The group members shown in the student overview come from /auth/me member context and GroupMembers.tsx. Changing the signup form does not recover human names that were never stored for older accounts. Staff account creation and role changes belong to administrative APIs, not a public role dropdown.

6.4 Password recovery

password_reset.py creates a hashed, expiring reset token, with a fifteen-minute token lifetime and an in-memory limit of five requests per hour per IP. Production responses generally avoid revealing whether an account exists. Development/demo branches can differ; local development reset links are specifically guarded. SMTP delivery uses STARTTLS when configured. A successful reset consumes the token, updates the Argon2id hash and revokes existing sessions.

Limitations to acknowledge: the rate limiter and development-link cache are process memory, so they are not shared between multiple workers and reset after restart. SMTP success is an external dependency. A generic success message is not proof that an email was delivered.

7. Academic administration and governance

7.1 Academic cycles and group policies

academic_cycles stores code, display name, academic year, session and registration state. academic_group_policies stores minimum/maximum group size, formation deadline and matching configuration. The cycle/policy APIs are in main.py and schemas/academic.py. AdminCycleManager.tsx is the visible management screen.

Opening registration requires a policy. Maximum group size cannot be reduced below an existing active group's size. The backend locks cycle/policy records for sensitive changes. Cycle and policy updates are audited. The database constrains valid years and session values, with unique year/session combinations when both are specified.

The current UI has removed the cycle-delete button, but a backend delete handler still exists. Removing a button is not equivalent to removing an endpoint. Database constraints and the endpoint's role/usage checks remain relevant.

7.2 User and group administration

AdminUsersRoles.tsx manages staff/account fields, roles and activity. AdminStudentGroups.tsx presents shared group accounts and member lists. main.py performs role changes, protects critical administrator access, revokes sessions when needed, and writes admin audit records. The current API, not the label on a browser button, determines what a user may do.

Group details are assembled from users, teams, team_members, current supervisor assignment and the latest related request. Group name display is commonly derived from the leader's full name because the Team model does not have a dedicated group_name column. A group_name property in a response is therefore not proof of a separately persisted field.

7.3 Deadlines, reminders, templates and rubrics

Academic deadlines have a parent and versions. The preview accepts local wall time, an IANA zone and, when needed, an explicit UTC offset; the inspected version model persists due_at_utc and time_zone. Do not claim that local wall time and UTC offset are separate stored columns. main.resolve_deadline_time uses ZoneInfo and checks round trips to detect nonexistent or ambiguous daylight-saving times rather than guessing an offset.

Reminder rules also have versioned configuration. Their existence does not prove that an independent periodic worker automatically sends scheduled messages: the Compose stack contains no Redis/Celery service or general scheduler. Distinguish a rule definition from an executed notification.

Template references and academic rules preserve version history; active versions are selected for proposal checking. Evaluation rubrics have parent, version and criterion tables. Governance APIs can exist even when their frontend menu entries are currently absent. AdminPortal.tsx contains branches for deadlines/reminders, but its current navigation groups do not expose every branch.

8. Supervisor request, capacity and assigned groups

8.1 Supervisor request lifecycle

```
DRAFT -> extraction review/confirmation -> SUBMITTED
SUBMITTED -> ACCEPTED
SUBMITTED -> REJECTED (reason required)
SUBMITTED -> CHANGES_REQUESTED (comment required)
CHANGES_REQUESTED -> revised extraction/confirmation -> SUBMITTED
Eligible open request -> WITHDRAWN
```

supervisor_requests stores title/summary, proposal type, extracted academic fields, the requesting shared user, team, cycle, chosen supervisor, document pointer, status, revision and decision timestamps. supervisor_request_documents stores document versions, original bytes, SHA-256, extracted text/fields and extraction state. supervisor_request_submission_note_versions preserves submitted notes. supervisor_request_status_history records actor, old/new status, comment and time.

The PDF selection component uploads a proposal, displays extracted fields, confirms extraction and submits the request. The backend owns eligibility checks and the accepted upload contract. A browser accept attribute is only a file-picker hint; it is not security validation.

8.2 Why acceptance is transactional

decide_request validates the selected supervisor and current request revision. For acceptance it locks the relevant records, verifies active capacity and the team/cycle relation, prevents an already active assignment, creates SupervisorAssignment, ensures the collaboration channel and links request.assignment_id. It then changes request status, writes decision history and audit events, and commits.

If any part fails, rollback prevents a partially accepted request without an assignment or an assignment without its history. A partial unique database index is the final guard against two active assignments for the same team. The revision check also prevents a supervisor acting on a stale browser copy.

8.3 Capacity calculation

Capacity policy is stored in supervisor_capacity_policies. _lock_capacity verifies the active supervisor and locks the selected annual/cycle policy. _capacity_used counts active assignments in the relevant academic year, including older assignments whose year is resolved through their cycle. It also counts accepted legacy requests not linked to an assignment. _ensure_capacity compares usage with max_group_capacity.

SupervisorCapacity.tsx displays a read-only breakdown. Administrators configure capacity. Accepted assignment rows now include the academic year/session when available. The live API can combine cycle-specific legacy policies with annual policies, so explain the actual lookup rather than claiming one unconditional row per supervisor.

8.4 Supervisor workspace implementation

supervisor_workspace.py provides /api/v1/supervisor/assigned-groups and /groups/{group_id}, plus members, proposal, project, documents, progress and conversation sections. /requests and /requests/{request_id} provide nested group information before a decision. Accept/reject/request-changes endpoints delegate to the existing transactional decision handler.

SupervisorAssignedTeams.tsx displays a group summary and opens SupervisorGroupDetails.tsx. The latter contains overview, members, proposal, project, documents and discussion tabs. SupervisorRequestsPdf.tsx embeds the same details during review. Primary labels show group names, emails and cycle names rather than raw internal IDs.

An assigned group's record is a joined response, not a duplicate database snapshot. It contains the real group, all members, assignment, supervisor, proposal request, required-document submissions and progress records. The related accepted request is prioritized using assignment_id; NULL-aware ordering matters because a later rejected request must not replace the assigned proposal in the display.

Completion is derived from recorded task completion, or from milestones when tasks are absent. The response provides a completion_basis. This derived display is different from a separately stored authoritative project percentage. Empty fields are displayed as not recorded/submitted rather than invented values.

8.5 Authorization and refresh

owned_group verifies an active assignment for the logged-in supervisor. owned_request verifies the selected supervisor and excludes drafts. Supervisors cannot use another group's URL to bypass these checks. A pending owned request may inspect related data through that request response without receiving general assigned-group access.

Assigned lists, selected group details and request views refresh on timers; group messages refresh more frequently. Acceptance removes the pending request and the next assigned-list load returns the group. The UI does not use a global live-event bus, so there can be a polling delay. Switching portals or remounting a view fetches fresh data.

9. Document requirements, versions and real files

9.1 Parent-child structure

- document_upload_requirements: title, instructions, cycle, optional assignee/group, permitted roles, extensions, size, due time, lifecycle and required/late settings.

- document_submissions: the requirement/group/student submission identity, status, late flag, reviewer feedback and timestamps.

- document_submission_versions: each uploaded filename, MIME type, size, uploader, version number and storage reference.

- Original required-document file bytes are written to DOCUMENT_STORAGE_ROOT rather than normally stored in the version content column.

Requirement lifecycle is DRAFT -> PUBLISHED -> CLOSED -> ARCHIVED, with explicit unpublish/reopen behavior. Only unused drafts may be deleted through the current API. A published/history-bearing requirement should be closed/archived according to its supported state transitions.

9.2 Upload and download flow

1. StudentRequiredDocumentsDashboard.tsx loads published/closed group requirements and existing submission/version data.

2. The form submits multipart content to the group requirement upload/resubmit endpoint.

3. The backend checks group/cycle visibility, requirement state, deadline/late policy, extension, MIME type, maximum bytes and implemented signature checks.

4. It obtains or creates the submission parent and calculates the next version.

5. It creates a random server storage name, writes a temporary file, renames it, writes version metadata and audits the operation.

6. Successful commit returns all versions. Failed persistence cleans up the temporary/new file in the relevant upload handler.

7. The student sees version history. Admin View submissions lists student/group context and all uploaded versions.

8. The download endpoint verifies the submission/version relationship and role/ownership before returning FileResponse with the original filename.

The storage path is derived and resolved under the configured root to block path traversal. Never trust a user filename as a direct disk path. Opening an arbitrary storage_reference from the database must still pass authorization before returning bytes.

9.3 File-type limitations to know before the demo

The active form now lists document, image, video, audio, spreadsheet, presentation, archive and any-supported-format categories. Pydantic accepts an extensive extension set. However, backend _validate does not implement successful signatures for every advertised format: several entries default to false and are not overridden, including legacy XLS/PPT, MP3 and WEBM in the inspected implementation. MIME checks are also strict and browsers can submit different or missing MIME types. Therefore, do not claim that every listed format is end-to-end supported without testing actual files.

Other practical constraints: Nginx has no explicit client_max_body_size in the inspected frontend/nginx.conf, so its own default request-body limit may reject a file before the backend's configured 10/100 MB allowance. A 413 response can originate at the proxy. The student upload UI also disables overdue uploads unconditionally in its current state, while the backend has a late_upload_allowed flag; late reopening requires a UI/backend consistency check before demonstration.

10. Proposal extraction, template compliance and AI

10.1 Separate proposal workflows

Workflow	Main record	Purpose
Supervisor selection	supervisor_requests + supervisor_request_documents	Ask a supervisor to accept the group and discuss the submitted proposal
Template compliance	proposal_compliance_submissions + proposal_compliance_reports	Compare an uploaded proposal against active template rules
Historical proposal reference	proposal_references + proposal_reference_documents	Advisory title lookup against reference material
Previous FYP archive	archived_projects + archived_project_documents	Historic project discovery and imported workbook data
Foundation proposal identity	proposals + proposal_versions	Minimal identity/version foundation; not the complete current content store
Historical m03 tables	m03_	Existing database objects absent from current ORM; verify before claiming they power an active feature

10.2 Extraction is local and deterministic

pdf_extraction.py uses pypdf for PDF text, PyMuPDF for measurable layout information and python-docx for DOCX structure/formatting. It computes SHA-256 and returns a structured extraction result with text, fields, page information and extraction metadata. Academic headings/labels are identified by rules and patterns. OCR is not a demonstrated general capability here; an image-only scanned PDF may not provide readable text.

DOCX can expose native formatting such as margins, font runs, paragraph spacing, styles, tables and headings. PDF layout measurement is an estimate from positioned text and document geometry; it is not equivalent to reading original editable Word settings. Some checks must remain UNVERIFIABLE when information is unavailable.

10.3 Compliance processing

AdminProposalTemplates.tsx uploads/activates a reference template through the module-3 API. The backend extracts measurable rules and stores the active reference version and rule payload. A student submits PDF/DOCX/TXT through ProposalComplianceWorkspace.tsx. The backend finds an active template for SHORT/LONG, records the next proposal_key/version, saves the original file, extracts fields and builds a report.

build_compliance_report checks rule-provided page layout, font formatting, paragraph properties, heading styles, tables/figures/captions, references, appendices, required sections and sequence. Required-heading matching uses difflib.SequenceMatcher with a threshold of 0.82. Numeric formatting comparisons use a tolerance of 0.05 when both values are numeric. Missing measurable values produce UNVERIFIABLE rather than a fabricated match.

Academic consistency compares token vocabularies between sections. Tokens are case-normalized, filtered and compared by intersection divided by the smaller token-set size. The inspected thresholds are 0.20 for PASS, 0.08 for PARTIAL, otherwise FAIL; missing text is UNVERIFIABLE. This is a lexical heuristic, not proof of semantic correctness or research originality.

The score is round(passed checks / total checks 100). A report is Compliant only when the logic finds no failures or partial/unverifiable checks; it can be Not Compliant or Review Required. The report JSON stores explanations, violations, sections and version history. Report endpoints expose original, annotated and report PDF outputs subject to access checks. Submission can lock the group's membership through team_lifecycle.lock_team.

The report engine is deterministic Python. build_change_summary labels its source AI_ASSISTED_DETERMINISTIC and model local-change-analyzer, but compares structured values locally. Do not describe this label as evidence that a language model generated the summary.

10.4 Optional AI provider layer

ai_provider.py supports configured Ollama and Groq clients, explicit model selection, timeouts, bounded retries, rate limiting and Pydantic JSON-response validation. The prompt tells the model to treat submitted text as data and remain advisory. Proposal review expects exactly seven required section findings in order with actionable text and confidence values.

Invalid configuration, timeouts, provider errors or invalid JSON produce explicit UNAVAILABLE/RATE_LIMITED or deterministic fallback metadata. Ollama can attempt a configured fallback model. API keys remain backend environment settings. Do not promise that a model is available simply because the Ollama container is running; the selected model must be downloaded and responding.

The agent-supervision backend persists plans and requires appropriate human approval before tool execution. Its supervisor navigation/module was removed at the user's request, but the backend router and supervision_agent_runs table remain. Hiding a UI feature does not delete its API or schema.

11. Previous-project Excel archive and knowledge search

11.1 Why this is project-oriented

AdminArchiveProjects.tsx presents one project card with related information, original import fields, editing and a centered delete confirmation. OldProjectArchiveImport.tsx handles the file selection/import action. ArchiveManagement.tsx is the active wrapper imported by AdminPortal.tsx.

The main database parent is archived_projects. A project title is display/search content, while the numeric primary key is its stable identity. The import grouping key combines cycle, normalized project title and resolved group/block context so repeated member rows can belong to the same project. Do not make title alone globally unique across every year and group.

11.2 Parser, grouping and provenance

parse_archive_workbook accepts XLS/XLSX, enforces the configured byte limit, extracts populated columns, removes blank values and known Excel errors, and applies the literal-email end boundary when an email column/value exists. XLSX is read in formula and data-only modes so formulas can be identified and excluded even when they have cached results. Without an email boundary it uses the last literal content row.

Current grouping helpers recognize explicit project-title aliases instead of blindly using the first populated cell as a title. _resolved_import_rows forward-fills merged project context within a worksheet block. _import_group_key combines the academic cycle, normalized title and group/block discriminator. _archive_import_group combines members and academic metadata. Existing matching projects can receive merged non-destructive data and deduplicated members.

archive_import_batches records original filename, uploader, processed/inserted/skipped/invalid counts, status and time. archived_projects.source_metadata preserves columns, cells, worksheet, source rows and row range. Flexible content is retained in JSONB; mapped metadata supports search and normal display. An imported worksheet with many member rows should not necessarily create the same number of projects.

Legacy XLS formula detection is less explicit than the XLSX path: xlrd generally returns cell values and the fallback formula-type comparison is not proof that every cached legacy formula can be excluded. State this as a format limitation and use the fixture tests to demonstrate supported behavior.

11.3 Search ranking

The inspected archive search is deterministic, not semantic/vector search. A normalized exact title match scores 100, a prefix/contained title match 80, keyword match 60, summary token overlap 40, scope/technology overlap 30 and remaining-gap overlap 20. Results sort by descending score and stable title/id ordering. The response explicitly reports semantic_search: unavailable and advisory: true.

Search filters include year/session/department/supervisor/technology and supported scope/stage parameters. Student-visible results require active, visible archive records. The pgvector extension's presence does not change this search into embedding similarity. An honest defence answer is that vector capability is installed but the current archive route uses explainable lexical scoring.

11.4 Edit, delete and historic documents

The editor preserves original workbook columns and updates stored cells through PATCH. Imported values can be remapped into structured fields for display. Archived project document versions are stored separately in archived_project_documents, including BYTEA content for original PDFs. Download authorization is checked before bytes are returned.

The current archive delete endpoint permanently deletes the selected archive and its stored document versions according to its implementation. The centered frontend confirmation is only UX; backend role/origin checks still enforce access. Visibility/soft-retirement operations also exist separately. Explain which operation the demonstrated button invokes.

12. Progress, meetings, discussions and notifications

12.1 Progress data and visual timeline

project_workflow.py stores milestones, tasks and supervision meetings. StudentProgressDashboard.tsx loads a permitted group, reads /project-progress/{team_id}, renders summary cards/timeline and submits create/update actions. A task may link to a milestone and an owner. Records include due dates and completion percentages.

The timeline is a CSS grid representation of dates and task/milestone spans. It is not an AI scheduler or a full critical-path scheduling engine. Completion updates are persisted through API calls. The supervisor workspace derives overall completion from task values, falling back to milestone values when appropriate.

_health counts overdue incomplete tasks/milestones: at least three produce Critical; any overdue item or a milestone below 50% produces Attention Required; otherwise On Track. This is a transparent heuristic. It does not infer project quality from code or proposal content.

Meetings record held_at, supervisor, minutes, decisions and optional action items. A stored meeting entry is supervision evidence; it is not proof that calendar invitations were emailed or a video call was created.

12.2 Three conversation-related concepts

- supervisor_request_messages is a request-specific conversation used before/after request decisions.

- collaboration_channels/participants/messages is an assignment-specific group discussion. It links to assignment_id, team_id and supervisor_id, so reassignment can archive an older channel instead of merging unrelated histories.

- group_messages and module2_ helpers are an earlier/general group communication and notification foundation. Check which frontend imports/API path the demonstration uses.

The explicit student/supervisor collaboration-message routes are now registered separately. An earlier invalid route pattern had caused 404 errors and was corrected. Backend authorization verifies group/assignment access independently of the URL role prefix.

Student group discussion navigation was intentionally removed earlier. The backend still supports authorized student discussion routes, and request conversation remains separately available. Do not promise an always-visible student discussion button if the current sidebar does not expose it.

12.3 Notifications

notifications.py lists recipient-scoped module2_notifications and marks individual notifications read. module2_collaboration.py contains supporting deadline/channel data. Read status is stored; polling is used by relevant UI components. A notification table is not an external email/push service and should be described separately from SMTP password recovery.

13. Evaluations, defence scheduling and results

13.1 Two related evaluation implementations

project_workflow.py provides defense_schedules, defense_panel_members, panel_evaluations and panel_evaluation_scores. It checks time ordering, student/evaluator scheduling conflicts, assignment scope and panel scoring state. EvaluatorWorkflow.tsx consumes these APIs. PanelScheduling.tsx exists but must be checked against the current coordinator composition before claiming it is reachable.

evaluations.py is a separate formal marks/publication API using evaluation_records, evaluation_audit_events, evaluation_corrections and evaluation_aggregation_configs. EvaluationMarksWorkspace.tsx exists, but its current reachability and role-shell integration should be checked in the frontend appendix. The two scoring stores are not automatically one synchronized table.

13.2 Formal evaluation categories

Category	Maximum marks	Responsible role in the inspected API
Final presentation	30	Assigned PANEL_MEMBER
Mid review	30	Assigned PANEL_MEMBER
Supervisor evaluation	10	Assigned SUPERVISOR
Project documentation	20	ADMIN / COORDINATOR
Proposal	10	ADMIN / COORDINATOR
Total	100	Consolidated result calculation

Formal state progresses through DRAFT, SUBMITTED, RETURNED, APPROVED and PUBLISHED according to the supported handlers. Marks must fit the category maximum. Target input identifies exactly one student or group. Admin correction/publishing creates history rather than pretending the originally submitted mark never existed.

Panel aggregation can use AVERAGE, MAXIMUM or MEDIAN according to active configuration. The result total sums the five category values. In the inspected grade mapping, 90+ gives A+/4.00, 85+ A/3.75, 80+ A-/3.50, 75+ B+/3.25, 70+ B/3.00, 66+ B-/2.75, 63+ C+/2.50, 60+ C/2.00, 55+ C-/1.50, otherwise F/0.00. Incomplete category data produces no final grade/GPA. Published student results select published marks only.

These are application rules in source, not a claim that they match every university's grading policy. Be prepared to point at schemas/evaluations.py and evaluations._grade and explain where a university-approved policy would be configured or changed.

14. Storage, deployment and operations

14.1 Where the real data lives

Data	Persistent location	Source configuration
PostgreSQL records	Named volume fyphub-p1-postgres-data mounted at /var/lib/postgresql/data	docker-compose.yml db service
Required-document uploads and compliance files	Named volume fyphub-p1-document-data mounted at /app/storage/document-requirements	DOCUMENT_STORAGE_ROOT and Compose backend volume
Supervisor-request PDFs and some historic/reference documents	BYTEA columns in their document tables	Corresponding ORM content fields and download handlers
Ollama model files	fyphub-p1-ollama-models mounted at /root/.ollama	Compose ollama service
Source code	Repository checkout	Git
Served browser assets	Frontend image/container /usr/share/nginx/html	frontend Dockerfile/Nginx
Private runtime settings	Root .env and container environment	config.py and Compose env_file

Document storage can include a feature subdirectory such as document-requirements/{requirement}/{actor}/random.ext beneath DOCUMENT_STORAGE_ROOT. Do not confuse the apparent repeated directory name with a second database. Keep the database and file volume consistent when backing up/restoring.

14.2 Containers and ports

PostgreSQL is reachable inside the Docker network as db:5432 and from this computer at 127.0.0.1:55732. The backend listens on container 8000 and is exposed on host 8010. The production frontend serves container port 80 on host 5173. Ollama uses 11434 inside the network. Container-local localhost is not the database container; use service DNS names within Compose.

The backend's health endpoint reports liveness. /health/ready checks SQL access and the vector extension; it does not check every table, migration, stored document or AI model. The migrate service runs alembic upgrade head. The backend depends on successful migration, model initialization and storage initialization under normal Compose startup.

14.3 Useful commands for defence preparation

Run these from the repository root unless stated otherwise. They are diagnostic examples; none of the documentation-generation commands reset the database.

```
docker compose ps -a
docker compose logs --tail=100 backend
docker compose logs --tail=100 migrate
Invoke-WebRequest http://localhost:5173
Invoke-WebRequest http://localhost:8010/health/ready
docker compose exec backend alembic heads
docker compose exec backend alembic current
```

Build the frontend from frontend/ using npm.cmd run build on Windows. The dev server uses npm.cmd run dev and Vite's configured /api proxy. A local source edit does not automatically change an already-built production Nginx container. Rebuild/redeploy deliberately and verify the served assets, especially if manual docker cp deployment was used during development.

14.4 Migration status found during inspection

The migration repository currently has three heads: fb1c2d3e4f5, qa1b2c3d4e5f and rf9a0b1c2d3e. The running alembic_version table reports rf9a0b1c2d3e. The migrate service is exited with code 255 while the existing application/database services are healthy.

Therefore a fresh docker compose up that runs alembic upgrade head must not be described as verified working from an empty database. Multiple heads make singular head ambiguous and may indicate branches requiring a reviewed merge. This documentation task does not stamp, merge, downgrade or reset the database. Existing application readiness does not resolve migration-history consistency.

Appendix E lists every source revision and parent. Appendix A compares ORM and installed tables. Historical m03_ and team_member_verifications tables remain in the running database although they are absent from current ORM metadata. They may be remnants of earlier modules; do not delete them during defence preparation without a data/migration review.

14.5 Backup and recovery principles

A PostgreSQL dump protects database records, including BYTEA files inside the database. It does not automatically protect files stored in the separate uploaded-files volume. Back up that volume too. Ollama models can be backed up separately or downloaded again if network/time permit. Source and environment configuration are separate assets; keep .env private.

Use pg_dump with the configured database/user. Do not display database passwords during a screen recording. Avoid docker compose down -v because it removes persistent volumes. A valid recovery rehearsal restores into an isolated environment and verifies schema, documents and login, rather than overwriting the working defence database.

15. Security, transactions and error handling

15.1 Controls present in the code

- Authentication resolves a hashed session and active account on the server.

- Role guards use the database role, not a role chosen by the browser.

- Group/request/document access checks restrict cross-group data access.

- Pydantic schemas normalize and validate field types, required values, ranges and allowed enums.

- SQLAlchemy uses parameterized expressions rather than concatenating user input into SQL statements.

- Row locks and partial unique constraints protect allocation/capacity workflows.

- Revision checks reject stale request decisions.

- Origin checks protect mutating browser requests within the configured deployment.

- File handlers enforce the implemented type/size/signature checks, random storage names and resolved root paths.

- Admin and evaluation events store actor, target, action, changed data and timestamps.

- Request middleware adds X-Request-ID and structured timing/status logs.

15.2 Error meanings you should explain

Status	Typical meaning in this application	Example
200 / 201	Successful read/update or creation	Group details / uploaded version
204	Successful deletion/action without response content	Delete unused draft
400	Invalid operation-specific input	Bad reset token
401	No valid authenticated session	Expired/revoked cookie
403	Authenticated but not permitted	Wrong role or document ownership
404	Missing or intentionally hidden inaccessible record	Another supervisor's assigned group
409	Workflow, capacity, concurrency or uniqueness conflict	Full capacity, closed requirement, stale revision
410	Metadata exists but stored file is absent	Download reference whose file is unavailable
413	Payload too large	Backend byte limit or proxy request limit
422	Request validation failure	Missing member name, invalid type/extension
500 / 503	Internal failure or dependency unavailable	Persistence exception / database readiness failure

An HTTP fetch rejection is not always a Python error. Network failure, backend startup, proxy resolution, cookie scope, CORS/origin rules, and body limits can all affect requests. Inspect the Network tab and request_id, then correlate Nginx/backend logs. A browser-rendered error message alone is insufficient evidence of its root cause.

15.3 Limits of the current implementation

Avoid claiming antivirus scanning, full OCR, automated plagiarism detection, distributed rate limiting, WebSocket delivery, production TLS termination, comprehensive fine-grained permissions or full microservice isolation unless the specific implementation is demonstrated. They are not established by the inspected code alone.

16. Testing evidence and how to present it

16.1 What the test directories cover

Backend tests include registration/auth/session revocation, password policy/reset, team constraints and compatibility helpers, cycle concurrency, user governance, deadlines, reminder/rubric/template versions, supervisor capacity/requests/assignments, PDF extraction, archive import/persistence, document visibility, group access/collaboration, proposal compliance, AI provider behavior, progress/scheduling, evaluations/results and observability.

The generated file atlas enumerates each test method. Many integration suites connect to fyphub_p1, wrap operations in an outer transaction and use savepoints so data can roll back. Read the test fixture before running it; do not assume every test is an isolated in-memory unit test. The root/backend Python may not have pytest installed even if Docker has the application dependencies. Some suites are unittest-based; others use pytest-style functions.

Frontend tests are Node .mjs sources for selected contracts/visibility. npm build runs TypeScript compilation and Vite production bundling. npm lint runs ESLint. These checks do not prove that every browser interaction was visually tested.

16.2 Verification claims for this guide

This documentation task performs read-only runtime schema/route inspection, source parsing/import tracing, and generated-file checks. It does not claim a fresh pass of every existing test. Earlier implementation work verified selected supervisor workspace integration cases and frontend builds, but the current source inventory is a separate snapshot. Record actual new test results if you run a defence rehearsal after this document is generated.

Example targeted commands, subject to the fixture/environment requirements:

```
Frontend directory: npm.cmd run build
Frontend directory: npm.cmd run lint
Backend directory: python -m unittest tests.test_supervisor_workspace
Backend directory: python -m pytest tests/test_archive_import.py
Repository root: python scripts/build_defence_guide.py --build
```

Tests are not always copied into the backend production image by its Dockerfile. If a container cannot import tests, that is a packaging/path issue rather than proof of a failing academic workflow. Use the documented project test environment or copy/mount tests deliberately.

17. Known gaps and honest defence boundaries

Finding	Evidence	What to say / what it affects
Multiple Alembic heads; migrate exited 255	alembic heads/current and Compose status	Existing stack works; clean schema bootstrap needs migration reconciliation
Legacy tables remain	Catalog has m03_ / team_member_verifications absent from ORM	Installed schema includes historical objects; current models are the active mapping
Agent UI removed, backend remains	Supervisor navigation versus registered agent router	Product exposure and backend capability are different
Archive search is lexical	archived_projects.search_archives response and scoring	Explainable deterministic search; no current embedding search
Several advertised file formats lack implemented signatures	document_requirements._validate	Verify actual demo file types; dropdown alone is not end-to-end support
Proxy upload limit not explicitly set	frontend/nginx.conf	Large uploads may be rejected before FastAPI
Some frontend navigation is state-driven	App.tsx and role shells	Refresh/deep links are only supported where coded
Coordinator sidebar selection does not switch all content	CoordinatorPortal.tsx fixed content composition	PanelScheduling source exists but current shell does not mount it by page state
Multiple evaluation/proposal workflows	Distinct models and routers	Do not claim automatic synchronization between independent stores
Supervisor project is assembled/derived	supervisor_workspace.workspace	No standalone general projects table; completion comes from tasks/milestones
Group display name is derived	Team model and group serializers	Response group name is usually built from leader data
In-memory rate limits	ai_provider.py, password_reset.py	Single-process protection; not a distributed limiter
CSS has layered overrides and !important	design-system.css plus feature styles	Current appearance depends on cascade/import order; consolidate for maintainability
Compliance uses lexical/format heuristics	compliance_engine.py	Useful automated checks; not a substitute for academic review or semantic proof
Student discussion entry removed earlier	StudentPortal phase list	Supported backend discussion does not imply visible navigation everywhere

This is a preparation report, not an authorization to repair these issues. Keep this table available for viva questions about limitations and future work. State the boundary clearly and describe the appropriate next engineering step.

18. Demonstration plan with source and database checkpoints

18.1 Pre-demo check

Verify frontend/backend health, known accounts/roles, the chosen academic cycle, configured group limits and supervisor capacity, a published document requirement, an active proposal template and the existence of the actual sample files. Prepare small supported files to avoid proxy/MIME surprises. Keep personal passwords and private .env values off screen.

18.2 Main demonstration path

1. Admin shows academic cycle and group policy. Explain academic_cycles/academic_group_policies and the registration guard.
2. Student signs up a group using member names, emails and registrations plus shared credentials. Explain users -> teams -> team_members.
3. Student logs in. Explain cookie/token hashing and show /auth/me response fields without exposing credentials.
4. Student submits a supervisor proposal request, inspects extraction, confirms and submits. Explain request/document/note/history tables.
5. Supervisor opens the request and checks members, proposal, documents and capacity before deciding.
6. Supervisor accepts. Explain transaction, row locks, unique active assignment and request.assignment_id.
7. Supervisor opens Assigned Students & Teams and a full group detail route. Explain the joined workspace response.
8. Student uploads a required document. Admin opens View submissions, and an authorized user downloads the original version. Explain metadata versus file bytes.
9. Student creates/updates milestone or task progress. Supervisor view picks it up on the next polling load. Explain completion/health derivation.
10. Show request conversation or supported group discussion and demonstrate a message appearing for the authorized recipient.

11. Admin imports a prepared historic workbook. Explain one project parent, multiple members, source provenance and skipped formulas/blanks.

12. Student searches the archive, opens a project and explains deterministic scoring and visibility.

13. Demonstrate compliance with a configured active template only if the selected sample meets the supported extraction contract.

14. Demonstrate evaluation through its actually mounted UI/API path; explain publication and audit if used. Avoid opening an unmounted placeholder as a promised finished feature.

18.3 Read-only SQL questions to practice

These example queries show relationships. Execute them only in an authorized database session. Do not select password_hash, token_hash or private uploaded bytes for a public demonstration.

```
-- Find groups and their shared login identity.
SELECT t.id, u.full_name, u.email, c.display_name, t.status
FROM teams t
JOIN users u ON u.id = t.shared_user_id
JOIN academic_cycles c ON c.id = t.academic_cycle_id
ORDER BY t.id DESC;

-- All group members, showing real stored names.
SELECT tm.team_id, tm.full_name, tm.registration_number,
       tm.university_email, tm.is_creator, tm.is_active
FROM team_members tm
ORDER BY tm.team_id, tm.is_creator DESC, tm.id;

-- Active allocations with readable supervisor/cycle values.
SELECT sa.team_id, u.full_name AS supervisor, c.display_name,
       sa.created_at AS assigned_at
FROM supervisor_assignments sa
JOIN users u ON u.id = sa.supervisor_id
JOIN academic_cycles c ON c.id = sa.academic_cycle_id
WHERE sa.ended_at IS NULL;

-- A requirement and its submitted versions.
SELECT r.title, s.group_id, s.status, v.version_number,
       v.original_filename, v.file_size, v.uploaded_at
FROM document_upload_requirements r
JOIN document_submissions s ON s.requirement_id = r.id
JOIN document_submission_versions v ON v.submission_id = s.id
ORDER BY r.id, s.id, v.version_number;

-- Project archive provenance without dumping original private cells.
SELECT p.title, p.academic_year, p.session_name,
       p.source_batch_id, b.original_file_name
FROM archived_projects p
LEFT JOIN archive_import_batches b ON b.batch_id = p.source_batch_id
WHERE p.is_active = TRUE;

-- Show installed migration state and installed extensions.
SELECT version_num FROM alembic_version;
SELECT extname, extversion FROM pg_extension ORDER BY extname;
```

19. Viva questions and accurate answers

19.1 Architecture and frontend

Q: What architecture does FYPHub use?

A: A modular monolith: a React SPA, one FastAPI backend with feature routers, PostgreSQL, file storage and optional AI. Containers separate deployment processes; academic modules remain in the same application.

Q: Where does execution start?

A: frontend/src/main.tsx mounts App.tsx. Backend Uvicorn imports app.main:app. main.py configures middleware, core handlers and feature routers.

Q: How does the frontend know which portal to show?

A: Login or /auth/me returns the persisted user role and group context. App.tsx selects the matching portal; backend permissions remain independently enforced.

Q: Is hiding a sidebar item security?

A: No. It is UI exposure. Authentication, roles and ownership are enforced by backend dependencies and queries, and some removed UI features still have registered endpoints.

Q: Is this React Router / Redux?

A: The inspected dependency list uses React state/effects and manual history logic. Do not claim React Router or Redux when they are not installed/used.

Q: How do changes appear automatically?

A: The relevant views refetch after actions and use polling timers. This is periodic HTTP refresh, not WebSocket push.

Q: Where is the system color defined?

A: main.tsx imports index.css followed by design-system.css. Feature CSS and utility classes also contribute. The final brand override uses royal blue #124BC4; sign-in/signup retains its photo and translucent form card.

19.2 Backend and security

Q: Why use FastAPI and Pydantic?

A: FastAPI provides HTTP routes, dependency injection and OpenAPI. Pydantic validates/normalizes request contracts before workflow logic executes. Database constraints provide a separate integrity layer.

Q: Are passwords stored in plaintext or reversibly encrypted?

A: Neither. security.py hashes passwords with Argon2id and verifies the hash. Responses exclude password_hash.

Q: Are login tokens JWTs?

A: No. The browser stores an opaque HttpOnly cookie, while auth_sessions stores a hash, expiry and revocation state.

Q: How do you stop one supervisor reading another group?

A: owned_group queries an active assignment matching both group and authenticated supervisor. The backend does not trust the group_id or frontend role label.

Q: Why use a revision number for request decisions?

A: It rejects an action based on stale data if the student or another action has changed the request since the supervisor loaded it.

Q: What prevents a race for the last supervisor slot?

A: Locked capacity/user/cycle records, a capacity recheck inside the transaction, and uniqueness constraints for active group assignment. Frontend availability is advisory and can become stale.

Q: What happens if assignment creation succeeds but history writing fails?

A: The transaction rolls back when the handler catches the failure, preserving consistency rather than committing a partial allocation.

Q: Is SQL injection prevented by hiding SQL from users?

A: No. SQLAlchemy parameterized queries keep values separate from SQL structure; validation and authorization are additional protections.

Q: How do you protect uploaded paths?

A: Random server-side filenames, sanitized display names and root-resolved storage references. Access checks happen before serving the original file.

19.3 Database and features

Q: Where is the database file for supervisor requests?

A: There is no separate supervisor database file. supervisor_requests and its related tables live in PostgreSQL, whose physical data is in the PostgreSQL volume. The model is backend/app/models/supervisor_request.py.

Q: Why separate submission and version tables?

A: The parent identifies one academic submission; child rows preserve every uploaded version and its metadata. Replacing a file should not erase the version history.

Q: Why do imported projects use JSONB?

A: Different historical workbooks have different columns. JSONB preserves the original flexible structure while mapped relational fields support search and stable parent identity.

Q: Can project titles act as primary keys?

A: No. Titles can repeat across years/groups or be edited. Numeric IDs are stable keys; normalized title plus scope/context is used for import matching and selected uniqueness rules.

Q: Does a PDF download come from React?

A: React requests an authorized backend endpoint. The backend returns actual bytes from file storage or a document BYTEA field, with the correct content type and filename.

Q: What is the difference between group_id and team_id?

A: They are naming aliases in several application contracts for the same teams identity. Check each column/relationship rather than assuming a separate groups table exists.

Q: Where is the project completion stored?

A: Tasks/milestones store completion_percent. Supervisor workspace completion_percentage is derived from those records; archived projects can separately store historic imported completion.

Q: Why is there more than one evaluation implementation?

A: The code contains a panel scheduling/scoring workflow and a separate formal marks/publication workflow. They have different parent tables and must be explained separately until explicitly integrated.

Q: What does a migration do?

A: It applies an ordered structural change to the database. Models describe intended mappings; migrations and installed alembic_version determine which schema transitions were applied.

Q: Does a healthy readiness endpoint guarantee all features are ready?

A: No. It verifies database connectivity and the vector extension. It does not validate every table, file, migration branch, SMTP server or AI model.

19.4 AI, extraction and algorithms

Q: Is proposal text extraction AI?

A: The current PDF/DOCX extraction is local deterministic parsing. A separate configured provider can generate advisory feedback; these are different stages.

Q: Is the compliance score an LLM opinion?

A: The compliance engine computes checks and a passed/total percentage with Python rules. Optional advisory AI review is stored separately.

Q: Can your system detect plagiarism?

A: The inspected title/reference search and lexical consistency checks are advisory. They are not a demonstrated full plagiarism or semantic originality engine.

Q: Why is pgvector installed if search is not vector-based?

A: The extension is part of the data platform/readiness requirement. The current archive route explicitly uses deterministic lexical ranking. Installing an extension does not implement embeddings or nearest-neighbor search.

Q: What happens when the AI service is down?

A: The provider layer returns explicit failure/fallback metadata; supported callers use deterministic fallback or an unavailable state. Normal authentication and database operations do not depend on successful AI feedback.

Q: Does temperature zero guarantee correct AI output?

A: No. Schema validation checks structure and constraints; academic correctness still requires review. Prompt instructions and confidence values do not guarantee truth.

Q: Why can a proposal check be UNVERIFIABLE?

A: Some formats do not expose a reliable measurable value, particularly editable formatting lost in PDFs. Marking uncertainty is more accurate than inventing a measurement.

Q: How does the archive avoid thousands of empty rows?

A: It strips blank/error/formula cells, ignores empty rows/columns, finds a content boundary and uses the last literal email where available. Grouping then combines multiple member rows under a project parent.

Q: Are all spreadsheet formulas safely excluded in every format?

A: The XLSX path explicitly compares formula/data-only readings. The legacy XLS reader has weaker formula visibility, so claim the tested behavior and state that format limitation.

19.5 Deployment, limitations and future work

Q: Why can localhost work in the browser but fail inside Docker?

A: localhost refers to the current machine/container. Services use Compose DNS names such as db and backend inside the Docker network.

Q: What must be backed up?

A: PostgreSQL plus uploaded-file storage; source/configuration and optionally model downloads are separate. A SQL dump alone does not include filesystem-only uploads.

Q: What are the most important current limitations?

A: Migration branching, partial/legacy UI integration, upload-format/signature gaps, state-driven navigation, lexical rather than semantic search, and separate evaluation/proposal stores. Point to the evidence table and a concrete future change.

Q: What would you improve for production?

A: Reconcile migration heads and test a clean install; consolidate theme/components and API error handling; align upload limits and validators; add distributed rate limiting/background jobs if needed; integrate overlapping workflows; add browser end-to-end coverage; configure

real-domain HTTPS/origin handling and verified backups.

20. Final revision checklist

- Can every team member explain the full browser -> API -> database/file flow for at least one major feature?
- Can you locate the model, schema, handler, active frontend file and relevant test in under one minute?
- Can you distinguish shared group accounts from member contact records?
- Can you explain the request, assignment and discussion foreign keys without relying on screenshots?
- Can you download the original submitted file and explain where its bytes reside?
- Can you show an actual negative authorization or stale/capacity check?
- Can you explain which algorithms are deterministic and which calls use an external/local model?
- Can you state the known migration and file-format gaps accurately?
- Is your demonstration data prepared, small enough for proxy limits, and free of private credentials on screen?
- Have you checked the current application after the last build rather than relying only on an old screenshot?

The remaining appendices are generated directly from the inspected source and running schema. Use them as the exact lookup reference for file locations, line numbers, API ownership, table columns and database constraints.

Appendix A - Inspection evidence and counts

Inspection: 2026-11-11 (Asia/Karachi). Git snapshot: f11c5064af86777e2867054de692baef3ca7e11a. The guide describes source and the running database as inspected, not a claim that every feature was tested during this documentation task.

Inspected source/configuration files in the generated inventory: 352. Live public tables: 70. ORM tables: 58. Registered application route entries: 264. Multiple methods/aliases can share one handler.

Database migration version(s): rf9a0b1c2d3e.

Migration repository heads: fb1c2d3e4f5 (head), qa1b2c3d4e5f (head), rf9a0b1c2d3e (head).

Extensions: plpgsql 1.0, vector 0.8.6.

Tables present in database but absent from current ORM metadata: alembic_version, m03_ai_runs, m03_compliance_findings, m03_compliance_reports, m03_proposal_access, m03_proposal_change_summaries, m03_proposal_documents, m03_proposal_records, m03_proposal_submissions, m03_reference_templates, m03_template_rule_sets, team_member_verifications. Historical/retired tables are not proof of active features.

ORM tables absent from the live database: None.

Appendix B - Complete feature database dictionary

This is generated from the running PostgreSQL catalog. No user rows, passwords, tokens, uploaded content or secrets are included. Locate the model file beside each table; the model declares the intended structure, while the catalog describes the actual installed database.

Table: academic_cycles

Model source: backend/app/models/academic.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".academic_cycles_id_seq'::regclass)
code	VARCHAR(100)	No	-
display_name	VARCHAR(255)	No	-
registration_open	BOOLEAN	No	false
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
academic_year	INTEGER	Yes	-
session	VARCHAR(16)	Yes	-

- Check ck_academic_cycles_code_normalized: code::text = upper(btrim(code::text)) AND code::text <> ''::text.

- Check ck_academic_cycles_display_name_nonblank: display_name::text = btrim(display_name::text) AND display_name::text <> ''::text.

- Check ck_academic_cycles_session_valid: session IS NULL OR (session::text = ANY (ARRAY['FALL'::character varying, 'SPRING'::character varying]::text[])).

- Check ck_academic_cycles_year_valid: academic_year IS NULL OR academic_year >= 1900 AND academic_year <= 2200.

- Index uq_academic_cycles_code_normalized; unique=True; columns/expressions [None]; options {'postgresql_include': []}.

- Index uq_academic_cycles_year_session; unique=True; columns/expressions ['academic_year', 'session']; options {'postgresql_where': '((academic_year IS NOT NULL) AND (session IS NOT NULL))', 'postgresql_include': []}.

Table: academic_deadline_versions

Model source: backend/app/models/academic_deadline.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".academic_deadline_versions_id_seq'::regclass)
academic_deadline_id	INTEGER	No	-
version_number	INTEGER	No	-
name	VARCHAR(160)	No	-
description	TEXT	Yes	-
due_at_utc	TIMESTAMP WITH TIME ZONE	No	-
time_zone	VARCHAR(100)	No	-
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_deadline_id -> academic_deadlines(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_academic_deadline_version: academic_deadline_id, version_number.

- Index uq_academic_deadline_version; unique=True; columns/expressions ['academic_deadline_id', 'version_number']; options {'postgresql_include': []}.

Table: academic_deadlines

Model source: backend/app/models/academic_deadline.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".academic_deadlines_id_seq'::regclass)
academic_cycle_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

Table: academic_group_policies

Model source: backend/app/models/academic.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".academic_group_policies_id_seq'::regclass)
academic_cycle_id	INTEGER	No	-
max_group_size	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
min_group_size	INTEGER	No	1
formation_deadline	TIMESTAMP WITH TIME ZONE	Yes	-
matching_weights	JSONB	No	'{}'::jsonb

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint academic_group_policies_academic_cycle_id_key: academic_cycle_id.

- Check ck_academic_group_policies_max_group_size_positive: max_group_size > 0.

- Index academic_group_policies_academic_cycle_id_key; unique=True; columns/expressions ['academic_cycle_id']; options {'postgresql_include': []}.

Table: academic_rule_versions

Model source: backend/app/models/template_reference.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".academic_rule_versions_id_seq'::regclass)
template_reference_version_id	INTEGER	No	-
version_number	INTEGER	No	-
provenance	VARCHAR(32)	No	-
rules	JSONB	No	-
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key template_reference_version_id -> template_reference_versions(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_academic_rule_version: template_reference_version_id, version_number.

- Index uq_academic_rule_version; unique=True; columns/expressions ['template_reference_version_id', 'version_number']; options {'postgresql_include': []}.

Table: admin_audit_events

Model source: backend/app/models/audit_event.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".admin_audit_events_id_seq'::regclass)
actor_id	INTEGER	No	-

Column	PostgreSQL type	Nullable	Default
affected_user_id	INTEGER	Yes	-
action	VARCHAR(64)	No	-
changed_values	JSONB	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
target_type	VARCHAR(32)	No	'USER'::character varying
target_id	INTEGER	No	-

- Foreign key actor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key affected_user_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_admin_audit_events_affected_user_id; unique=False; columns/expressions ['affected_user_id']; options {'postgresql_include': []}.

- Index ix_admin_audit_events_created_at; unique=False; columns/expressions ['created_at']; options {'postgresql_include': []}.

Table: alembic_version

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: version_num.

Column	PostgreSQL type	Nullable	Default
version_num	VARCHAR(32)	No	-

Table: archive_import_batches

Model source: backend/app/models/archive_import_batch.py.

Primary key: batch_id.

Column	PostgreSQL type	Nullable	Default
batch_id	VARCHAR(64)	No	-
original_file_name	VARCHAR(255)	No	-
uploaded_by_id	INTEGER	No	-
uploaded_at	TIMESTAMP WITH TIME ZONE	No	now()
total_rows	INTEGER	No	0
inserted_rows	INTEGER	No	0
skipped_rows	INTEGER	No	0
invalid_rows	INTEGER	No	0
import_status	VARCHAR(32)	No	'PROCESSING'::character varying

- Foreign key uploaded_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

Table: archived_project_documents

Model source: backend/app/models/archived_project.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".archived_project_documents_id_seq'::regclass)
archived_project_id	INTEGER	No	-
document_type	VARCHAR(32)	No	-
display_name	VARCHAR(255)	No	-
storage_key	VARCHAR(512)	No	-
file_hash	VARCHAR(64)	No	-
mime_type	VARCHAR(100)	No	-
file_size	INTEGER	No	-
content	BYTEA	No	-
version	INTEGER	No	-
uploaded_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

Column	PostgreSQL type	Nullable	Default
is_active	BOOLEAN	No	true

- Foreign key archived_project_id -> archived_projects(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key uploaded_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_archived_project_documents_project_type; unique=False; columns/expressions ['archived_project_id', 'document_type', 'version']; options {'postgresql_include': []}.

- Index uq_archived_project_active_document_type; unique=True; columns/expressions ['archived_project_id', 'document_type']; options {'postgresql_where': '(is_active = true)', 'postgresql_include': []}.

Table: archived_projects

Model source: backend/app/models/archived_project.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".archived_projects_id_seq'::regclass)
title	VARCHAR(255)	No	-
normalized_title	VARCHAR(255)	No	-
abstract_summary	TEXT	No	-
keywords	VARCHAR(100)[]	No	'{}'::character varying[]
academic_cycle_id	INTEGER	Yes	-
academic_year	VARCHAR(64)	No	-
department_program	VARCHAR(255)	No	-
project_scope	VARCHAR(255)	No	-
completion_stage	VARCHAR(100)	No	-
implemented_scope	TEXT	No	-
remaining_gaps	TEXT	No	-
future_work	TEXT	No	-
supervisor_display_reference	VARCHAR(255)	Yes	-
source_label	VARCHAR(64)	No	-
source_metadata	JSONB	No	'{}'::jsonb
is_active	BOOLEAN	No	true
archived_at	TIMESTAMP WITH TIME ZONE	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
project_code	VARCHAR(120)	Yes	-
group_leader_name	VARCHAR(255)	Yes	-
group_leader_registration_number	VARCHAR(120)	Yes	-
group_members	JSONB	No	-
supervisor_name	VARCHAR(255)	Yes	-
session_name	VARCHAR(16)	Yes	-
problem_statement	TEXT	Yes	-
objectives	TEXT	Yes	-
methodology	TEXT	Yes	-
technologies_used	TEXT	Yes	-
main_features	TEXT	Yes	-
project_status	VARCHAR(100)	Yes	-
completion_percentage	INTEGER	Yes	-
outcomes_results	TEXT	Yes	-
repository_url	VARCHAR(1000)	Yes	-
report_url	VARCHAR(1000)	Yes	-
presentation_url	VARCHAR(1000)	Yes	-
external_reference	VARCHAR(1000)	Yes	-
notes	TEXT	Yes	-

Column	PostgreSQL type	Nullable	Default
is_visible_to_students	BOOLEAN	No	true
normalized_group_name	VARCHAR(255)	No	-
group_name	VARCHAR(255)	Yes	-
source_batch_id	VARCHAR(64)	Yes	-
source_row_number	INTEGER	Yes	-

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_archived_projects_active_cycle; unique=False; columns/expressions ['is_active', 'academic_year']; options {'postgresql_include': []}.

- Index ix_archived_projects_active_scope; unique=False; columns/expressions ['is_active', 'project_scope']; options {'postgresql_include': []}.

- Index ix_archived_projects_cycle_code; unique=False; columns/expressions ['academic_cycle_id', 'project_code']; options {'postgresql_include': []}.

- Index ix_archived_projects_normalized_title; unique=False; columns/expressions ['normalized_title']; options {'postgresql_include': []}.

- Index ix_archived_projects_source_batch_id; unique=False; columns/expressions ['source_batch_id']; options {'postgresql_include': []}.

- Index uq_archived_project_active_title_year_group; unique=True; columns/expressions ['normalized_title', 'academic_year', ' normalized_group_name']; options {'postgresql_where': '(is_active = true)', 'postgresql_include': []}.

Table: auth_sessions

Model source: backend/app/models/auth_session.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".auth_sessions_id_seq'::regclass)
user_id	INTEGER	No	-
token_hash	VARCHAR(64)	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
expires_at	TIMESTAMP WITH TIME ZONE	No	-
revoked_at	TIMESTAMP WITH TIME ZONE	Yes	-

- Foreign key user_id -> users(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint auth_sessions_token_hash_key: token_hash.

- Index auth_sessions_token_hash_key; unique=True; columns/expressions ['token_hash']; options {'postgresql_include': []}.

- Index ix_auth_sessions_expires_at; unique=False; columns/expressions ['expires_at']; options {'postgresql_include': []}.

- Index ix_auth_sessions_user_id; unique=False; columns/expressions ['user_id']; options {'postgresql_include': []}.

Table: collaboration_channels

Model source: backend/app/models/module2_collaboration.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".collaboration_channels_id_seq'::regclass)
team_id	INTEGER	No	-
supervisor_id	INTEGER	No	-
academic_year	INTEGER	No	-
session	VARCHAR(16)	No	-
assignment_id	INTEGER	No	-
status	VARCHAR(16)	No	'ACTIVE'::character varying
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key assignment_id -> supervisor_assignments(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key supervisor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_collaboration_channels_supervisor_status; unique=False; columns/expressions ['supervisor_id', 'status']; options {'postgresql_include': []}.

- Index uq_collaboration_channel_active_assignment; unique=True; columns/expressions ['team_id', 'supervisor_id', 'assignment_id']; options {'postgresql_where': "((status)::text = 'ACTIVE'::text)", 'postgresql_include': []}.

Table: collaboration_messages

Model source: backend/app/models/module2_collaboration.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".collaboration_messages_id_seq'::regclass)
channel_id	INTEGER	No	-
sender_user_id	INTEGER	No	-
message_text	TEXT	No	-
attachment_reference	VARCHAR(500)	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
deleted_at	TIMESTAMP WITH TIME ZONE	Yes	-

- Foreign key channel_id -> collaboration_channels(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key sender_user_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_collaboration_messages_channel_created; unique=False; columns/expressions ['channel_id', 'created_at', 'id']; options {'postgresql_include': []}.

Table: collaboration_participants

Model source: backend/app/models/module2_collaboration.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".collaboration_participants_id_seq'::regclass)
channel_id	INTEGER	No	-
user_id	INTEGER	Yes	-
team_member_id	INTEGER	Yes	-
participant_role	VARCHAR(24)	No	-
joined_at	TIMESTAMP WITH TIME ZONE	No	now()
is_active	BOOLEAN	No	true

- Foreign key channel_id -> collaboration_channels(id); options: {'ondelete': 'CASCADE'}.

- Foreign key team_member_id -> team_members(id); options: {'ondelete': 'CASCADE'}.

- Foreign key user_id -> users(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint uq_collaboration_participant_user: channel_id, user_id.

- Index ix_collaboration_participants_channel_active; unique=False; columns/expressions ['channel_id', 'is_active']; options {'postgresql_include': []}.

- Index uq_collaboration_participant_user; unique=True; columns/expressions ['channel_id', 'user_id']; options {'postgresql_include': []}.

Table: dashboard_requirements

Model source: backend/app/models/dashboard_requirement.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".dashboard_requirements_id_seq'::regclass)
academic_deadline_id	INTEGER	Yes	-
academic_cycle_id	INTEGER	No	-
title	VARCHAR(200)	No	-

Column	PostgreSQL type	Nullable	Default
description	TEXT	Yes	-
requirement_type	VARCHAR(24)	No	-
module	VARCHAR(120)	No	-
priority	VARCHAR(16)	No	'MEDIUM'::character varying
role_visibility	JSONB	No	'[]'::jsonb
assigned_user_id	INTEGER	Yes	-
assigned_group_id	INTEGER	Yes	-
due_at	TIMESTAMP WITH TIME ZONE	No	-
completion_percent	INTEGER	No	0
status	VARCHAR(20)	No	'UPCOMING'::character varying
is_required	BOOLEAN	No	true
completed_at	TIMESTAMP WITH TIME ZONE	Yes	-
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.
- Foreign key academic_deadline_id -> academic_deadlines(id); options: {'ondelete': 'RESTRICT'}.
- Foreign key assigned_group_id -> teams(id); options: {'ondelete': 'RESTRICT'}.
- Foreign key assigned_user_id -> users(id); options: {'ondelete': 'RESTRICT'}.
- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.
- Index ix_dashboard_requirements_cycle; unique=False; columns/expressions ['academic_cycle_id']; options {'postgresql_include': []}.
- Index ix_dashboard_requirements_status_due; unique=False; columns/expressions ['status', 'due_at']; options {'postgresql_include': []}.
- Index ix_dashboard_requirements_user; unique=False; columns/expressions ['assigned_user_id']; options {'postgresql_include': []}.

Table: defense_panel_members

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".defense_panel_members_id_seq'::regclass)
schedule_id	INTEGER	No	-
user_id	INTEGER	No	-
assigned_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key assigned_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.
- Foreign key schedule_id -> defense_schedules(id); options: {'ondelete': 'CASCADE'}.
- Foreign key user_id -> users(id); options: {'ondelete': 'RESTRICT'}.
- Unique constraint uq_defense_panel_member: schedule_id, user_id.
- Index uq_defense_panel_member; unique=True; columns/expressions ['schedule_id', 'user_id']; options {'postgresql_include': []}.

Table: defense_schedules

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".defense_schedules_id_seq'::regclass)
team_id	INTEGER	No	-
academic_cycle_id	INTEGER	No	-
created_by_id	INTEGER	No	-

Column	PostgreSQL type	Nullable	Default
start_at	TIMESTAMP WITH TIME ZONE	No	-
end_at	TIMESTAMP WITH TIME ZONE	No	-
room	VARCHAR(255)	Yes	-
online_link	VARCHAR(1000)	Yes	-
status	VARCHAR(16)	No	'SCHEDULED'::character varying
conflict_note	TEXT	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Check ck_defense_schedules_status: status::text = ANY (ARRAY['SCHEDULED'::character varying, 'RESCHEDULED'::character varying, ' CANCELLED'::character varying]::text[]).

- Check ck_defense_schedules_time_order: end_at > start_at.

- Index ix_defense_schedules_time; unique=False; columns/expressions ['start_at', 'end_at']; options {'postgresql_include': []}.

Table: document_submission_versions

Model source: backend/app/models/document_requirement.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".document_submission_versions_id_seq'::regclass)
submission_id	INTEGER	No	-
version_number	INTEGER	No	-
original_filename	VARCHAR(255)	No	-
stored_filename	VARCHAR(255)	No	-
mime_type	VARCHAR(120)	No	-
file_size	INTEGER	No	-
storage_reference	VARCHAR(500)	No	-
content	BYTEA	Yes	-
uploaded_by_id	INTEGER	No	-
uploaded_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key submission_id -> document_submissions(id); options: {'ondelete': 'CASCADE'}.

- Foreign key uploaded_by_id -> users(id); options: {}.

- Unique constraint document_submission_versions_stored_filename_key: stored_filename.

- Unique constraint uq_document_submission_version: submission_id, version_number.

- Index document_submission_versions_stored_filename_key; unique=True; columns/expressions ['stored_filename']; options {'postgresql_include': []}.

- Index ix_document_versions_submission; unique=False; columns/expressions ['submission_id']; options {'postgresql_include': []}.

- Index uq_document_submission_version; unique=True; columns/expressions ['submission_id', 'version_number']; options {'postgresql_include': []}.

Table: document_submissions

Model source: backend/app/models/document_requirement.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".document_submissions_id_seq'::regclass)
requirement_id	INTEGER	No	-
student_id	INTEGER	No	-

Column	PostgreSQL type	Nullable	Default
status	VARCHAR(20)	No	-
is_late	BOOLEAN	No	false
review_comment	TEXT	Yes	-
reviewed_by_id	INTEGER	Yes	-
reviewed_at	TIMESTAMP WITH TIME ZONE	Yes	-
uploaded_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
group_id	INTEGER	Yes	-
academic_cycle_id	INTEGER	Yes	-
uploaded_by_id	INTEGER	Yes	-

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key group_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key requirement_id -> document_upload_requirements(id); options: {'ondelete': 'CASCADE'}.

- Foreign key reviewed_by_id -> users(id); options: {}.

- Foreign key student_id -> users(id); options: {}.

- Foreign key uploaded_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_document_submission_requirement_student: requirement_id, student_id.

- Index ix_document_submissions_student; unique=False; columns/expressions ['student_id']; options {'postgresql_include': []}.

- Index uq_document_submission_requirement_group; unique=True; columns/expressions ['requirement_id', 'group_id']; options {'postgresql_where': '(group_id IS NOT NULL)', 'postgresql_include': []}.

- Index uq_document_submission_requirement_student; unique=True; columns/expressions ['requirement_id', 'student_id']; options {'postgresql_include': []}.

Table: document_upload_requirements

Model source: backend/app/models/document_requirement.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".document_upload_requirements_id_seq'::regclass)
title	VARCHAR(200)	No	-
description	TEXT	Yes	-
academic_cycle_id	INTEGER	No	-
module	VARCHAR(120)	No	-
visible_to_roles	JSONB	No	-
assigned_user_id	INTEGER	Yes	-
assigned_group_id	INTEGER	Yes	-
allowed_file_types	JSONB	No	-
max_file_size_mb	INTEGER	No	-
due_at	TIMESTAMP WITH TIME ZONE	No	-
reminder_days_before	INTEGER	No	-
is_required	BOOLEAN	No	true
status	VARCHAR(16)	No	'DRAFT'::character varying
late_upload_allowed	BOOLEAN	No	false
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_cycle_id -> academic_cycles(id); options: {}.

- Foreign key assigned_group_id -> teams(id); options: {}.

- Foreign key assigned_user_id -> users(id); options: {}.

- Foreign key created_by_id -> users(id); options: {}.

- Index ix_document_requirements_assigned_user; unique=False; columns/expressions ['assigned_user_id']; options {'postgresql_include': []}.

- Index ix_document_requirements_status_due; unique=False; columns/expressions ['status', 'due_at']; options {'postgresql_include': []}.

Table: evaluation_aggregation_configs

Model source: backend/app/models/evaluation.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_aggregation_configs_id_seq'::regclass)
evaluation_category	VARCHAR(32)	No	-
method	VARCHAR(16)	No	'AVERAGE'::character varying
is_active	BOOLEAN	No	true
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_evaluation_aggregation_category: evaluation_category.

- Check ck_evaluation_aggregation_category: evaluation_category::text = ANY (ARRAY['FINAL_PRESENTATION'::character varying, ' MID_REVIEW'::character varying]::text[]).

- Check ck_evaluation_aggregation_method: method::text = ANY (ARRAY['AVERAGE'::character varying, 'MAXIMUM'::character varying, ' MEDIAN'::character varying]::text[]).

- Index uq_evaluation_aggregation_category; unique=True; columns/expressions ['evaluation_category']; options {'postgresql_include': []}.

Table: evaluation_audit_events

Model source: backend/app/models/evaluation.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_audit_events_id_seq'::regclass)
evaluation_id	INTEGER	No	-
actor_id	INTEGER	No	-
action	VARCHAR(48)	No	-
previous_marks	INTEGER	Yes	-
new_marks	INTEGER	Yes	-
reason	TEXT	Yes	-
changed_values	JSONB	No	'{}'::jsonb
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key actor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key evaluation_id -> evaluation_records(id); options: {'ondelete': 'CASCADE'}.

- Index ix_evaluation_audit_record_created; unique=False; columns/expressions ['evaluation_id', 'created_at']; options {'postgresql_include': []}.

Table: evaluation_corrections

Model source: backend/app/models/evaluation.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_corrections_id_seq'::regclass)
evaluation_id	INTEGER	No	-
authorized_by_id	INTEGER	No	-
previous_marks	INTEGER	No	-
new_marks	INTEGER	No	-

Column	PostgreSQL type	Nullable	Default
reason	TEXT	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key authorized_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key evaluation_id -> evaluation_records(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_evaluation_corrections_evaluation; unique=False; columns/expressions ['evaluation_id', 'created_at']; options {'postgresql_include': []}.

Table: evaluation_records

Model source: backend/app/models/evaluation.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_records_id_seq'::regclass)
student_id	INTEGER	Yes	-
group_id	INTEGER	Yes	-
evaluator_id	INTEGER	No	-
evaluator_role	VARCHAR(32)	No	-
evaluation_category	VARCHAR(32)	No	-
maximum_marks	INTEGER	No	-
obtained_marks	INTEGER	Yes	-
feedback	TEXT	Yes	-
status	VARCHAR(16)	No	'DRAFT'::character varying
returned_reason	TEXT	Yes	-
submitted_at	TIMESTAMP WITH TIME ZONE	Yes	-
approved_at	TIMESTAMP WITH TIME ZONE	Yes	-
published_at	TIMESTAMP WITH TIME ZONE	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key evaluator_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key group_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key student_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_evaluation_group_category_evaluator: group_id, evaluation_category, evaluator_id.

- Unique constraint uq_evaluation_student_category_evaluator: student_id, evaluation_category, evaluator_id.

- Check ck_evaluation_category: evaluation_category::text = ANY (ARRAY['FINAL_PRESENTATION'::character varying, ' MID_REVIEW'::character varying, 'SUPERVISOR_EVALUATION'::character varying, 'PROJECT_DOCUMENTATION'::character varying, ' PROPOSAL'::character varying]::text[]).

- Check ck_evaluation_marks_range: obtained_marks IS NULL OR obtained_marks >= 0 AND obtained_marks <= maximum_marks.

- Check ck_evaluation_maximum: maximum_marks = ANY (ARRAY[30, 10, 20]).

- Check ck_evaluation_one_target: student_id IS NOT NULL AND group_id IS NULL OR student_id IS NULL AND group_id IS NOT NULL.

- Check ck_evaluation_status: status::text = ANY (ARRAY['DRAFT'::character varying, 'SUBMITTED'::character varying, ' RETURNED'::character varying, 'APPROVED'::character varying, 'PUBLISHED'::character varying]::text[]).

- Index ix_evaluation_records_category_status; unique=False; columns/expressions ['evaluation_category', 'status']; options {'postgresql_include': []}.

- Index ix_evaluation_records_evaluator; unique=False; columns/expressions ['evaluator_id', 'status']; options {'postgresql_include': []}.

- Index ix_evaluation_records_group; unique=False; columns/expressions ['group_id']; options {'postgresql_include': []}.

- Index ix_evaluation_records_student; unique=False; columns/expressions ['student_id']; options {'postgresql_include': []}.

- Index uq_evaluation_group_category_evaluator; unique=True; columns/expressions ['group_id', 'evaluation_category', 'evaluator_id']; options {'postgresql_include': []}.

- Index uq_evaluation_student_category_evaluator; unique=True; columns/expressions ['student_id', 'evaluation_category', 'evaluator_id']; options {'postgresql_include': []}.

Table: evaluation_rubric_criteria

Model source: backend/app/models/evaluation_rubric.py.

Primary key: id.

Column-name drift: model-only []; database-only ['max_score', 'weight'].

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_rubric_criteria_id_seq'::regclass)
rubric_version_id	INTEGER	No	-
name	VARCHAR(160)	No	-
description	TEXT	Yes	-
display_order	INTEGER	No	-
max_score	INTEGER	No	10
weight	DOUBLE PRECISION	No	'1'::double precision

- Foreign key rubric_version_id -> evaluation_rubric_versions(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_evaluation_rubric_criterion_order: rubric_version_id, display_order.

- Check ck_rubric_criterion_max_score: max_score >= 1 AND max_score <= 1000.

- Check ck_rubric_criterion_weight: weight > 0::double precision AND weight <= 100::double precision.

- Index uq_evaluation_rubric_criterion_order; unique=True; columns/expressions ['rubric_version_id', 'display_order']; options {'postgresql_include': []}.

Table: evaluation_rubric_versions

Model source: backend/app/models/evaluation_rubric.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_rubric_versions_id_seq'::regclass)
rubric_id	INTEGER	No	-
version_number	INTEGER	No	-
name	VARCHAR(160)	No	-
description	TEXT	Yes	-
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key rubric_id -> evaluation_rubrics(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_evaluation_rubric_version: rubric_id, version_number.

- Index uq_evaluation_rubric_version; unique=True; columns/expressions ['rubric_id', 'version_number']; options {'postgresql_include': []}.

Table: evaluation_rubrics

Model source: backend/app/models/evaluation_rubric.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".evaluation_rubrics_id_seq'::regclass)
created_at	TIMESTAMP WITH TIME ZONE	No	now()

Table: group_messages

Model source: backend/app/models/group_message.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".group_messages_id_seq'::regclass)
team_id	INTEGER	No	-
channel	VARCHAR(32)	No	-
author_id	INTEGER	No	-
author_kind	VARCHAR(16)	No	-
body	TEXT	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
thread_root_id	INTEGER	Yes	-
mentions	JSONB	No	'[]'::jsonb
attachment_name	VARCHAR(255)	Yes	-
attachment_url	VARCHAR(500)	Yes	-

- Foreign key author_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_group_messages_team_channel_id; unique=False; columns/expressions ['team_id', 'channel', 'id']; options {'postgresql_include': []}.

Table: m03_ai_runs

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_ai_runs_id_seq'::regclass)
proposal_id	VARCHAR(128)	No	-
report_id	VARCHAR(128)	Yes	-
check_name	VARCHAR(64)	No	-
status	VARCHAR(32)	No	-
runtime	VARCHAR(128)	No	-
model	VARCHAR(128)	Yes	-
result	JSONB	No	'{}'::jsonb
error_message	TEXT	Yes	-
started_at	TIMESTAMP WITH TIME ZONE	No	now()
finished_at	TIMESTAMP WITH TIME ZONE	Yes	-

- Index ix_m03_ai_runs_proposal; unique=False; columns/expressions ['proposal_id']; options {'postgresql_include': []}.

Table: m03_compliance_findings

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_compliance_findings_id_seq'::regclass)
report_id	VARCHAR(128)	No	-
check_name	VARCHAR(64)	No	-
severity	VARCHAR(16)	No	-
rule_reference	VARCHAR(255)	No	-
student_value	JSONB	No	'{}'::jsonb
baseline_value	JSONB	No	'{}'::jsonb
evidence	JSONB	No	'{}'::jsonb
remediation	TEXT	No	-
confidence	VARCHAR(24)	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Index ix_m03_findings_report; unique=False; columns/expressions ['report_id']; options {'postgresql_include': []}.

Table: m03_compliance_reports

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_compliance_reports_id_seq'::regclass)
report_id	VARCHAR(128)	No	-
proposal_id	VARCHAR(128)	No	-
proposal_document_id	VARCHAR(128)	No	-
baseline_template_id	VARCHAR(128)	No	-
proposal_type	VARCHAR(16)	No	-
report_version	INTEGER	No	-
status	VARCHAR(32)	No	-
check_1	JSONB	No	'{}'::jsonb
check_2	JSONB	No	'{}'::jsonb
check_3	JSONB	No	'{}'::jsonb
severity_counts	JSONB	No	'{}'::jsonb
remediation	JSONB	No	'[]'::jsonb
advisory_summary	TEXT	No	-
runtime_metadata	JSONB	No	'{}'::jsonb
generated_by_user_id	VARCHAR(128)	No	-
generated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Unique constraint m03_compliance_reports_report_id_key: report_id.

- Unique constraint uq_m03_report_proposal_version: proposal_id, report_version.

- Index ix_m03_compliance_reports_proposal; unique=False; columns/expressions ['proposal_id', 'report_version']; options {'postgresql_include': []}.

- Index m03_compliance_reports_report_id_key; unique=True; columns/expressions ['report_id']; options {'postgresql_include': []}.

- Index uq_m03_report_proposal_version; unique=True; columns/expressions ['proposal_id', 'report_version']; options {'postgresql_include': []}.

Table: m03_proposal_access

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_proposal_access_id_seq'::regclass)
proposal_id	VARCHAR(128)	No	-
user_id	INTEGER	No	-
granted_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Unique constraint uq_m03_proposal_access_user: proposal_id, user_id.

- Index ix_m03_proposal_access_proposal_id; unique=False; columns/expressions ['proposal_id']; options {'postgresql_include': []}.

- Index ix_m03_proposal_access_user_id; unique=False; columns/expressions ['user_id']; options {'postgresql_include': []}.

- Index uq_m03_proposal_access_user; unique=True; columns/expressions ['proposal_id', 'user_id']; options {'postgresql_include': []}.

Table: m03_proposal_change_summaries

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_proposal_change_summaries_id_seq'::regclass)
document_id	INTEGER	No	-

Column	PostgreSQL type	Nullable	Default
status	VARCHAR(24)	No	-
summary	TEXT	No	-
runtime_metadata	JSONB	No	-

- Unique constraint uq_m03_change_summary_document: document_id.

- Index uq_m03_change_summary_document; unique=True; columns/expressions ['document_id']; options {'postgresql_include': []}.

Table: m03_proposal_documents

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_proposal_documents_id_seq'::regclass)
proposal_id	VARCHAR(128)	No	-
proposal_type	VARCHAR(16)	No	-
owner_user_id	VARCHAR(128)	No	-
team_id	VARCHAR(128)	Yes	-
filename	VARCHAR(255)	No	-
storage_key	VARCHAR(512)	No	-
sha256	VARCHAR(64)	No	-
mime_type	VARCHAR(100)	No	-
file_size	INTEGER	No	-
content	BYTEA	No	-
version	INTEGER	No	-
extraction_status	VARCHAR(24)	No	-
ast	JSONB	No	'{}'::jsonb
extraction_warnings	JSONB	No	'[]'::jsonb
is_active	BOOLEAN	No	true
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Unique constraint uq_m03_proposal_document_version: proposal_id, version.

- Index ix_m03_proposal_documents_owner; unique=False; columns/expressions ['owner_user_id', 'is_active']; options {'postgresql_include': []}.

- Index uq_m03_proposal_document_version; unique=True; columns/expressions ['proposal_id', 'version']; options {'postgresql_include': []}.

Table: m03_proposal_records

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	VARCHAR(128)	No	-
owner_user_id	VARCHAR(128)	No	-
proposal_type	VARCHAR(16)	No	-
title	VARCHAR(255)	No	-
status	VARCHAR(24)	No	'DRAFT'::character varying
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Index ix_m03_proposal_records_owner_user_id; unique=False; columns/expressions ['owner_user_id']; options {'postgresql_include': []}.

Table: m03_proposal_submissions

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_proposal_submissions_id_seq'::regclass)
proposal_id	VARCHAR(128)	No	-
document_id	INTEGER	No	-
version	INTEGER	No	-
action	VARCHAR(24)	No	-
actor_id	VARCHAR(128)	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Unique constraint uq_m03_submission_document: proposal_id, document_id.

- Index ix_m03_proposal_submissions_proposal_id; unique=False; columns/expressions ['proposal_id']; options {'postgresql_include': []}.

- Index uq_m03_submission_document; unique=True; columns/expressions ['proposal_id', 'document_id']; options {'postgresql_include': []}.

Table: m03_reference_templates

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_reference_templates_id_seq'::regclass)
template_type	VARCHAR(16)	No	-
filename	VARCHAR(255)	No	-
storage_key	VARCHAR(512)	No	-
sha256	VARCHAR(64)	No	-
mime_type	VARCHAR(100)	No	-
file_size	INTEGER	No	-
content	BYTEA	No	-
version	INTEGER	No	-
extraction_status	VARCHAR(24)	No	-
extracted_rules	JSONB	No	'{}'::jsonb
extraction_warnings	JSONB	No	'[]'::jsonb
confidence	VARCHAR(24)	Yes	-
is_active	BOOLEAN	No	false
uploaded_by_user_id	VARCHAR(128)	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
activated_at	TIMESTAMP WITH TIME ZONE	Yes	-
version_kind	VARCHAR(24)	No	'REFERENCE'::character varying
parent_template_id	INTEGER	Yes	-
template_metadata	JSONB	No	'{}'::jsonb
provenance	VARCHAR(64)	No	'Observed reference property'::character varying
activation_mode	VARCHAR(16)	Yes	-

- Unique constraint uq_m03_reference_template_type_version: template_type, version.

- Index ix_m03_reference_templates_active_type; unique=False; columns/expressions ['template_type', 'is_active']; options {'postgresql_include': []}.

- Index ix_m03_reference_templates_parent; unique=False; columns/expressions ['parent_template_id']; options {'postgresql_include': []}.

- Index uq_m03_reference_template_type_version; unique=True; columns/expressions ['template_type', 'version']; options {'postgresql_include': []}.

Table: m03_template_rule_sets

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".m03_template_rule_sets_id_seq'::regclass)
reference_template_id	INTEGER	No	-
rule_key	VARCHAR(128)	No	-
rule_value	JSONB	No	'{}'::jsonb
source	VARCHAR(255)	No	-
confidence	VARCHAR(24)	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
rule_version	INTEGER	No	1
content_hash	VARCHAR(64)	No	''::character varying
provenance	VARCHAR(64)	No	'Observed reference property'::character varying

- Unique constraint uq_m03_template_rule_version_key: reference_template_id, rule_version, rule_key.

- Index ix_m03_template_rule_sets_template; unique=False; columns/expressions ['reference_template_id']; options {'postgresql_include': []}.

- Index ix_m03_template_rule_sets_version; unique=False; columns/expressions ['reference_template_id', 'rule_version']; options {'postgresql_include': []}.

- Index uq_m03_template_rule_version_key; unique=True; columns/expressions ['reference_template_id', 'rule_version', 'rule_key']; options {'postgresql_include': []}.

Table: module2_channel_reads

Model source: backend/app/models/module2_collaboration.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".module2_channel_reads_id_seq'::regclass)
user_id	INTEGER	No	-
team_id	INTEGER	No	-
channel	VARCHAR(32)	No	-
last_read_message_id	INTEGER	Yes	-
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key team_id -> teams(id); options: {'ondelete': 'CASCADE'}.

- Foreign key user_id -> users(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint uq_module2_channel_read: user_id, team_id, channel.

- Index uq_module2_channel_read; unique=True; columns/expressions ['user_id', 'team_id', 'channel']; options {'postgresql_include': []}.

Table: module2_chatbot_sessions

Model source: backend/app/models/module2_collaboration.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".module2_chatbot_sessions_id_seq'::regclass)
team_id	INTEGER	No	-
user_id	INTEGER	No	-
title	VARCHAR(200)	No	'Group assistant'::character varying
context	JSON	No	'{}'::json
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key team_id -> teams(id); options: {'ondelete': 'CASCADE'}.

- Foreign key user_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_module2_chatbot_sessions_team_created; unique=False; columns/expressions ['team_id', 'created_at']; options {'postgresql_include': []}.

Table: module2_notifications

Model source: backend/app/models/module2_collaboration.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".module2_notifications_id_seq'::regclass)
recipient_user_id	INTEGER	No	-
kind	VARCHAR(64)	No	-
payload	JSON	No	'{}'::json
read_at	TIMESTAMP WITH TIME ZONE	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key recipient_user_id -> users(id); options: {'ondelete': 'CASCADE'}.

- Index ix_module2_notifications_recipient_created; unique=False; columns/expressions ['recipient_user_id', 'created_at']; options {'postgresql_include': []}.

Table: panel_evaluation_scores

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".panel_evaluation_scores_id_seq'::regclass)
evaluation_id	INTEGER	No	-
criterion_key	VARCHAR(160)	No	-
score	INTEGER	No	-
max_score	INTEGER	No	-
feedback	TEXT	Yes	-

- Foreign key evaluation_id -> panel_evaluations(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint uq_panel_evaluation_score_criterion: evaluation_id, criterion_key.

- Check ck_panel_evaluation_scores_range: score >= 0 AND max_score > 0 AND score <= max_score.

- Index uq_panel_evaluation_score_criterion; unique=True; columns/expressions ['evaluation_id', 'criterion_key']; options {'postgresql_include': []}.

Table: panel_evaluations

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".panel_evaluations_id_seq'::regclass)
schedule_id	INTEGER	No	-
evaluator_id	INTEGER	No	-
status	VARCHAR(24)	No	'DRAFT'::character varying
overall_score	INTEGER	Yes	-
overall_feedback	TEXT	Yes	-
decision	VARCHAR(64)	Yes	-
disagreement_detected	BOOLEAN	No	false
resolution_note	TEXT	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key evaluator_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key schedule_id -> defense_schedules(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_panel_evaluation_schedule_evaluator: schedule_id, evaluator_id.

- Check ck_panel_evaluations_score: overall_score IS NULL OR overall_score >= 0 AND overall_score <= 100.

- Check ck_panel_evaluations_status: status::text = ANY (ARRAY['DRAFT'::character varying, 'SUBMITTED'::character varying, 'RECHECK_REQUESTED'::character varying, 'RESOLVED'::character varying]::text[]).

- Index uq_panel_evaluation_schedule_evaluator; unique=True; columns/expressions ['schedule_id', 'evaluator_id']; options {'postgresql_include': []}.

Table: password_reset_tokens

Model source: backend/app/models/password_reset_token.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".password_reset_tokens_id_seq'::regclass)
user_id	INTEGER	No	-
token_hash	VARCHAR(64)	No	-
expires_at	TIMESTAMP WITH TIME ZONE	No	-
used_at	TIMESTAMP WITH TIME ZONE	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
requested_ip	VARCHAR(64)	Yes	-

- Foreign key user_id -> users(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint uq_password_reset_tokens_token_hash: token_hash.

- Index ix_password_reset_tokens_expires_at; unique=False; columns/expressions ['expires_at']; options {'postgresql_include': []}.

- Index ix_password_reset_tokens_user_id; unique=False; columns/expressions ['user_id']; options {'postgresql_include': []}.

- Index uq_password_reset_tokens_token_hash; unique=True; columns/expressions ['token_hash']; options {'postgresql_include': []}.

Table: project_milestones

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".project_milestones_id_seq'::regclass)
team_id	INTEGER	No	-
created_by_id	INTEGER	No	-
title	VARCHAR(255)	No	-
description	TEXT	Yes	-
due_at	TIMESTAMP WITH TIME ZONE	Yes	-
completion_percent	INTEGER	No	0
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Check ck_project_milestones_completion: completion_percent >= 0 AND completion_percent <= 100.

- Index ix_project_milestones_team_due; unique=False; columns/expressions ['team_id', 'due_at']; options {'postgresql_include': []}.

Table: project_tasks

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".project_tasks_id_seq'::regclass)
team_id	INTEGER	No	-
milestone_id	INTEGER	Yes	-

Column	PostgreSQL type	Nullable	Default
owner_user_id	INTEGER	Yes	-
created_by_id	INTEGER	No	-
title	VARCHAR(255)	No	-
description	TEXT	Yes	-
due_at	TIMESTAMP WITH TIME ZONE	Yes	-
completion_percent	INTEGER	No	0
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key milestone_id -> project_milestones(id); options: {'ondelete': 'SET NULL'}.

- Foreign key owner_user_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Check ck_project_tasks_completion: completion_percent >= 0 AND completion_percent <= 100.

- Index ix_project_tasks_team_due; unique=False; columns/expressions ['team_id', 'due_at']; options {'postgresql_include': []}.

Table: proposal_ai_review_runs

Model source: backend/app/models/proposal_ai_review.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposal_ai_review_runs_id_seq'::regclass)
proposal_id	VARCHAR(128)	No	-
status	VARCHAR(32)	No	-
runtime	VARCHAR(128)	No	-
model	VARCHAR(128)	Yes	-
result	JSONB	No	-
error_message	TEXT	Yes	-
started_at	TIMESTAMP WITH TIME ZONE	No	now()
finished_at	TIMESTAMP WITH TIME ZONE	Yes	-

- Index ix_proposal_ai_review_runs_proposal; unique=False; columns/expressions ['proposal_id']; options {'postgresql_include': []}.

Table: proposal_compliance_reports

Model source: backend/app/models/proposal_compliance.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposal_compliance_reports_id_seq'::regclass)
submission_id	INTEGER	No	-
status	VARCHAR(32)	No	-
score	INTEGER	No	-
total_checks	INTEGER	No	-
report	JSONB	No	'{}'::jsonb
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key submission_id -> proposal_compliance_submissions(id); options: {'ondelete': 'CASCADE'}.

- Unique constraint uq_proposal_compliance_report_submission: submission_id.

- Index ix_proposal_compliance_report_created; unique=False; columns/expressions ['created_at']; options {'postgresql_include': []}.

- Index uq_proposal_compliance_report_submission; unique=True; columns/expressions ['submission_id']; options {'postgresql_include': []}.

Table: proposal_compliance_submissions

Model source: backend/app/models/proposal_compliance.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposal_compliance_submissions_id_seq'::regclass)
proposal_key	VARCHAR(64)	No	-
owner_user_id	INTEGER	No	-
team_id	INTEGER	Yes	-
proposal_type	VARCHAR(5)	No	-
title	VARCHAR(200)	No	-
version	INTEGER	No	1
status	VARCHAR(32)	No	'SUBMITTED'::character varying
processing_status	VARCHAR(32)	No	'QUEUED'::character varying
filename	VARCHAR(255)	No	-
content_type	VARCHAR(100)	No	-
file_size	INTEGER	No	-
sha256	VARCHAR(64)	No	-
page_count	INTEGER	No	-
storage_key	VARCHAR(500)	No	-
extracted_text	TEXT	No	-
extracted_fields	JSONB	No	'{}'::jsonb
template_reference_version_id	INTEGER	Yes	-
error_message	TEXT	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key template_reference_version_id -> template_reference_versions(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key owner_user_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_proposal_compliance_submission_version: proposal_key, version.

- Check ck_proposal_compliance_submission_type: proposal_type::text = ANY (ARRAY['SHORT'::character varying, 'LONG'::character varying]::text[]).

- Check ck_proposal_compliance_submission_version_positive: version > 0.

- Index ix_proposal_compliance_submission_key; unique=False; columns/expressions ['proposal_key', 'version']; options {'postgresql_include': []}.

- Index ix_proposal_compliance_submission_owner; unique=False; columns/expressions ['owner_user_id', 'created_at']; options {'postgresql_include': []}.

- Index uq_proposal_compliance_submission_version; unique=True; columns/expressions ['proposal_key', 'version']; options {'postgresql_include': []}.

Table: proposal_reference_documents

Model source: backend/app/models/proposal_reference.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposal_reference_documents_id_seq'::regclass)
reference_id	INTEGER	No	-
filename	VARCHAR(255)	No	-
content_type	VARCHAR(100)	No	-
file_size	INTEGER	No	-
sha256	VARCHAR(64)	No	-
content	BYTEA	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key reference_id -> proposal_references(id); options: {'ondelete': 'RESTRICT'}.

- Index uq_proposal_reference_document_reference; unique=True; columns/expressions ['reference_id']; options {'postgresql_include': []}.

Table: proposal_references

Model source: backend/app/models/proposal_reference.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposal_references_id_seq'::regclass)
title	VARCHAR(255)	No	-
normalized_title	VARCHAR(255)	No	-
short_summary	TEXT	No	-
academic_cycle_id	INTEGER	Yes	-
academic_year	VARCHAR(32)	No	-
category_scope	VARCHAR(255)	No	-
source_label	VARCHAR(64)	No	-
source_metadata	JSONB	No	'{}'::jsonb
is_active	BOOLEAN	No	true
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_proposal_references_cycle_active; unique=False; columns/expressions ['academic_cycle_id', 'is_active']; options {'postgresql_include': []}.

- Index ix_proposal_references_normalized_title; unique=False; columns/expressions ['normalized_title']; options {'postgresql_include': []}.

Table: proposal_versions

Model source: backend/app/models/proposal.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposal_versions_id_seq'::regclass)
proposal_id	INTEGER	No	-
version_number	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key proposal_id -> proposals(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_proposal_version_number: proposal_id, version_number.

- Check ck_proposal_version_positive: version_number > 0.

- Index uq_proposal_version_number; unique=True; columns/expressions ['proposal_id', 'version_number']; options {'postgresql_include': []}.

Table: proposals

Model source: backend/app/models/proposal.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".proposals_id_seq'::regclass)
team_id	INTEGER	No	-
proposal_type	VARCHAR(5)	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Check ck_proposals_type: proposal_type::text = ANY (ARRAY['SHORT'::character varying, 'LONG'::character varying]::text[]).

Table: reminder_rule_versions

Model source: backend/app/models/reminder_rule.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".reminder_rule_versions_id_seq'::regclass)
rule_id	INTEGER	No	-
version_number	INTEGER	No	-
deadline_version_id	INTEGER	No	-
name	VARCHAR(160)	No	-
description	TEXT	Yes	-
category	VARCHAR(16)	No	-
direction	VARCHAR(6)	No	-
elapsed_minutes	INTEGER	No	-
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key deadline_version_id -> academic_deadline_versions(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key rule_id -> reminder_rules(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_reminder_rule_version: rule_id, version_number.

- Check ck_reminder_rule_category: category::text = ANY (ARRAY['REMINDER'::character varying, 'ESCALATION'::character varying]::text[]).

- Check ck_reminder_rule_direction: direction::text = ANY (ARRAY['BEFORE'::character varying, 'AFTER'::character varying]::text[]).

- Check ck_reminder_rule_minutes: elapsed_minutes > 0.

- Index uq_reminder_rule_version; unique=True; columns/expressions ['rule_id', 'version_number']; options {'postgresql_include': []}.

Table: reminder_rules

Model source: backend/app/models/reminder_rule.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".reminder_rules_id_seq'::regclass)
academic_cycle_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

Table: supervision_agent_runs

Model source: backend/app/models/agent_supervision.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervision_agent_runs_id_seq'::regclass)
team_id	INTEGER	No	-
proposal_request_id	INTEGER	Yes	-
created_by_id	INTEGER	No	-
approved_by_id	INTEGER	Yes	-
state	VARCHAR(32)	No	'IDLE'::character varying
approval_status	VARCHAR(24)	No	'PENDING'::character varying
plan	JSONB	No	'{}'::jsonb
tool_calls	JSONB	No	'[]'::jsonb
result	JSONB	No	'{}'::jsonb
error	TEXT	Yes	-
approved_at	TIMESTAMP WITH TIME ZONE	Yes	-

Column	PostgreSQL type	Nullable	Default
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key approved_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key proposal_request_id -> supervisor_requests(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_supervision_agent_runs_team_created; unique=False; columns/expressions ['team_id', 'created_at']; options {'postgresql_include': []}.

Table: supervision_meetings

Model source: backend/app/models/project_workflow.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervision_meetings_id_seq'::regclass)
team_id	INTEGER	No	-
supervisor_id	INTEGER	No	-
created_by_id	INTEGER	No	-
held_at	TIMESTAMP WITH TIME ZONE	No	-
minutes	TEXT	No	-
decisions	TEXT	Yes	-
action_items	JSONB	No	'[]'::jsonb
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key supervisor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id -> teams(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_supervision_meetings_team_held; unique=False; columns/expressions ['team_id', 'held_at']; options {'postgresql_include': []}.

Table: supervisor_assignments

Model source: backend/app/models/supervisor_assignment.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_assignments_id_seq'::regclass)
team_id	INTEGER	No	-
academic_cycle_id	INTEGER	No	-
supervisor_id	INTEGER	No	-
assigned_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
ended_at	TIMESTAMP WITH TIME ZONE	Yes	-
ended_by_id	INTEGER	Yes	-
academic_year	INTEGER	Yes	-
session	VARCHAR(16)	Yes	-

- Foreign key assigned_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key ended_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key supervisor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id, academic_cycle_id -> teams(id, academic_cycle_id); options: {'ondelete': 'RESTRICT'}.

- Check ck_supervisor_assignments_session_valid: session IS NULL OR (session::text = ANY (ARRAY['FALL'::character varying, 'SPRING'::character varying]::text[])).

- Check ck_supervisor_assignments_year_valid: academic_year IS NULL OR academic_year >= 1900 AND academic_year <= 2200.

- Index ix_supervisor_assignments_supervisor_year_session; unique=False; columns/expressions ['supervisor_id', 'academic_year', 'session']; options {'postgresql_include': []}.

- Index uq_supervisor_assignment_active_team; unique=True; columns/expressions ['team_id']; options {'postgresql_where': '(ended_at IS NULL)', 'postgresql_include': []}.

Table: supervisor_capacity_policies

Model source: backend/app/models/academic.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_capacity_policies_id_seq'::regclass)
supervisor_id	INTEGER	No	-
academic_cycle_id	INTEGER	Yes	-
max_group_capacity	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
academic_year	INTEGER	Yes	-

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key supervisor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_supervisor_capacity_policies_supervisor_year: supervisor_id, academic_year.

- Check ck_supervisor_capacity_policies_capacity_positive: max_group_capacity > 0.

- Check ck_supervisor_capacity_policies_year_valid: academic_year IS NULL OR academic_year >= 1900 AND academic_year <= 2200.

- Index uq_supervisor_capacity_policies_supervisor_year; unique=True; columns/expressions ['supervisor_id', 'academic_year']; options {'postgresql_include': []}.

Table: supervisor_request_documents

Model source: backend/app/models/supervisor_request.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_request_documents_id_seq'::regclass)
request_id	INTEGER	No	-
version	INTEGER	No	-
filename	VARCHAR(255)	No	-
content_type	VARCHAR(100)	No	-
file_size	INTEGER	No	-
sha256	VARCHAR(64)	No	-
page_count	INTEGER	No	-
content	BYTEA	No	-
extracted_text	TEXT	No	-
extracted_fields	JSONB	No	-
extraction_status	VARCHAR(16)	No	-
extraction_runtime	VARCHAR(100)	No	-
extracted_at	TIMESTAMP WITH TIME ZONE	No	-
confirmed_at	TIMESTAMP WITH TIME ZONE	Yes	-
confirmed_by_id	INTEGER	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key confirmed_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key request_id -> supervisor_requests(id); options: {'ondelete': 'RESTRICT'}.

- Check ck_supervisor_request_documents_extraction_status: extraction_status::text = ANY (ARRAY['PENDING'::character varying, ' EXTRACTED'::character varying, 'FAILED'::character varying, 'CONFIRMED'::character varying]::text[]).

- Check ck_supervisor_request_documents_file_size_positive: file_size > 0.

- Check ck_supervisor_request_documents_one_page: page_count = 1.

- Check ck_supervisor_request_documents_version_positive: version > 0.

- Index uq_supervisor_request_document_version; unique=True; columns/expressions ['request_id', 'version']; options {'postgresql_include': []}.

Table: supervisor_request_messages

Model source: backend/app/models/supervisor_request_conversation.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_request_messages_id_seq'::regclass)
request_id	INTEGER	No	-
sender_id	INTEGER	No	-
body	TEXT	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key request_id -> supervisor_requests(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key sender_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Index ix_supervisor_request_messages_request_id; unique=False; columns/expressions ['request_id', 'id']; options {'postgresql_include': []}.

Table: supervisor_request_status_history

Model source: backend/app/models/supervisor_request.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_request_status_history_id_seq'::regclass)
request_id	INTEGER	No	-
actor_id	INTEGER	No	-
from_status	VARCHAR(32)	Yes	-
to_status	VARCHAR(32)	No	-
comment	VARCHAR(2000)	Yes	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key actor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key request_id -> supervisor_requests(id); options: {'ondelete': 'CASCADE'}.

- Check ck_supervisor_request_history_to_status: to_status::text = ANY (ARRAY['DRAFT'::character varying, 'SUBMITTED'::character varying, 'CHANGES_REQUESTED'::character varying, 'ACCEPTED'::character varying, 'REJECTED'::character varying, 'WITHDRAWN'::character varying]::text[]).

- Index ix_supervisor_request_history_request_created; unique=False; columns/expressions ['request_id', 'created_at']; options {'postgresql_include': []}.

Table: supervisor_request_submission_note_versions

Model source: backend/app/models/supervisor_request.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_request_submission_note_versions_id_seq'::regclass)
request_id	INTEGER	No	-
version	INTEGER	No	-
body	TEXT	No	-
created_by_id	INTEGER	No	-
submitted_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key request_id -> supervisor_requests(id); options: {'ondelete': 'RESTRICT'}.

- Check ck_supervisor_request_note_body_nonblank: body = btrim(body) AND body <> ''::text.

- Check ck_supervisor_request_note_version_positive: version > 0.

- Index ix_supervisor_request_note_request; unique=False; columns/expressions ['request_id', 'submitted_at']; options {'postgresql_include': []}.

- Index uq_supervisor_request_note_version; unique=True; columns/expressions ['request_id', 'version']; options {'postgresql_include': []}.

Table: supervisor_requests

Model source: backend/app/models/supervisor_request.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".supervisor_requests_id_seq'::regclass)
academic_cycle_id	INTEGER	No	-
requester_user_id	INTEGER	No	-
team_id	INTEGER	Yes	-
supervisor_id	INTEGER	No	-
assignment_id	INTEGER	Yes	-
title	VARCHAR(200)	Yes	-
summary	TEXT	Yes	-
status	VARCHAR(32)	No	'DRAFT'::character varying
revision	INTEGER	No	1
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
submitted_at	TIMESTAMP WITH TIME ZONE	Yes	-
reviewed_at	TIMESTAMP WITH TIME ZONE	Yes	-
reviewer_id	INTEGER	Yes	-
document_id	INTEGER	Yes	-
extraction_status	VARCHAR(16)	No	'PENDING'::character varying
extraction_confirmed_at	TIMESTAMP WITH TIME ZONE	Yes	-
extraction_confirmed_by_id	INTEGER	Yes	-
submission_note	TEXT	Yes	-
proposal_type	VARCHAR(5)	No	'SHORT'::character varying
problem_statement	TEXT	Yes	-
objectives	TEXT	Yes	-
scope	TEXT	Yes	-
methodology	TEXT	Yes	-

- Foreign key document_id -> supervisor_request_documents(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key extraction_confirmed_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key academic_cycle_id -> academic_cycles(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key assignment_id -> supervisor_assignments(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key requester_user_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key reviewer_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key supervisor_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key team_id, academic_cycle_id -> teams(id, academic_cycle_id); options: {'ondelete': 'RESTRICT'}.

- Check ck_supervisor_requests_extraction_status: extraction_status::text = ANY (ARRAY['PENDING'::character varying, 'EXTRACTED'::character varying, 'FAILED'::character varying, 'CONFIRMED'::character varying]::text[]).

- Check ck_supervisor_requests_revision_positive: revision > 0.

- Check ck_supervisor_requests_status: status::text = ANY (ARRAY['DRAFT'::character varying, 'SUBMITTED'::character varying, 'CHANGES_REQUESTED'::character varying, 'ACCEPTED'::character varying, 'REJECTED'::character varying, 'WITHDRAWN'::character varying]::text[]).

- Check ck_supervisor_requests_summary_nonblank: summary IS NULL OR summary = btrim(summary) AND summary <> ''::text.

- Check ck_supervisor_requests_title_nonblank: title IS NULL OR title::text = btrim(title::text) AND title::text <> ''::text.

- Index ix_supervisor_requests_cycle_status; unique=False; columns/expressions ['academic_cycle_id', 'status']; options {'postgresql_include': []}.

- Index ix_supervisor_requests_extraction_status; unique=False; columns/expressions ['extraction_status']; options {'postgresql_include': []}.

- Index ix_supervisor_requests_supervisor_status; unique=False; columns/expressions ['supervisor_id', 'status']; options {'postgresql_include': []}.

- Index uq_supervisor_requests_active_owner_supervisor; unique=True; columns/expressions ['academic_cycle_id', 'requester_user_id', 'supervisor_id']; options {'postgresql_where': "((status)::text = ANY ((ARRAY['DRAFT'::character varying, 'SUBMITTED'::character varying, 'CHANGES_REQUESTED'::character varying, 'ACCEPTED'::character varying])::text[]))", 'postgresql_include': []}.

Table: team_member_verifications

Model source: No current ORM class; migration/legacy infrastructure table.

Primary key: member_id.

Column	PostgreSQL type	Nullable	Default
member_id	INTEGER	No	nextval('"public".team_member_verifications_member_id_seq'::regclass)
code_hash	VARCHAR(255)	Yes	-
expires_at	TIMESTAMP WITH TIME ZONE	Yes	-
sent_at	TIMESTAMP WITH TIME ZONE	Yes	-
verified_at	TIMESTAMP WITH TIME ZONE	Yes	-
attempts	INTEGER	No	0
window_started_at	TIMESTAMP WITH TIME ZONE	Yes	-
requests	INTEGER	No	0

Table: team_members

Model source: backend/app/models/team.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".team_members_id_seq'::regclass)
team_id	INTEGER	No	-
academic_cycle_id	INTEGER	No	-
full_name	VARCHAR(255)	No	-
registration_number	VARCHAR(100)	No	-
university_email	VARCHAR(320)	No	-
is_creator	BOOLEAN	No	false
is_active	BOOLEAN	No	true
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Foreign key team_id, academic_cycle_id -> teams(id, academic_cycle_id); options: {}.

- Unique constraint uq_team_members_registration: team_id, registration_number.

- Check ck_team_members_email: university_email::text = lower(btrim(university_email::text)) AND university_email::text <> ''::text.

- Check ck_team_members_name: full_name::text = btrim(full_name::text) AND full_name::text <> ''::text.

- Check ck_team_members_registration: registration_number::text = upper(btrim(registration_number::text)) AND registration_number::text <> ''::text.

- Index uq_active_cycle_contact_email; unique=True; columns/expressions ['academic_cycle_id', None]; options {'postgresql_where': 'is_active', 'postgresql_include': []}.

- Index uq_active_cycle_registration; unique=True; columns/expressions ['academic_cycle_id', None]; options {'postgresql_where': 'is_active', 'postgresql_include': []}.

- Index uq_team_creator; unique=True; columns/expressions ['team_id']; options {'postgresql_where': 'is_creator', 'postgresql_include': []}.

- Index uq_team_members_registration; unique=True; columns/expressions ['team_id', 'registration_number']; options {'postgresql_include': []}.

Table: teams

Model source: backend/app/models/team.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".teams_id_seq'::regclass)
academic_cycle_id	INTEGER	No	-
shared_user_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()
status	VARCHAR(16)	No	'forming'::character varying
locked_at	TIMESTAMP WITH TIME ZONE	Yes	-
lock_reason	VARCHAR(64)	Yes	-

- Foreign key academic_cycle_id -> academic_cycles(id); options: {}.

- Foreign key shared_user_id -> users(id); options: {}.

- Unique constraint teams_shared_user_id_key: shared_user_id.

- Unique constraint uq_teams_id_cycle: id, academic_cycle_id.

- Index teams_shared_user_id_key; unique=True; columns/expressions ['shared_user_id']; options {'postgresql_include': []}.

- Index uq_teams_id_cycle; unique=True; columns/expressions ['id', 'academic_cycle_id']; options {'postgresql_include': []}.

Table: template_reference_versions

Model source: backend/app/models/template_reference.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".template_reference_versions_id_seq'::regclass)
template_reference_id	INTEGER	No	-
version_number	INTEGER	No	-
title	VARCHAR(160)	No	-
provenance	VARCHAR(32)	No	-
source_note	VARCHAR(500)	No	-
created_by_id	INTEGER	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()
is_active	BOOLEAN	No	false
activated_at	TIMESTAMP WITH TIME ZONE	Yes	-
activated_by_id	INTEGER	Yes	-

- Foreign key activated_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key created_by_id -> users(id); options: {'ondelete': 'RESTRICT'}.

- Foreign key template_reference_id -> template_references(id); options: {'ondelete': 'RESTRICT'}.

- Unique constraint uq_template_reference_version: template_reference_id, version_number.

- Index uq_template_reference_version; unique=True; columns/expressions ['template_reference_id', 'version_number']; options {'postgresql_include': []}.

Table: template_references

Model source: backend/app/models/template_reference.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".template_references_id_seq'::regclass)
template_type	template_type	No	-
created_at	TIMESTAMP WITH TIME ZONE	No	now()

- Unique constraint template_references_template_type_key: template_type.

- Index template_references_template_type_key; unique=True; columns/expressions ['template_type']; options {'postgresql_include': []}.

Table: users

Model source: backend/app/models/user.py.

Primary key: id.

Column	PostgreSQL type	Nullable	Default
id	INTEGER	No	nextval('"public".users_id_seq'::regclass)
full_name	VARCHAR(255)	No	-
email	VARCHAR(320)	No	-
registration_number	VARCHAR(100)	Yes	-
password_hash	VARCHAR(255)	No	-
role	user_role	No	-
is_active	BOOLEAN	No	true
created_at	TIMESTAMP WITH TIME ZONE	No	now()
updated_at	TIMESTAMP WITH TIME ZONE	No	now()

- Unique constraint users_registration_number_key: registration_number.

- Check ck_users_email_normalized: email::text = lower(btrim(email::text)) AND email::text <> ''::text.

- Check ck_users_registration_number_normalized: registration_number IS NULL OR registration_number::text = upper(btrim(registration_number::text)) AND registration_number::text <> ''::text.

- Check ck_users_student_registration_required: role <> 'STUDENT'::user_role OR registration_number IS NOT NULL.

- Index uq_users_email_normalized; unique=True; columns/expressions [None]; options {'postgresql_include': []}.

- Index users_registration_number_key; unique=True; columns/expressions ['registration_number']; options {'postgresql_include': []}.

Appendix C - Complete registered backend API map

Routes come from the running FastAPI application. Guards show dependency-based checks; ownership, state and transaction checks also occur inside handlers and helpers. A route with no listed guard may be public or perform authorization internally. Use the source reference to explain it. Browser page URLs and these API URLs are different layers.

/usr/local/lib/python3.12/site-packages/fastapi/applications.py

Method	API path	Handler / line	Dependency guards
GET, HEAD	/openapi.json	openapi line 1145	See handler
GET, HEAD	/docs	swagger_ui_html line 1160	See handler
GET, HEAD	/docs/oauth2-redirect	swagger_ui_redirect line 1178	See handler
GET, HEAD	/redoc	redoc_html line 1188	See handler

- GET,HEAD /openapi.json: body None; response None; query parameters [].

- GET,HEAD /docs: body None; response None; query parameters [].

- GET,HEAD /docs/oauth2-redirect: body None; response None; query parameters [].

- GET,HEAD /redoc: body None; response None; query parameters [].

backend/app/academic_cycle_overview.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/academic-cycles/{cycle_id}/overview	cycle_overview line 37	role_guard: ADMIN, COORDINATOR; get_current_user; get_db

- GET /api/v1/academic-cycles/{cycle_id}/overview: body None; response <class 'dict'>; query parameters [].

backend/app/agent_supervision.py

Method	API path	Handler / line	Dependency guards
POST	/api/v1/agent-supervision/{team_id}/runs	start_run line 40	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
GET	/api/v1/agent-supervision/{team_id}/runs	list_runs line 59	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
POST	/api/v1/agent-supervision/runs/{run_id}/approval	approve_run line 63	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/agent-supervision/runs/{run_id}/execute	execute_run line 71	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/agent-supervision/runs/{run_id}/reminder/approve	approve_reminder line 92	role_guard: SUPERVISOR; get_current_user; get_db

- POST /api/v1/agent-supervision/{team_id}/runs: body None; response None; query parameters [].

- GET /api/v1/agent-supervision/{team_id}/runs: body None; response None; query parameters [].

- POST /api/v1/agent-supervision/runs/{run_id}/approval: body <class 'app.agent_supervision.Approval'>; response None; query parameters [].

- POST /api/v1/agent-supervision/runs/{run_id}/execute: body None; response None; query parameters [].

- POST /api/v1/agent-supervision/runs/{run_id}/reminder/approve: body <class 'app.agent_supervision.Approval'>; response None; query parameters [].

backend/app/archived_projects.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/archived-projects/search	search_archives line 668	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
GET	/api/v1/archived-projects/admin/list	admin_archive_list line 723	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/archived-projects/admin/detail/{project_id}	admin_archive_detail line 728	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/archived-projects/{project_id}	read_archive line 736	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db

Method	API path	Handler / line	Dependency guards
GET	/api/v1/archived-projects/{project_id}/documents/{document_id}	read_archive_document line 741	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
POST	/api/v1/archived-projects/admin	create_archive line 754	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/archived-projects/admin/{project_id}	update_archive line 770	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/archived-projects/admin/{project_id}	delete_archive line 804	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/archived-projects/admin/{project_id}/rollback/{event_id}	rollback_archive_metadata line 823	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/archived-projects/admin/{project_id}/active	set_archive_active line 845	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/archived-projects/admin/{project_id}/documents	upload_archive_document line 857	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/archived-projects/admin/{project_id}/documents/history	archive_document_history line 878	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/archived-projects/admin/{project_id}/documents/{document_id}	delete_archive_document line 886	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/archived-projects/admin/{project_id}/visibility	set_archive_visibility line 1099	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/archive/import/preview	archive_import_preview line 909	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/archive/projects/import	archive_import_confirm line 917	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/archive/import/confirm	archive_import_confirm line 917	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/archive/projects	admin_archive_projects line 1006	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/archive/projects/repair-imported	repair_imported_archive line 1029	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/archive/projects/{project_id}	admin_archive_project line 1067	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/admin/archive/projects/{project_id}	admin_update_archive line 1074	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/archive/projects/{project_id}	admin_delete_archive line 1079	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/archive/import-batches	archive_import_batches line 1084	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/archive/import-batches/{batch_id}	archive_import_batch_detail line 1091	role_guard: ADMIN; get_current_user; get_db

- GET /api/v1/archived-projects/search: body None; response None; query parameters [{'name': 'q', 'required': False}, {'name': 'academic_year', 'required': False}, {'name': 'session_name', 'required': False}, {'name': 'department', 'required': False}, {'name': 'supervisor', 'required': False}, {'name': 'technology', 'required': False}, {'name': 'project_scope', 'required': False}, {'name': 'completion_stage', 'required': False}].

- GET /api/v1/archived-projects/admin/list: body None; response None; query parameters [].

- GET /api/v1/archived-projects/admin/detail/{project_id}: body None; response None; query parameters [].

- GET /api/v1/archived-projects/{project_id}: body None; response None; query parameters [].

- GET /api/v1/archived-projects/{project_id}/documents/{document_id}: body None; response None; query parameters [].

- POST /api/v1/archived-projects/admin: body <class 'app.archived_projects.ArchiveCreate'>; response None; query parameters [].

- PATCH /api/v1/archived-projects/admin/{project_id}: body <class 'app.archived_projects.ArchiveUpdate'>; response None; query parameters [].

- DELETE /api/v1/archived-projects/admin/{project_id}: body None; response None; query parameters []. Permanently remove an archive record and all of its stored document versions.

- POST /api/v1/archived-projects/admin/{project_id}/rollback/{event_id}: body None; response None; query parameters [].

- POST /api/v1/archived-projects/admin/{project_id}/active: body <class 'fastapi._compat.v2.Body_set_archive_active_api_v1_archived_projects_admin__project_id__active_post'>; response None; query parameters [].

- POST /api/v1/archived-projects/admin/{project_id}/documents: body <class 'fastapi._compat.v2.Body_upload_archive_document_api_v1_archived_projects_admin__project_id__documents_post'>; response None; query parameters [].

- GET /api/v1/archived-projects/admin/{project_id}/documents/history: body None; response None; query parameters [].

- DELETE /api/v1/archived-projects/admin/{project_id}/documents/{document_id}: body None; response None; query parameters [].

- POST /api/v1/archived-projects/admin/{project_id}/visibility: body <class 'fastapi._compat.v2.Body_set_archive_visibility_api_v1_archived_projects_admin__project_id__visibility_post'>; response None; query parameters [].

- POST /api/v1/admin/archive/import/preview: body <class 'fastapi._compat.v2.Body_archive_import_preview_api_v1_admin_archive_import_preview_post'>; response None; query parameters [].

- POST /api/v1/admin/archive/projects/import: body <class 'fastapi._compat.v2.Body_archive_import_confirm_api_v1_admin_archive_projects_import_post'>; response None; query parameters [].

- POST /api/v1/admin/archive/import/confirm: body <class 'fastapi._compat.v2.Body_archive_import_confirm_api_v1_admin_archive_import_confirm_post'>; response None; query parameters [].

- GET /api/v1/admin/archive/projects: body None; response dict[str, object]; query parameters [{'name': 'search', 'required': False}, {'name': 'academic_cycle_id', 'required': False}, {'name': 'session_name', 'required': False}, {'name': 'supervisor', 'required': False}, {'name': 'domain', 'required': False}, {'name': 'source_batch_id', 'required': False}, {'name': 'page', 'required': False}, {'name': 'page_size', 'required': False}, {'name': 'sort', 'required': False}, {'name': 'direction', 'required': False}].

- POST /api/v1/admin/archive/projects/repair-imported: body None; response dict[str, object]; query parameters []. Safely consolidate obvious duplicate imported parents.

- GET /api/v1/admin/archive/projects/{project_id}: body None; response <class 'dict'>; query parameters [].

- PATCH /api/v1/admin/archive/projects/{project_id}: body <class 'app.archived_projects.ArchiveUpdate'>; response <class 'dict'>; query parameters [].

- DELETE /api/v1/admin/archive/projects/{project_id}: body None; response None; query parameters [].

- GET /api/v1/admin/archive/import-batches: body None; response dict[str, object]; query parameters [{'name': 'page', 'required': False}, {'name': 'page_size', 'required': False}].

- GET /api/v1/admin/archive/import-batches/{batch_id}: body None; response dict[str, object]; query parameters [].

backend/app/audit_history.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/admin/audit-events	list_events line 88	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/audit-events/{event_id}	event_detail line 112	role_guard: ADMIN; get_current_user; get_db

- GET /api/v1/admin/audit-events: body None; response None; query parameters [{'name': 'actor_id', 'required': False}, {'name': 'action', 'required': False}, {'name': 'target_type', 'required': False}, {'name': 'target_id', 'required': False}, {'name': 'from_at', 'required': False}, {'name': 'to_at', 'required': False}, {'name': 'page', 'required': False}, {'name': 'page_size', 'required': False}].

- GET /api/v1/admin/audit-events/{event_id}: body None; response None; query parameters [].

backend/app/collaboration_channels.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/student/collaboration/channels	student_channels line 105	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/supervisor/collaboration/channels	supervisor_channels line 110	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/collaboration/channels/{channel_id}	collaboration_channel line 115	get_current_user; get_db
GET	/api/v1/student/collaboration/channels/{channel_id}	collaboration_channel line 115	get_current_user; get_db
GET	/api/v1/supervisor/collaboration/channels/{channel_id}/messages	collaboration_messages line 122	get_current_user; get_db
GET	/api/v1/student/collaboration/channels/{channel_id}/messages	collaboration_messages line 122	get_current_user; get_db
POST	/api/v1/supervisor/collaboration/channels/{channel_id}/messages	send_collaboration_message line 130	get_current_user; get_db
POST	/api/v1/student/collaboration/channels/{channel_id}/messages	send_collaboration_message line 130	get_current_user; get_db

- GET /api/v1/student/collaboration/channels: body None; response None; query parameters [].

- GET /api/v1/supervisor/collaboration/channels: body None; response None; query parameters [].

- GET /api/v1/supervisor/collaboration/channels/{channel_id}: body None; response None; query parameters [].

- GET /api/v1/student/collaboration/channels/{channel_id}: body None; response None; query parameters [].

- GET /api/v1/supervisor/collaboration/channels/{channel_id}/messages: body None; response None; query parameters [].

- GET /api/v1/student/collaboration/channels/{channel_id}/messages: body None; response None; query parameters [].

- POST /api/v1/supervisor/collaboration/channels/{channel_id}/messages: body <class 'app.collaboration_channels.MessageWrite'>; response None; query parameters [].

- POST /api/v1/student/collaboration/channels/{channel_id}/messages: body <class 'app.collaboration_channels.MessageWrite'>; response None; query parameters [].

backend/app/coordinator_operations.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/coordinator/supervisor-capacity	supervisor_capacity line 15	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/coordinator/audit-history	audit_history line 52	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/coordinator/proposal-templates	proposal_templates line 65	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/coordinator/evaluation-criteria	evaluation_criteria line 77	role_guard: ADMIN, COORDINATOR; get_current_user; get_db

- GET /api/v1/coordinator/supervisor-capacity: body None; response None; query parameters [].

- GET /api/v1/coordinator/audit-history: body None; response None; query parameters [].

- GET /api/v1/coordinator/proposal-templates: body None; response None; query parameters [].

- GET /api/v1/coordinator/evaluation-criteria: body None; response None; query parameters [].

backend/app/dashboard_requirements.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/dashboard/deadlines	dashboard_requirements line 34	get_current_user; get_db
GET	/api/v1/dashboard/requirements	dashboard_requirements line 34	get_current_user; get_db
GET	/api/v1/dashboard/summary	dashboard_summary line 42	get_current_user; get_db
POST	/api/v1/dashboard/requirements	create_requirement line 47	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
PATCH	/api/v1/dashboard/requirements/{requirement_id}	update_requirement line 56	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
DELETE	/api/v1/dashboard/requirements/{requirement_id}	delete_requirement line 65	role_guard: ADMIN, COORDINATOR; get_current_user; get_db

- GET /api/v1/dashboard/deadlines: body None; response list[app.schemas.dashboard_requirements.DashboardRequirementOut]; query parameters [].

- GET /api/v1/dashboard/requirements: body None; response list[app.schemas.dashboard_requirements.DashboardRequirementOut]; query parameters [].

- GET /api/v1/dashboard/summary: body None; response None; query parameters [].

- POST /api/v1/dashboard/requirements: body <class 'app.schemas.dashboard_requirements.DashboardRequirementWrite'>; response <class 'app.schemas.dashboard_requirements.DashboardRequirementOut'>; query parameters [].

- PATCH /api/v1/dashboard/requirements/{requirement_id}: body <class 'app.schemas.dashboard_requirements.DashboardRequirementWrite'>; response <class 'app.schemas.dashboard_requirements.DashboardRequirementOut'>; query parameters [].

- DELETE /api/v1/dashboard/requirements/{requirement_id}: body None; response None; query parameters [].

backend/app/document_requirements.py

Method	API path	Handler / line	Dependency guards
POST	/api/v1/document-requirements	create_requirement line 87	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
DELETE	/api/v1/document-requirements/{requirement_id}	delete_requirement line 112	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/document-requirements	list_requirements line 132	get_current_user; get_db
GET	/api/v1/document-requirements/{requirement_id}	get_requirement line 137	get_current_user; get_db
PATCH	/api/v1/document-requirements/{requirement_id}	update_requirement line 145	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/publish	publish line 162	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/unpublish	unpublish line 164	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/close	close line 166	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db

Method	API path	Handler / line	Dependency guards
POST	/api/v1/document-requirements/{requirement_id}/archive	archive line 168	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/reopen	reopen line 170	validate_mutating_origin; role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/document-requirements/student	student_requirements line 210	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/upload	upload line 216	validate_mutating_origin; role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/document-requirements/{requirement_id}/submission	submission line 246	get_current_user; get_db
GET	/api/v1/document-requirements/{requirement_id}/submissions	all_submissions line 254	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/document-requirements/{requirement_id}/submission/versions	versions line 273	get_current_user; get_db
GET	/api/v1/document-requirements/submissions/{submission_id}/versions/{version_id}/download	download line 279	get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/submissions/{submission_id}/approve	approve line 298	validate_mutating_origin; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/submissions/{submission_id}/reject	reject line 300	validate_mutating_origin; get_current_user; get_db
POST	/api/v1/document-requirements/{requirement_id}/submissions/{submission_id}/request-replacement	replacement line 302	validate_mutating_origin; get_current_user; get_db
GET	/api/v1/student/group/requirements/{requirement_id}	get_requirement line 137	get_current_user; get_db
GET	/api/v1/student/group/requirements	student_requirements line 210	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/student/document-requirements	student_requirements line 210	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/student/group/requirements/{requirement_id}/resubmit	upload line 216	validate_mutating_origin; role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/student/group/requirements/{requirement_id}/upload	upload line 216	validate_mutating_origin; role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/student/group/requirements/{requirement_id}/submission	submission line 246	get_current_user; get_db

- POST /api/v1/document-requirements: body <class 'app.schemas.document_requirements.RequirementWrite'>; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- DELETE /api/v1/document-requirements/{requirement_id}: body None; response None; query parameters []. Remove an unused draft requirement while preserving published/history records.

- GET /api/v1/document-requirements: body None; response list[app.schemas.document_requirements.RequirementOut]; query parameters [].

- GET /api/v1/document-requirements/{requirement_id}: body None; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- PATCH /api/v1/document-requirements/{requirement_id}: body <class 'app.schemas.document_requirements.RequirementPatch'>; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/publish: body None; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/unpublish: body None; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/close: body None; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/archive: body None; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/reopen: body None; response <class 'app.schemas.document_requirements.RequirementOut'>; query parameters [].

- GET /api/v1/document-requirements/student: body None; response list[app.schemas.document_requirements.RequirementOut]; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/upload: body <class 'fastapi._compat.v2.Body_upload_api_v1_document_requirements__requirement_id__upload_post'>; response <class 'app.schemas.document_requirements.SubmissionOut'>; query parameters [].

- GET /api/v1/document-requirements/{requirement_id}/submission: body None; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [].

- GET /api/v1/document-requirements/{requirement_id}/submissions: body None; response None; query parameters []. Return every student's submission and every uploaded version for an admin requirement view.

- GET /api/v1/document-requirements/{requirement_id}/submission/versions: body None; response list[app.schemas.document_requirements.VersionOut]; query parameters [].

- GET /api/v1/document-requirements/submissions/{submission_id}/versions/{version_id}/download: body None; response None; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/submissions/{submission_id}/approve: body None; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [].

- POST /api/v1/document-requirements/{requirement_id}/submissions/{submission_id}/reject: body None; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [{'name': 'comment', 'required': False}].

- POST /api/v1/document-requirements/{requirement_id}/submissions/{submission_id}/request-replacement: body None; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [{'name': 'comment', 'required': False}].

- GET /api/v1/student/group/requirements/{requirement_id}: body None; response <class ' app.schemas.document_requirements.RequirementOut'>; query parameters [].

- GET /api/v1/student/group/requirements: body None; response list[app.schemas.document_requirements.RequirementOut]; query parameters [].

- GET /api/v1/student/document-requirements: body None; response list[app.schemas.document_requirements.RequirementOut]; query parameters [].

- POST /api/v1/student/group/requirements/{requirement_id}/resubmit: body <class ' fastapi._compat.v2.Body_upload_api_v1_student_group_requirements__requirement_id__resubmit_post'>; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [].

- POST /api/v1/student/group/requirements/{requirement_id}/upload: body <class ' fastapi._compat.v2.Body_upload_api_v1_student_group_requirements__requirement_id__upload_post'>; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [].

- GET /api/v1/student/group/requirements/{requirement_id}/submission: body None; response <class ' app.schemas.document_requirements.SubmissionOut'>; query parameters [].

backend/app/evaluations.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/evaluations/assignments	evaluation_assignments line 141	get_current_user; get_db
GET	/api/v1/evaluations/my	my_evaluations line 148	get_current_user; get_db
POST	/api/v1/evaluations	create_evaluation line 200	get_current_user; get_db
PATCH	/api/v1/evaluations/{evaluation_id}	update_evaluation line 216	get_current_user; get_db
POST	/api/v1/evaluations/{evaluation_id}/submit	submit_evaluation line 232	get_current_user; get_db
POST	/api/v1/evaluations/{evaluation_id}/return	return_evaluation line 253	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/evaluations/{evaluation_id}/approve	approve_evaluation line 271	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/evaluations/{evaluation_id}/correct	correct_evaluation line 287	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
POST	/api/v1/evaluations/{evaluation_id}/publish	publish_evaluation line 309	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/evaluations/{evaluation_id}/audit	evaluation_audit line 325	get_current_user; get_db
GET	/api/v1/admin/evaluations	admin_evaluations line 336	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/admin/evaluations/aggregation	list_aggregation line 348	role_guard: ADMIN; get_current_user; get_db
PUT	/api/v1/admin/evaluations/aggregation	update_aggregation line 359	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/student/results	student_results line 430	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/admin/results/group/{group_id}	admin_group_results line 435	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/admin/results/{student_id}	admin_results line 445	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/evaluations/{evaluation_id}/report	evaluation_report line 476	get_current_user; get_db
GET	/api/v1/evaluations/{evaluation_id}/report.pdf	evaluation_report_pdf line 487	get_current_user; get_db

- GET /api/v1/evaluations/assignments: body None; response list[app.schemas.evaluations.EvaluationAssignmentOut]; query parameters [].

- GET /api/v1/evaluations/my: body None; response list[app.schemas.evaluations.EvaluationOut]; query parameters [{'name': ' evaluation_status', 'required': False}, {'name': 'category', 'required': False}].

- POST /api/v1/evaluations: body <class 'app.schemas.evaluations.EvaluationWrite'>; response <class ' app.schemas.evaluations.EvaluationOut'>; query parameters [].

- PATCH /api/v1/evaluations/{evaluation_id}: body <class 'app.schemas.evaluations.EvaluationWrite'>; response <class ' app.schemas.evaluations.EvaluationOut'>; query parameters [].

- POST /api/v1/evaluations/{evaluation_id}/submit: body None; response <class 'app.schemas.evaluations.EvaluationOut'>; query parameters [].

- POST /api/v1/evaluations/{evaluation_id}/return: body <class 'app.schemas.evaluations.EvaluationReturn'>; response <class ' app.schemas.evaluations.EvaluationOut'>; query parameters [].

- POST /api/v1/evaluations/{evaluation_id}/approve: body None; response <class 'app.schemas.evaluations.EvaluationOut'>; query parameters [].

- POST /api/v1/evaluations/{evaluation_id}/correct: body <class 'app.schemas.evaluations.EvaluationCorrectionWrite'>; response <class ' app.schemas.evaluations.EvaluationOut'>; query parameters [].

- POST /api/v1/evaluations/{evaluation_id}/publish: body None; response <class 'app.schemas.evaluations.EvaluationOut'>; query parameters [].

- GET /api/v1/evaluations/{evaluation_id}/audit: body None; response list[dict]; query parameters [].

- GET /api/v1/admin/evaluations: body None; response list[app.schemas.evaluations.EvaluationOut]; query parameters [{'name': ' evaluation_status', 'required': False}, {'name': 'category', 'required': False}, {'name': 'evaluator_id', 'required': False}].

- GET /api/v1/admin/evaluations/aggregation: body None; response list[app.schemas.evaluations.EvaluationAggregationOut]; query parameters [].

- PUT /api/v1/admin/evaluations/aggregation: body <class 'app.schemas.evaluations.EvaluationAggregationWrite'>; response <class ' app.schemas.evaluations.EvaluationAggregationOut'>; query parameters [].

- GET /api/v1/student/results: body None; response <class 'app.schemas.evaluations.ConsolidatedResultOut'>; query parameters [].

- GET /api/v1/admin/results/group/{group_id}: body None; response list[app.schemas.evaluations.ConsolidatedResultOut]; query parameters [].

- GET /api/v1/admin/results/{student_id}: body None; response <class 'app.schemas.evaluations.ConsolidatedResultOut'>; query parameters [].

- GET /api/v1/evaluations/{evaluation_id}/report: body None; response <class 'dict'>; query parameters [].

- GET /api/v1/evaluations/{evaluation_id}/report.pdf: body None; response None; query parameters [].

backend/app/group_ai.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/group-ai/{team_id}	history line 84	get_current_user; get_db
POST	/api/v1/group-ai/{team_id}	chat line 91	get_current_user; get_db

- GET /api/v1/group-ai/{team_id}: body None; response None; query parameters [{'name': 'before_id', 'required': False}].

- POST /api/v1/group-ai/{team_id}: body <class 'app.group_channels.MessageWrite'>; response None; query parameters [].

backend/app/group_channels.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/group-channels/teams	teams line 57	get_current_user; get_db
GET	/api/v1/group-channels/{team_id}/{channel}	history line 66	get_current_user; get_db
POST	/api/v1/group-channels/{team_id}/{channel}	send line 73	get_current_user; get_db
POST	/api/v1/group-channels/{team_id}/files	upload_file line 92	get_current_user; get_db
GET	/api/v1/group-channels/{team_id}/files/{stored_name}	download_file line 112	get_current_user; get_db

- GET /api/v1/group-channels/teams: body None; response None; query parameters [].

- GET /api/v1/group-channels/{team_id}/{channel}: body None; response None; query parameters [{'name': 'before_id', 'required': False}].

- POST /api/v1/group-channels/{team_id}/{channel}: body <class 'app.group_channels.MessageWrite'>; response None; query parameters [].

- POST /api/v1/group-channels/{team_id}/files: body <class 'fastapi._compat.v2.Body_upload_file_api_v1_group_channels__team_id__files_pos t'>; response None; query parameters []. Store a team-scoped resource and publish it into Files & Resources.

- GET /api/v1/group-channels/{team_id}/files/{stored_name}: body None; response None; query parameters [].

backend/app/main.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/academic-cycles	list_cycles line 168	get_db
GET	/api/v1/academic-cycles/{cycle_id}/group-policy	get_group_policy line 172	get_db
GET	/api/v1/admin/time-zones	list_time_zones line 320	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/academic-deadlines/preview	preview_academic_deadline line 326	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/academic-deadlines	list_academic_deadlines line 337	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/academic-deadlines/{deadline_id}	academic_deadline_detail line 372	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/academic-deadlines	create_academic_deadline line 384	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/academic-deadlines/{deadline_id}/versions	create_academic_deadline_version line 414	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/academic-deadlines/{deadline_id}	delete_academic_deadline line 446	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/template-references	list_template_references line 477	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/template-references/{reference_id}	template_reference_detail line 487	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/template-references	create_template_reference line 516	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/template-references/{reference_id}/versions	create_template_version line 540	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/template-reference-versions/{version_id}/rules	create_rule_version line 547	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/template-reference-versions/{version_id}	delete_template_reference_version line 556	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/evaluation-rubrics	list_evaluation_rubrics line 650	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/evaluation-rubrics/{rubric_id}	evaluation_rubric_detail line 685	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/evaluation-rubrics	create_evaluation_rubric line 697	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/evaluation-rubrics/{rubric_id}/versions	create_evaluation_rubric_version line 721	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/academic-cycles	create_cycle line 758	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/admin/academic-cycles/{cycle_id}	update_cycle line 775	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/academic-cycles/{cycle_id}	delete_cycle line 795	role_guard: ADMIN; get_current_user; get_db
PUT	/api/v1/admin/academic-cycles/{cycle_id}/group-policy	set_group_policy line 814	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/academic-cycles/{cycle_id}/group-policy	delete_group_policy line 833	role_guard: ADMIN; get_current_user; get_db
GET	/health	health line 849	See handler
GET	/health/ready	readiness line 855	See handler
POST	/api/v1/auth/login	login line 879	get_db
GET	/api/v1/auth/me	current_user line 1003	get_current_user; get_db
GET	/api/v1/admin/me	admin_current_user line 1018	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/registration-mode	admin_registration_mode line 1037	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/users	list_admin_users line 1186	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/users	create_admin_user line 1216	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/admin/users/{user_id}/status	change_admin_user_status line 1249	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/admin/users/{user_id}/role	change_admin_user_role line 1285	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/users/{user_id}	delete_admin_user line 1327	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/supervisor-capacities	list_supervisor_capacities line 1357	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/supervisor/capacity	supervisor_capacity_workspace line 1367	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/admin/supervisor-capacities/{policy_id}	read_supervisor_capacity line 1409	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/supervisor-capacities	create_supervisor_capacity line 1419	role_guard: ADMIN; get_current_user; get_db
PUT	/api/v1/admin/supervisor-capacities/{policy_id}	update_supervisor_capacity line 1453	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/supervisor-capacities/{policy_id}	delete_supervisor_capacity line 1491	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/auth/logout	logout line 1522	get_db
POST	/api/v1/auth/register	register_student line 1543	get_db
POST	/api/v1/team-registration	register_team line 1563	get_db
POST	/api/v1/auth/group-signup	register_student_group line 1813	get_db
POST	/api/v1/student/group-registration	register_student_group line 1813	get_db

Method	API path	Handler / line	Dependency guards
GET	/api/v1/student/group/me	student_group_me line 1911	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/student/group/members	student_group_members line 1919	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/admin/student-groups	list_student_groups line 1928	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/student-groups/{group_id}	get_student_group line 1945	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/admin/student-groups/{group_id}/activate	activate_student_group line 1978	role_guard: ADMIN; get_current_user; get_db
PATCH	/api/v1/admin/student-groups/{group_id}/deactivate	deactivate_student_group line 1983	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/student-groups/{group_id}	delete_student_group line 1988	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/reminder-rules/preview	preview_reminder_rule line 2057	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/reminder-rules	list_reminder_rules line 2063	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/reminder-rules/{rule_id}	read_reminder_rule line 2073	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/reminder-rules	create_reminder_rule line 2081	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/reminder-rules/{rule_id}/versions	create_reminder_rule_version line 2103	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/admin/reminder-rules/{rule_id}	delete_reminder_rule line 2126	role_guard: ADMIN; get_current_user; get_db

- GET /api/v1/academic-cycles: body None; response list[app.schemas.academic.CycleOut]; query parameters [].

- GET /api/v1/academic-cycles/{cycle_id}/group-policy: body None; response <class 'app.schemas.academic.PolicyOut'>; query parameters [].

- GET /api/v1/admin/time-zones: body None; response list[str]; query parameters [].

- POST /api/v1/admin/academic-deadlines/preview: body <class 'app.schemas.academic_deadlines.DeadlinePreviewRequest'>; response <class 'app.schemas.academic_deadlines.DeadlinePreviewOut'>; query parameters [].

- GET /api/v1/admin/academic-deadlines: body None; response list[app.schemas.academic_deadlines.DeadlineListOut]; query parameters [{'name': 'academic_cycle_id', 'required': False}].

- GET /api/v1/admin/academic-deadlines/{deadline_id}: body None; response <class 'app.schemas.academic_deadlines.DeadlineDetailOut'>; query parameters [].

- POST /api/v1/admin/academic-deadlines: body <class 'app.schemas.academic_deadlines.DeadlineCreate'>; response <class ' app.schemas.academic_deadlines.DeadlineDetailOut'>; query parameters [].

- POST /api/v1/admin/academic-deadlines/{deadline_id}/versions: body <class 'app.schemas.academic_deadlines.DeadlineWrite'>; response <class 'app.schemas.academic_deadlines.DeadlineVersionOut'>; query parameters [].

- DELETE /api/v1/admin/academic-deadlines/{deadline_id}: body None; response None; query parameters []. Remove a deadline and its recorded versions when no rule references it.

- GET /api/v1/admin/template-references: body None; response list[app.schemas.templates.TemplateListOut]; query parameters [].

- GET /api/v1/admin/template-references/{reference_id}: body None; response <class 'app.schemas.templates.TemplateDetail'>; query parameters [].

- POST /api/v1/admin/template-references: body <class 'app.schemas.templates.TemplateCreate'>; response <class ' app.schemas.templates.TemplateDetail'>; query parameters [].

- POST /api/v1/admin/template-references/{reference_id}/versions: body <class 'app.schemas.templates.TemplateVersionCreate'>; response <class 'app.schemas.templates.TemplateOut'>; query parameters [].

- POST /api/v1/admin/template-reference-versions/{version_id}/rules: body <class 'app.schemas.templates.RuleCreate'>; response <class ' app.schemas.templates.RuleOut'>; query parameters [].

- DELETE /api/v1/admin/template-reference-versions/{version_id}: body None; response None; query parameters [].

- GET /api/v1/admin/evaluation-rubrics: body None; response list[app.schemas.evaluation_rubrics.RubricListOut]; query parameters [].

- GET /api/v1/admin/evaluation-rubrics/{rubric_id}: body None; response <class 'app.schemas.evaluation_rubrics.RubricDetailOut'>; query parameters [].

- POST /api/v1/admin/evaluation-rubrics: body <class 'app.schemas.evaluation_rubrics.RubricVersionWrite'>; response <class ' app.schemas.evaluation_rubrics.RubricDetailOut'>; query parameters [].

- POST /api/v1/admin/evaluation-rubrics/{rubric_id}/versions: body <class 'app.schemas.evaluation_rubrics.RubricVersionWrite'>; response <class 'app.schemas.evaluation_rubrics.RubricVersionOut'>; query parameters [].

- POST /api/v1/admin/academic-cycles: body <class 'app.schemas.academic.CycleCreate'>; response <class ' app.schemas.academic.CycleOut'>; query parameters [].

- PATCH /api/v1/admin/academic-cycles/{cycle_id}: body <class 'app.schemas.academic.CycleUpdate'>; response <class ' app.schemas.academic.CycleOut'>; query parameters [].

- DELETE /api/v1/admin/academic-cycles/{cycle_id}: body None; response None; query parameters [].

- PUT /api/v1/admin/academic-cycles/{cycle_id}/group-policy: body <class 'app.schemas.academic.PolicyUpdate'>; response <class ' app.schemas.academic.PolicyOut'>; query parameters [].

- DELETE /api/v1/admin/academic-cycles/{cycle_id}/group-policy: body None; response None; query parameters [].

- GET /health: body None; response dict[str, str]; query parameters []. Report application liveness only; this does not check database connectivity.

- GET /health/ready: body None; response dict[str, str]; query parameters []. Report readiness only when the database and vector extension are available.

- POST /api/v1/auth/login: body <class 'app.schemas.auth.LoginRequest'>; response <class 'app.schemas.auth.CurrentUserResponse'>; query parameters []. Authenticate an active account and issue a new opaque cookie session.

- GET /api/v1/auth/me: body None; response <class 'app.schemas.auth.CurrentUserResponse'>; query parameters []. Return the authenticated active account's safe fields.

- GET /api/v1/admin/me: body None; response <class 'app.schemas.auth.CurrentUserResponse'>; query parameters []. Return the current account only when its persisted role is ADMIN.

- GET /api/v1/admin/registration-mode: body None; response dict[str, str / None]; query parameters []. Expose demo-mode status only to authenticated administrators.

- GET /api/v1/admin/users: body None; response <class 'app.schemas.admin_users.AdminUserPage'>; query parameters [{'name': 'search', ' required': False}, {'name': 'role', 'required': False}, {'name': 'is_active', 'required': False}, {'name': 'page', 'required': False}, {'name': 'page_size', ' required': False}].

- POST /api/v1/admin/users: body <class 'app.schemas.admin_users.AdminUserCreate'>; response <class ' app.schemas.admin_users.AdminUserResponse'>; query parameters [].

- PATCH /api/v1/admin/users/{user_id}/status: body <class 'app.schemas.admin_users.AdminUserStatusChange'>; response <class ' app.schemas.admin_users.AdminUserResponse'>; query parameters [].

- PATCH /api/v1/admin/users/{user_id}/role: body <class 'app.schemas.admin_users.AdminUserRoleChange'>; response <class ' app.schemas.admin_users.AdminUserResponse'>; query parameters [].

- DELETE /api/v1/admin/users/{user_id}: body None; response None; query parameters []. Permanently remove an unused account; linked records stay protected by FK guards.

- GET /api/v1/admin/supervisor-capacities: body None; response list[app.schemas.supervisor_capacity.SupervisorCapacityOut]; query parameters [].

- GET /api/v1/supervisor/capacity: body None; response list[dict[str, object]]; query parameters []. Return the signed-in supervisor's read-only annual capacity and assignment detail.

- GET /api/v1/admin/supervisor-capacities/{policy_id}: body None; response <class ' app.schemas.supervisor_capacity.SupervisorCapacityOut'>; query parameters [].

- POST /api/v1/admin/supervisor-capacities: body <class 'app.schemas.supervisor_capacity.SupervisorCapacityWrite'>; response <class ' app.schemas.supervisor_capacity.SupervisorCapacityOut'>; query parameters [].

- PUT /api/v1/admin/supervisor-capacities/{policy_id}: body <class 'app.schemas.supervisor_capacity.SupervisorCapacityWrite'>; response <class 'app.schemas.supervisor_capacity.SupervisorCapacityOut'>; query parameters [].

- DELETE /api/v1/admin/supervisor-capacities/{policy_id}: body None; response None; query parameters [].

- POST /api/v1/auth/logout: body None; response None; query parameters []. Revoke the presented session when present and always clear its browser cookie.

- POST /api/v1/auth/register: body <class 'app.schemas.auth.RegistrationRequest'>; response <class ' app.schemas.auth.RegistrationResponse'>; query parameters []. Reject the retired individual public signup flow.

- POST /api/v1/team-registration: body <class 'app.schemas.team_registration.TeamRegistrationRequest'>; response <class ' app.schemas.team_registration.TeamRegistrationResponse'>; query parameters []. Atomically create one shared student account, team, and member contacts.

- POST /api/v1/auth/group-signup: body <class 'app.schemas.team_registration.StudentGroupRegistrationRequest'>; response <class ' app.schemas.team_registration.TeamRegistrationResponse'>; query parameters []. Create one shared STUDENT group account from member contacts only.

- POST /api/v1/student/group-registration: body <class 'app.schemas.team_registration.StudentGroupRegistrationRequest'>; response <class ' app.schemas.team_registration.TeamRegistrationResponse'>; query parameters []. Create one shared STUDENT group account from member contacts only.

- GET /api/v1/student/group/me: body None; response dict[str, object]; query parameters [].

- GET /api/v1/student/group/members: body None; response dict[str, object]; query parameters [].

- GET /api/v1/admin/student-groups: body None; response dict[str, object]; query parameters [{'name': 'search', 'required': False}, {'name': ' is_active', 'required': False}].

- GET /api/v1/admin/student-groups/{group_id}: body None; response dict[str, object]; query parameters [].

- PATCH /api/v1/admin/student-groups/{group_id}/activate: body None; response dict[str, object]; query parameters [].

- PATCH /api/v1/admin/student-groups/{group_id}/deactivate: body None; response dict[str, object]; query parameters [].

- DELETE /api/v1/admin/student-groups/{group_id}: body None; response None; query parameters [].

- POST /api/v1/admin/reminder-rules/preview: body <class 'app.schemas.reminder_rules.RulePreview'>; response None; query parameters [].

- GET /api/v1/admin/reminder-rules: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': False}].

- GET /api/v1/admin/reminder-rules/{rule_id}: body None; response None; query parameters [].

- POST /api/v1/admin/reminder-rules: body <class 'app.schemas.reminder_rules.RuleCreate'>; response None; query parameters [].

- POST /api/v1/admin/reminder-rules/{rule_id}/versions: body <class 'app.schemas.reminder_rules.RuleWrite'>; response None; query parameters [].

- DELETE /api/v1/admin/reminder-rules/{rule_id}: body None; response None; query parameters [].

backend/app/module2_collaboration.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/module-2/workspace	workspace line 51	get_current_user; get_db
POST	/api/v1/module-2/teams/{team_id}/transfer-leadership	transfer_leadership line 66	get_current_user; get_db
POST	/api/v1/module-2/teams/{team_id}/dissolve	dissolve line 82	get_current_user; get_db
POST	/api/v1/module-2/teams/{team_id}/leave	leave_team line 96	get_current_user; get_db
GET	/api/v1/module-2/notifications	notifications line 122	get_current_user; get_db
POST	/api/v1/module-2/notifications/{notification_id}/read	mark_notification_read line 131	get_current_user; get_db
POST	/api/v1/module-2/teams/{team_id}/{channel}/read	mark_channel_read line 141	get_current_user; get_db
GET	/api/v1/module-2/teams/{team_id}/unread	unread_counts line 154	get_current_user; get_db
GET	/api/v1/module-2/teams/{team_id}/proposal-versions	proposal_versions line 167	get_current_user; get_db
GET	/api/v1/module-2/teams/{team_id}/tasks	tasks line 174	get_current_user; get_db
GET	/api/v1/module-2/teams/{team_id}/deadlines	deadlines line 181	get_current_user; get_db

- GET /api/v1/module-2/workspace: body None; response None; query parameters [].

- POST /api/v1/module-2/teams/{team_id}/transfer-leadership: body <class 'app.module2_collaboration.TransferLeadership'>; response None; query parameters [].

- POST /api/v1/module-2/teams/{team_id}/dissolve: body None; response None; query parameters [].

- POST /api/v1/module-2/teams/{team_id}/leave: body <class 'app.module2_collaboration.LeaveTeam'>; response None; query parameters [].

- GET /api/v1/module-2/notifications: body None; response None; query parameters [{'name': 'unread_only', 'required': False}].

- POST /api/v1/module-2/notifications/{notification_id}/read: body None; response None; query parameters [].

- POST /api/v1/module-2/teams/{team_id}/{channel}/read: body None; response None; query parameters [{'name': 'message_id', 'required': False}].

- GET /api/v1/module-2/teams/{team_id}/unread: body None; response None; query parameters [].

- GET /api/v1/module-2/teams/{team_id}/proposal-versions: body None; response None; query parameters [].

- GET /api/v1/module-2/teams/{team_id}/tasks: body None; response None; query parameters [].

- GET /api/v1/module-2/teams/{team_id}/deadlines: body None; response None; query parameters [].

backend/app/modules/module_03_proposal/router.py

Method	API path	Handler / line	Dependency guards
POST	/api/v1/proposal-compliance/templates/upload	upload_template line 184	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/proposal-compliance/templates	list_templates line 225	role_guard: ADMIN, COORDINATOR, SUPERVISOR, PANEL_MEMBER; get_current_user; get_db
POST	/api/v1/proposal-compliance/templates/{version_id}/activate	activate_template line 231	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/proposal-compliance/templates/{version_id}/deactivate	deactivate_template line 249	role_guard: ADMIN; get_current_user; get_db
DELETE	/api/v1/proposal-compliance/templates/{version_id}	delete_template line 264	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/proposal-compliance/templates/{version_id}/rules	save_template_rules line 283	role_guard: ADMIN; get_current_user; get_db

Method	API path	Handler / line	Dependency guards
POST	/api/v1/proposal-compliance/submissions	submit_proposal line 319	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposal-compliance/submissions/{submission_id}/submit	submit_draft line 401	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposal-compliance/submissions/{submission_id}/withdraw	withdraw_submission line 415	role_guard: STUDENT; get_current_user; get_db
PATCH	/api/v1/proposal-compliance/submissions/{submission_id}/status	update_submission_status line 431	role_guard: ADMIN, COORDINATOR, SUPERVISOR, PANEL_MEMBER; get_current_user; get_db
GET	/api/v1/proposal-compliance/submissions/mine	my_submissions line 450	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/proposal-compliance/submissions/{submission_id}	get_submission line 458	get_current_user; get_db
GET	/api/v1/proposal-compliance/reports	report_queue line 467	role_guard: ADMIN, COORDINATOR, SUPERVISOR, PANEL_MEMBER; get_current_user; get_db
GET	/api/v1/proposal-compliance/reports/{submission_id}	get_report line 473	get_current_user; get_db
POST	/api/v1/proposal-compliance/reports/{submission_id}/run	rerun_report line 484	get_current_user; get_db
GET	/api/v1/proposal-compliance/reports/{submission_id}/original	get_original line 497	get_current_user; get_db
GET	/api/v1/proposal-compliance/reports/{submission_id}/annotated	get_annotated_document line 508	get_current_user; get_db
GET	/api/v1/proposal-compliance/reports/{submission_id}/pdf	download_report_pdf line 527	get_current_user; get_db

- POST /api/v1/proposal-compliance/templates/upload: body <class 'fastapi._compat.v2.Body_upload_template_api_v1_proposal_compliance_templates_upload_post'>; response None; query parameters [].

- GET /api/v1/proposal-compliance/templates: body None; response None; query parameters [].

- POST /api/v1/proposal-compliance/templates/{version_id}/activate: body None; response None; query parameters [].

- POST /api/v1/proposal-compliance/templates/{version_id}/deactivate: body None; response None; query parameters []. Stop using a baseline without deleting its immutable version history.

- DELETE /api/v1/proposal-compliance/templates/{version_id}: body None; response None; query parameters []. Delete an unused uploaded template version and its extracted rule snapshot.

- POST /api/v1/proposal-compliance/templates/{version_id}/rules: body dict[str, typing.Any]; response None; query parameters []. Save an admin-confirmed rule review as a new immutable rule snapshot.

- POST /api/v1/proposal-compliance/submissions: body <class 'fastapi._compat.v2.Body_submit_proposal_api_v1_proposal_compliance_submissions_post'>; response None; query parameters [].

- POST /api/v1/proposal-compliance/submissions/{submission_id}/submit: body None; response None; query parameters [].

- POST /api/v1/proposal-compliance/submissions/{submission_id}/withdraw: body None; response None; query parameters []. Withdraw a student-owned draft/submission without deleting its history.

- PATCH /api/v1/proposal-compliance/submissions/{submission_id}/status: body <class 'app.modules.module_03_proposal.router.ProposalStatusUpdate'>; response None; query parameters []. Allow an authorized reviewer to record a lifecycle decision.

- GET /api/v1/proposal-compliance/submissions/mine: body None; response None; query parameters [].

- GET /api/v1/proposal-compliance/submissions/{submission_id}: body None; response None; query parameters [].

- GET /api/v1/proposal-compliance/reports: body None; response None; query parameters [].

- GET /api/v1/proposal-compliance/reports/{submission_id}: body None; response None; query parameters [].

- POST /api/v1/proposal-compliance/reports/{submission_id}/run: body None; response None; query parameters []. Explicitly re-run a report against the current active baseline.

- GET /api/v1/proposal-compliance/reports/{submission_id}/original: body None; response None; query parameters [].

- GET /api/v1/proposal-compliance/reports/{submission_id}/annotated: body None; response None; query parameters []. Return a safe annotated copy; the student's original file is never modified.

- GET /api/v1/proposal-compliance/reports/{submission_id}/pdf: body None; response None; query parameters [].

backend/app/notifications.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/notifications	notifications line 14	get_current_user; get_db
POST	/api/v1/notifications/{notification_id}/read	mark_notification_read line 23	get_current_user; get_db

- GET /api/v1/notifications: body None; response None; query parameters [{'name': 'unread_only', 'required': False}].

- POST /api/v1/notifications/{notification_id}/read: body None; response None; query parameters [].

backend/app/password_reset.py

Method	API path	Handler / line	Dependency guards
POST	/api/v1/auth/forgot-password	forgot_password line 116	get_db
GET	/api/v1/auth/password-reset-status	password_reset_status line 166	See handler
GET	/api/v1/auth/dev/password-reset-link	development_password_reset_link line 175	See handler
POST	/api/v1/auth/reset-password	reset_password line 191	get_db

- POST /api/v1/auth/forgot-password: body <class 'app.schemas.password_reset.ForgotPasswordRequest'>; response <class ' app.schemas.password_reset.PasswordResetMessage'>; query parameters []. Create a reset token without revealing whether an account exists.

- GET /api/v1/auth/password-reset-status: body None; response dict[str, bool]; query parameters []. Expose only the non-sensitive delivery mode needed by the local recovery UI.

- GET /api/v1/auth/dev/password-reset-link: body None; response dict[str, str]; query parameters [{'name': 'identifier', 'required': True}]. Return a local reset link only when the backend is explicitly in development mode.

- POST /api/v1/auth/reset-password: body <class 'app.schemas.password_reset.ResetPasswordRequest'>; response <class ' app.schemas.password_reset.PasswordResetMessage'>; query parameters []. Consume a valid reset token and revoke every existing browser session.

backend/app/project_workflow.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/project-progress/{team_id}	project_progress line 109	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN, PANEL_MEMBER; get_current_user; get_db
POST	/api/v1/project-progress/{team_id}/milestones	create_milestone line 118	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
PATCH	/api/v1/project-progress/milestones/{milestone_id}	update_milestone line 127	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
POST	/api/v1/project-progress/{team_id}/tasks	create_task line 138	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
PATCH	/api/v1/project-progress/tasks/{task_id}	update_task line 151	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
GET	/api/v1/project-progress/{team_id}/meetings	list_meetings line 164	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN, PANEL_MEMBER; get_current_user; get_db
POST	/api/v1/project-progress/{team_id}/meetings	create_meeting line 171	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
GET	/api/v1/panel-schedules	list_schedules line 227	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN, PANEL_MEMBER; get_current_user; get_db
GET	/api/v1/panel-schedules/options	schedule_options line 235	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/panel-schedules/{schedule_id}	schedule_detail line 243	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN, PANEL_MEMBER; get_current_user; get_db
POST	/api/v1/panel-schedules	create_schedule line 251	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
PATCH	/api/v1/panel-schedules/{schedule_id}	reschedule line 265	role_guard: ADMIN, COORDINATOR; get_current_user; get_db
GET	/api/v1/panel-evaluations/schedule/{schedule_id}	list_evaluations line 290	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN, PANEL_MEMBER; get_current_user; get_db
POST	/api/v1/panel-evaluations	save_evaluation line 298	role_guard: PANEL_MEMBER; get_current_user; get_db

Method	API path	Handler / line	Dependency guards
POST	/api/v1/panel-evaluations/{evaluation_id}/resolution	resolve_evaluation line 320	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db

- GET /api/v1/project-progress/{team_id}: body None; response <class 'dict'>; query parameters [].

- POST /api/v1/project-progress/{team_id}/milestones: body <class 'app.schemas.project_workflow.MilestoneWrite'>; response <class 'dict'>; query parameters [].

- PATCH /api/v1/project-progress/milestones/{milestone_id}: body <class 'app.schemas.project_workflow.MilestoneWrite'>; response <class ' dict'>; query parameters [].

- POST /api/v1/project-progress/{team_id}/tasks: body <class 'app.schemas.project_workflow.TaskWrite'>; response <class 'dict'>; query parameters [].

- PATCH /api/v1/project-progress/tasks/{task_id}: body <class 'app.schemas.project_workflow.TaskWrite'>; response <class 'dict'>; query parameters [].

- GET /api/v1/project-progress/{team_id}/meetings: body None; response list[dict]; query parameters [].

- POST /api/v1/project-progress/{team_id}/meetings: body <class 'app.schemas.project_workflow.MeetingWrite'>; response <class 'dict'>; query parameters [].

- GET /api/v1/panel-schedules: body None; response list[dict]; query parameters [{'name': 'academic_cycle_id', 'required': False}].

- GET /api/v1/panel-schedules/options: body None; response <class 'dict'>; query parameters [].

- GET /api/v1/panel-schedules/{schedule_id}: body None; response <class 'dict'>; query parameters [].

- POST /api/v1/panel-schedules: body <class 'app.schemas.project_workflow.ScheduleWrite'>; response <class 'dict'>; query parameters [].

- PATCH /api/v1/panel-schedules/{schedule_id}: body <class 'app.schemas.project_workflow.ScheduleReschedule'>; response <class 'dict'>; query parameters [].

- GET /api/v1/panel-evaluations/schedule/{schedule_id}: body None; response list[dict]; query parameters [].

- POST /api/v1/panel-evaluations: body <class 'app.schemas.project_workflow.EvaluationWrite'>; response <class 'dict'>; query parameters [].

- POST /api/v1/panel-evaluations/{evaluation_id}/resolution: body <class 'app.schemas.project_workflow.ResolutionWrite'>; response <class ' dict'>; query parameters [].

backend/app/proposal_reference.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/proposal-references/search	search_references line 49	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db
GET	/api/v1/proposal-references/{reference_id}/document	reference_document line 77	role_guard: STUDENT, SUPERVISOR, COORDINATOR, ADMIN; get_current_user; get_db

- GET /api/v1/proposal-references/search: body None; response None; query parameters [{'name': 'q', 'required': True}].

- GET /api/v1/proposal-references/{reference_id}/document: body None; response None; query parameters [].

backend/app/supervisor_assignments.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/admin/supervisor-assignments/teams	teams line 24	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/admin/supervisor-assignments	listing line 28	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/supervisor-assignments	assign line 31	role_guard: ADMIN; get_current_user; get_db
POST	/api/v1/admin/supervisor-assignments/{assignment_id}/end	end line 72	role_guard: ADMIN; get_current_user; get_db

- GET /api/v1/admin/supervisor-assignments/teams: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': True}].

- GET /api/v1/admin/supervisor-assignments: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': True}].

- POST /api/v1/admin/supervisor-assignments: body <class 'app.supervisor_assignments.AssignmentWrite'>; response None; query parameters [].

- POST /api/v1/admin/supervisor-assignments/{assignment_id}/end: body None; response None; query parameters [].

backend/app/supervisor_request_conversations.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/supervisor-requests/{request_id}/conversation	conversation_history line 51	get_current_user; get_db
POST	/api/v1/supervisor-requests/{request_id}/conversation	send_conversation_message line 58	get_current_user; get_db

- GET /api/v1/supervisor-requests/{request_id}/conversation: body None; response None; query parameters [].

- POST /api/v1/supervisor-requests/{request_id}/conversation: body <class 'app.supervisor_request_conversations.RequestMessageWrite'>; response None; query parameters [].

backend/app/supervisor_requests.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/supervisor-requests/cycles	active_cycles line 476	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/supervisor-requests/supervisors	eligible_supervisors line 485	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/supervisor-requests/mine	my_requests line 521	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/supervisor-requests	create_request line 540	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/supervisor-requests/draft/{request_id}/document	update_draft_document line 631	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/supervisor-requests/{request_id}/confirm	confirm_extraction line 696	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/supervisor-requests/{request_id}/submit	submit_request line 734	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/supervisor-requests/{request_id}/withdraw	withdraw_request line 799	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/supervisor-requests/supervisor/queue	supervisor_queue line 831	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/supervisor-requests/{request_id}/decision	decide_request line 848	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor-requests/document/{request_id}	proposal_document line 920	get_current_user; get_db
GET	/api/v1/supervisor-requests/admin/monitor	admin_monitor line 971	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/supervisor-requests/coordinator/monitor	coordinator_monitor line 986	role_guard: COORDINATOR; get_current_user; get_db
GET	/api/v1/supervisor-requests/detail/{request_id}	read_request line 1001	get_current_user; get_db
GET	/api/v1/proposals/cycles	active_cycles line 476	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/proposals/supervisors	eligible_supervisors line 485	role_guard: STUDENT; get_current_user; get_db
GET	/api/v1/proposals/mine	my_requests line 521	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposals	create_request line 540	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposals/draft/{request_id}/document	update_draft_document line 631	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposals/{request_id}/confirm	confirm_extraction line 696	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposals/{request_id}/submit	submit_request line 734	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposals/{request_id}/withdraw	withdraw_request line 799	role_guard: STUDENT; get_current_user; get_db
POST	/api/v1/proposals/{request_id}/decision	decide_request line 848	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/proposals/document/{request_id}	proposal_document line 920	get_current_user; get_db
GET	/api/v1/proposals/admin/monitor	admin_monitor line 971	role_guard: ADMIN; get_current_user; get_db
GET	/api/v1/proposals/coordinator/monitor	coordinator_monitor line 986	role_guard: COORDINATOR; get_current_user; get_db
GET	/api/v1/proposals/{request_id}	read_request line 1001	get_current_user; get_db
POST	/api/v1/proposals/{request_id}/ai-review	run_proposal_ai_review line 1020	get_current_user; get_db
GET	/api/v1/proposals/{request_id}/ai-review	latest_proposal_ai_review line 1072	get_current_user; get_db

- GET /api/v1/supervisor-requests/cycles: body None; response None; query parameters [].

- GET /api/v1/supervisor-requests/supervisors: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': True}].

- GET /api/v1/supervisor-requests/mine: body None; response None; query parameters [].

- POST /api/v1/supervisor-requests: body <class 'fastapi._compat.v2.Body_create_request_api_v1_supervisor_requests_post'>; response None; query parameters [].

- POST /api/v1/supervisor-requests/draft/{request_id}/document: body <class ' fastapi._compat.v2.Body_update_draft_document_api_v1_supervisor_requests_draft__request_id__document_post'>; response None; query parameters [].

- POST /api/v1/supervisor-requests/{request_id}/confirm: body <class 'app.supervisor_requests.RevisionAction'>; response None; query parameters [].

- POST /api/v1/supervisor-requests/{request_id}/submit: body <class 'app.supervisor_requests.RevisionAction'>; response None; query parameters [].

- POST /api/v1/supervisor-requests/{request_id}/withdraw: body <class 'app.supervisor_requests.RevisionAction'>; response None; query parameters [].

- GET /api/v1/supervisor-requests/supervisor/queue: body None; response None; query parameters [{'name': 'status_filter', 'required': False}].

- POST /api/v1/supervisor-requests/{request_id}/decision: body <class 'app.supervisor_requests.DecisionAction'>; response None; query parameters [].

- GET /api/v1/supervisor-requests/document/{request_id}: body None; response None; query parameters [].

- GET /api/v1/supervisor-requests/admin/monitor: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': False}, {'name': 'supervisor_id', 'required': False}, {'name': 'status_filter', 'required': False}, {'name': 'from_date', 'required': False}, {'name': ' to_date', 'required': False}].

- GET /api/v1/supervisor-requests/coordinator/monitor: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': False}, {'name': 'supervisor_id', 'required': False}, {'name': 'status_filter', 'required': False}, {'name': 'from_date', 'required': False}, {'name': ' to_date', 'required': False}].

- GET /api/v1/supervisor-requests/detail/{request_id}: body None; response None; query parameters [].

- GET /api/v1/proposals/cycles: body None; response None; query parameters [].

- GET /api/v1/proposals/supervisors: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': True}].

- GET /api/v1/proposals/mine: body None; response None; query parameters [].

- POST /api/v1/proposals: body <class 'fastapi._compat.v2.Body_create_request_api_v1_proposals_post'>; response None; query parameters [].

- POST /api/v1/proposals/draft/{request_id}/document: body <class 'fastapi._compat.v2.Body_update_draft_document_api_v1_proposals_draft_ _request_id__document_post'>; response None; query parameters [].

- POST /api/v1/proposals/{request_id}/confirm: body <class 'app.supervisor_requests.RevisionAction'>; response None; query parameters [].

- POST /api/v1/proposals/{request_id}/submit: body <class 'app.supervisor_requests.RevisionAction'>; response None; query parameters [].

- POST /api/v1/proposals/{request_id}/withdraw: body <class 'app.supervisor_requests.RevisionAction'>; response None; query parameters [].

- POST /api/v1/proposals/{request_id}/decision: body <class 'app.supervisor_requests.DecisionAction'>; response None; query parameters [].

- GET /api/v1/proposals/document/{request_id}: body None; response None; query parameters [].

- GET /api/v1/proposals/admin/monitor: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': False}, {'name': ' supervisor_id', 'required': False}, {'name': 'status_filter', 'required': False}, {'name': 'from_date', 'required': False}, {'name': 'to_date', 'required': False}].

- GET /api/v1/proposals/coordinator/monitor: body None; response None; query parameters [{'name': 'academic_cycle_id', 'required': False}, {'name': 'supervisor_id', 'required': False}, {'name': 'status_filter', 'required': False}, {'name': 'from_date', 'required': False}, {'name': 'to_date', ' required': False}].

- GET /api/v1/proposals/{request_id}: body None; response None; query parameters [].

- POST /api/v1/proposals/{request_id}/ai-review: body None; response <class 'app.supervisor_requests.ProposalAIReviewResponse'>; query parameters [].

- GET /api/v1/proposals/{request_id}/ai-review: body None; response None; query parameters [].

backend/app/supervisor_workspace.py

Method	API path	Handler / line	Dependency guards
GET	/api/v1/supervisor/assigned-groups	assigned_groups line 88	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/groups/{group_id}	group_detail line 94	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/groups/{group_id}/progress	group_section line 100	role_guard: SUPERVISOR; get_current_user; get_db

Method	API path	Handler / line	Dependency guards
GET	/api/v1/supervisor/groups/{group_id}/documents	group_section line 100	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/groups/{group_id}/project	group_section line 100	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/groups/{group_id}/proposal	group_section line 100	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/groups/{group_id}/members	group_section line 100	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/requests	requests line 109	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/requests/{request_id}	request_detail line 115	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/supervisor/requests/{request_id}/request-changes	decision line 130	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/supervisor/requests/{request_id}/reject	decision line 130	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/supervisor/requests/{request_id}/accept	decision line 130	role_guard: SUPERVISOR; get_current_user; get_db
GET	/api/v1/supervisor/groups/{group_id}/conversation	conversation line 146	role_guard: SUPERVISOR; get_current_user; get_db
POST	/api/v1/supervisor/groups/{group_id}/conversation	send_message line 152	role_guard: SUPERVISOR; get_current_user; get_db

- GET /api/v1/supervisor/assigned-groups: body None; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}: body None; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}/progress: body None; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}/documents: body None; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}/project: body None; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}/proposal: body None; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}/members: body None; response None; query parameters [].

- GET /api/v1/supervisor/requests: body None; response None; query parameters [].

- GET /api/v1/supervisor/requests/{request_id}: body None; response None; query parameters [].

- POST /api/v1/supervisor/requests/{request_id}/request-changes: body <class 'app.supervisor_workspace.DecisionInput'>; response None; query parameters [].

- POST /api/v1/supervisor/requests/{request_id}/reject: body <class 'app.supervisor_workspace.DecisionInput'>; response None; query parameters [].

- POST /api/v1/supervisor/requests/{request_id}/accept: body <class 'app.supervisor_workspace.DecisionInput'>; response None; query parameters [].

- GET /api/v1/supervisor/groups/{group_id}/conversation: body None; response None; query parameters [].

- POST /api/v1/supervisor/groups/{group_id}/conversation: body <class 'app.collaboration_channels.MessageWrite'>; response None; query parameters [].

Appendix D - Complete source-file navigation atlas

Every discovered application, migration, test and script source file is listed below. Line numbers refer to this snapshot. Frontend reachability means an import path exists from main.tsx; a reachable module may still have conditional or inaccessible branches. A file not reachable through static relative imports is an alternative/helper candidate, not automatically safe to delete.

backend/app/__init__.py

Lines: 1. FYPHub backend application package.

backend/app/academic_cycle_overview.py

Lines: 115. Authorized, read-only academic-cycle dashboard aggregation.

Symbol	Line	Purpose / source docstring
_user	31	Implementation entry; inspect this symbol for details
cycle_overview	38	Implementation entry; inspect this symbol for details

backend/app/agent_supervision.py

Lines: 101. Controlled supervision-agent planning and approval workflow.

Symbol	Line	Purpose / source docstring
Approval	17	Implementation entry; inspect this symbol for details
_team	21	Implementation entry; inspect this symbol for details
_access	26	Implementation entry; inspect this symbol for details
_out	34	Implementation entry; inspect this symbol for details
_accepted_proposal	37	Implementation entry; inspect this symbol for details
start_run	41	Implementation entry; inspect this symbol for details
list_runs	60	Implementation entry; inspect this symbol for details
approve_run	64	Implementation entry; inspect this symbol for details
execute_run	72	Implementation entry; inspect this symbol for details
approve_reminder	93	Implementation entry; inspect this symbol for details

backend/app/ai_provider.py

Lines: 129. Backend-only, schema-validated advisory AI clients with deterministic fallback.

Symbol	Line	Purpose / source docstring
ProposalReviewFinding	12	Implementation entry; inspect this symbol for details
ProposalProviderPayload	23	Implementation entry; inspect this symbol for details
PlanItem	36	Implementation entry; inspect this symbol for details
PlanMilestone	39	Implementation entry; inspect this symbol for details
SupervisionPlanPayload	42	Implementation entry; inspect this symbol for details
provider_config	45	Implementation entry; inspect this symbol for details
validate_provider_configuration	48	Implementation entry; inspect this symbol for details
_allowed	60	Implementation entry; inspect this symbol for details
_run_ai_json_once	64	Implementation entry; inspect this symbol for details
run_ai_json	120	Implementation entry; inspect this symbol for details
run_groq_json	128	Implementation entry; inspect this symbol for details

backend/app/archived_projects.py

Lines: 1108. Authorized logical FYP archive management and deterministic knowledge search.

Symbol	Line	Purpose / source docstring
_cell_text	31	Implementation entry; inspect this symbol for details
_clean_cell	38	Return only literal cell content; discard formulas and Excel errors.

Symbol	Line	Purpose / source docstring
_is_email_value	50	Implementation entry; inspect this symbol for details
_last_content_row	54	Find the true end of a worksheet, preferring the last literal email row.
_xls_value	80	Implementation entry; inspect this symbol for details
parse_archive_workbook	90	Extract every populated column without imposing an archive schema.
_column_token	175	Implementation entry; inspect this symbol for details
_mapped_value	179	Implementation entry; inspect this symbol for details
_project_title	196	Read a project title only from an explicit project-title column.
_column_values	218	Implementation entry; inspect this symbol for details
_member_rows	228	Extract member data from one source row without treating it as a project.
_same_member	258	Implementation entry; inspect this symbol for details
_merge_member_values	266	Implementation entry; inspect this symbol for details
_first_non_empty	282	Implementation entry; inspect this symbol for details
_unique_text	290	Implementation entry; inspect this symbol for details
_import_display	302	Implementation entry; inspect this symbol for details
_apply_imported_cells	308	Apply an edited imported row while preserving its original Excel columns.
_archive_import_project	352	Build one archive parent from one Excel row.
_resolved_import_rows	395	Forward-fill merged Excel cells and return rows plus safe orphan rows.
_merge_source_cells	443	Preserve all original columns while exposing a single editable value per key.
_group_mapped_value	462	Implementation entry; inspect this symbol for details
_import_row_errors	466	Implementation entry; inspect this symbol for details
_archive_import_group	476	Create one parent archive entity for all rows sharing one safe key.
_import_group_key	526	Implementation entry; inspect this symbol for details
_merge_imported_project	533	Merge one grouped import into an existing project without overwriting data.
normalize_title	557	Implementation entry; inspect this symbol for details
clean_filename	561	Implementation entry; inspect this symbol for details
ArchiveCreate	567	Implementation entry; inspect this symbol for details
ArchiveUpdate	593	Implementation entry; inspect this symbol for details
_error	617	Implementation entry; inspect this symbol for details
_audit	621	Implementation entry; inspect this symbol for details
_document_out	627	Implementation entry; inspect this symbol for details
_project_out	631	Implementation entry; inspect this symbol for details
_active_project	642	Implementation entry; inspect this symbol for details
_metadata_snapshot	649	Implementation entry; inspect this symbol for details
_validate_document	653	Implementation entry; inspect this symbol for details
_next_version	663	Implementation entry; inspect this symbol for details
search_archives	669	Implementation entry; inspect this symbol for details
admin_archive_list	724	Implementation entry; inspect this symbol for details
admin_archive_detail	729	Implementation entry; inspect this symbol for details
read_archive	737	Implementation entry; inspect this symbol for details
read_archive_document	742	Implementation entry; inspect this symbol for details
create_archive	755	Implementation entry; inspect this symbol for details
update_archive	771	Implementation entry; inspect this symbol for details
delete_archive	805	Permanently remove an archive record and all of its stored document versions.
rollback_archive_metadata	824	Implementation entry; inspect this symbol for details
set_archive_active	846	Implementation entry; inspect this symbol for details
upload_archive_document	858	Implementation entry; inspect this symbol for details
archive_document_history	879	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
delete_archive_document	887	Implementation entry; inspect this symbol for details
_read_archive_upload	903	Implementation entry; inspect this symbol for details
archive_import_preview	910	Implementation entry; inspect this symbol for details
archive_import_confirm	919	Implementation entry; inspect this symbol for details
admin_archive_projects	1007	Implementation entry; inspect this symbol for details
repair_imported_archive	1030	Safely consolidate obvious duplicate imported parents.
admin_archive_project	1068	Implementation entry; inspect this symbol for details
admin_update_archive	1075	Implementation entry; inspect this symbol for details
admin_delete_archive	1080	Implementation entry; inspect this symbol for details
archive_import_batches	1085	Implementation entry; inspect this symbol for details
archive_import_batch_detail	1092	Implementation entry; inspect this symbol for details
set_archive_visibility	1100	Implementation entry; inspect this symbol for details

backend/app/audit_history.py

Lines: 119. Read-only, allowlisted administrator audit history.

Symbol	Line	Purpose / source docstring
scalar	54	Implementation entry; inspect this symbol for details
safe_changes	57	Implementation entry; inspect this symbol for details
output	74	Implementation entry; inspect this symbol for details
parse_time	77	Implementation entry; inspect this symbol for details
list_events	89	Implementation entry; inspect this symbol for details
event_detail	113	Implementation entry; inspect this symbol for details

backend/app/auth.py

Lines: 203. Reusable cookie-session authentication dependencies and helpers.

Symbol	Line	Purpose / source docstring
log_temporary_auth_diagnostic	31	TEMPORARY: emit safe metadata while diagnosing post-login access failures.
utc_now	46	Implementation entry; inspect this symbol for details
authentication_error	50	Implementation entry; inspect this symbol for details
validate_mutating_origin	54	Allow same-origin/non-browser calls, but reject a supplied foreign browser Origin.
session_cookie_secure	67	Mark session cookies Secure exactly when the request is served over HTTPS.
set_session_cookie	72	Implementation entry; inspect this symbol for details
clear_session_cookie	84	Implementation entry; inspect this symbol for details
get_current_user	94	Resolve an active user through a current, non-revoked hashed cookie session.
require_roles	176	Create a dependency that permits only the listed roles after authentication.

backend/app/collaboration_channels.py

Lines: 137. Private assignment-scoped Module 2 collaboration channels.

Symbol	Line	Purpose / source docstring
MessageWrite	15	Implementation entry; inspect this symbol for details
_active_channel	28	Implementation entry; inspect this symbol for details
_student_member	35	Implementation entry; inspect this symbol for details
_authorize	41	Implementation entry; inspect this symbol for details
_accepted_request	56	Implementation entry; inspect this symbol for details
ensure_collaboration_channel	60	Implementation entry; inspect this symbol for details
_channel_out	81	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
_message_out	91	Implementation entry; inspect this symbol for details
_list_channels	96	Implementation entry; inspect this symbol for details
student_channels	106	Implementation entry; inspect this symbol for details
supervisor_channels	111	Implementation entry; inspect this symbol for details
collaboration_channel	117	Implementation entry; inspect this symbol for details
collaboration_messages	124	Implementation entry; inspect this symbol for details
send_collaboration_message	132	Implementation entry; inspect this symbol for details

backend/app/config.py

Lines: 103. Configuration for the FYPHub backend.

Symbol	Line	Purpose / source docstring
Settings	15	Implementation entry; inspect this symbol for details
get_settings	59	Implementation entry; inspect this symbol for details

backend/app/coordinator_operations.py

Lines: 81. Read-only operational data exposed specifically to coordinators.

Symbol	Line	Purpose / source docstring
supervisor_capacity	16	Implementation entry; inspect this symbol for details
audit_history	53	Implementation entry; inspect this symbol for details
proposal_templates	66	Implementation entry; inspect this symbol for details
evaluation_criteria	78	Implementation entry; inspect this symbol for details

backend/app/dashboard_requirements.py

Lines: 71. Role-scoped deadlines and requirements for all dashboards.

Symbol	Line	Purpose / source docstring
_status	13	Implementation entry; inspect this symbol for details
_due_source	19	Implementation entry; inspect this symbol for details
_out	25	Implementation entry; inspect this symbol for details
_visible	29	Implementation entry; inspect this symbol for details
dashboard_requirements	36	Implementation entry; inspect this symbol for details
dashboard_summary	43	Implementation entry; inspect this symbol for details
create_requirement	48	Implementation entry; inspect this symbol for details
update_requirement	57	Implementation entry; inspect this symbol for details
delete_requirement	66	Implementation entry; inspect this symbol for details

backend/app/database.py

Lines: 36. Synchronous SQLAlchemy database resources.

Symbol	Line	Purpose / source docstring
Base	11	Shared declarative base for future application models.
get_db	30	Provide a request-scoped session that is always closed.

backend/app/document_requirements.py

Lines: 303.

Symbol	Line	Purpose / source docstring
_audit	25	Implementation entry; inspect this symbol for details
_student_cycle_ids	29	Resolve the student's current cycle without trusting a client-supplied cycle.

Symbol	Line	Purpose / source docstring
_student_team_ids	43	Implementation entry; inspect this symbol for details
_student_group	52	Implementation entry; inspect this symbol for details
_current_active_cycle_id	56	Implementation entry; inspect this symbol for details
_visible	60	Implementation entry; inspect this symbol for details
_out	74	Implementation entry; inspect this symbol for details
_now	79	Implementation entry; inspect this symbol for details
_stored_path	80	Implementation entry; inspect this symbol for details
create_requirement	88	Implementation entry; inspect this symbol for details
delete_requirement	113	Remove an unused draft requirement while preserving published/history records.
list_requirements	133	Implementation entry; inspect this symbol for details
get_requirement	139	Implementation entry; inspect this symbol for details
update_requirement	146	Implementation entry; inspect this symbol for details
_state	153	Implementation entry; inspect this symbol for details
publish	163	Implementation entry; inspect this symbol for details
unpublish	165	Implementation entry; inspect this symbol for details
close	167	Implementation entry; inspect this symbol for details
archive	169	Implementation entry; inspect this symbol for details
reopen	171	Implementation entry; inspect this symbol for details
_submission	173	Implementation entry; inspect this symbol for details
_validate	184	Implementation entry; inspect this symbol for details
_submission_out	206	Implementation entry; inspect this symbol for details
student_requirements	213	Implementation entry; inspect this symbol for details
upload	219	Implementation entry; inspect this symbol for details
submission	248	Implementation entry; inspect this symbol for details
all_submissions	255	Return every student's submission and every uploaded version for an admin requirement view.
versions	274	Implementation entry; inspect this symbol for details
download	280	Implementation entry; inspect this symbol for details
_review	292	Implementation entry; inspect this symbol for details
approve	299	Implementation entry; inspect this symbol for details
reject	301	Implementation entry; inspect this symbol for details
replacement	303	Implementation entry; inspect this symbol for details

backend/app/evaluations.py

Lines: 495. Role-scoped FYP evaluation, marks, audit, reporting, and result APIs.

Symbol	Line	Purpose / source docstring
_error	56	Implementation entry; inspect this symbol for details
_role_categories	60	Implementation entry; inspect this symbol for details
_target_group_ids	70	Implementation entry; inspect this symbol for details
_panel_can_access	82	Implementation entry; inspect this symbol for details
_actor_can_access_group	90	Implementation entry; inspect this symbol for details
_record_target	100	Implementation entry; inspect this symbol for details
_record_out	104	Implementation entry; inspect this symbol for details
_audit	108	Implementation entry; inspect this symbol for details
_validate_category_owner	112	Implementation entry; inspect this symbol for details
_resolve_target	117	Implementation entry; inspect this symbol for details
_target_filter	132	Implementation entry; inspect this symbol for details
_assignment_out	136	Implementation entry; inspect this symbol for details
evaluation_assignments	142	Implementation entry; inspect this symbol for details
my_evaluations	149	Implementation entry; inspect this symbol for details
_save_record	163	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
create_evaluation	201	Implementation entry; inspect this symbol for details
update_evaluation	217	Implementation entry; inspect this symbol for details
submit_evaluation	233	Implementation entry; inspect this symbol for details
return_evaluation	254	Implementation entry; inspect this symbol for details
approve_evaluation	272	Implementation entry; inspect this symbol for details
correct_evaluation	288	Implementation entry; inspect this symbol for details
publish_evaluation	310	Implementation entry; inspect this symbol for details
evaluation_audit	326	Implementation entry; inspect this symbol for details
admin_evaluations	337	Implementation entry; inspect this symbol for details
list_aggregation	349	Implementation entry; inspect this symbol for details
update_aggregation	360	Implementation entry; inspect this symbol for details
_student_group	377	Implementation entry; inspect this symbol for details
_aggregation_method	381	Implementation entry; inspect this symbol for details
_category_result	386	Implementation entry; inspect this symbol for details
_grade	408	Implementation entry; inspect this symbol for details
_result	421	Implementation entry; inspect this symbol for details
student_results	431	Implementation entry; inspect this symbol for details
admin_group_results	436	Implementation entry; inspect this symbol for details
admin_results	446	Implementation entry; inspect this symbol for details
_report_lines	453	Implementation entry; inspect this symbol for details
_simple_pdf	465	Implementation entry; inspect this symbol for details
evaluation_report	477	Implementation entry; inspect this symbol for details
evaluation_report_pdf	488	Implementation entry; inspect this symbol for details

backend/app/group_ai.py

Lines: 105. Team-scoped, advisory local Ollama chat.

Symbol	Line	Purpose / source docstring
NoRedirect	17	Implementation entry; inspect this symbol for details
require_team	21	Implementation entry; inspect this symbol for details
scoped_context	28	Implementation entry; inspect this symbol for details
unavailable	42	Implementation entry; inspect this symbol for details
generate	45	Implementation entry; inspect this symbol for details
history	85	Implementation entry; inspect this symbol for details
chat	92	Implementation entry; inspect this symbol for details

backend/app/group_channels.py

Lines: 118.

Symbol	Line	Purpose / source docstring
Channel	16	Implementation entry; inspect this symbol for details
MessageWrite	24	Implementation entry; inspect this symbol for details
permission	36	Implementation entry; inspect this symbol for details
output	50	Implementation entry; inspect this symbol for details
_file_path	54	Implementation entry; inspect this symbol for details
teams	58	Implementation entry; inspect this symbol for details
history	67	Implementation entry; inspect this symbol for details
send	74	Implementation entry; inspect this symbol for details
upload_file	93	Store a team-scoped resource and publish it into Files & Resources.
download_file	113	Implementation entry; inspect this symbol for details

backend/app/main.py

Lines: 2188.

Symbol	Line	Purpose / source docstring
_workflow_name	61	Implementation entry; inspect this symbol for details
StructuredRequestLoggingMiddleware	69	Implementation entry; inspect this symbol for details
request_validation_error	106	Expose consistent, field-addressable request validation failures.
registration_error	125	Return a public, structured error without database implementation details.
normalize_team_member	135	Apply the established identifier conventions and field-length rules.
list_cycles	169	Implementation entry; inspect this symbol for details
get_group_policy	173	Implementation entry; inspect this symbol for details
cycle_error	180	Implementation entry; inspect this symbol for details
locked_cycle_and_policy	184	Use the same cycle-then-policy lock order as team registration.
largest_active_team_size	193	Implementation entry; inspect this symbol for details
deadline_error	198	Implementation entry; inspect this symbol for details
formatted_utc_offset	202	Implementation entry; inspect this symbol for details
require_timezone_data	209	Implementation entry; inspect this symbol for details
resolve_deadline_time	216	Resolve a wall time without guessing a zone, gap, or overlapping offset.
deadline_version_out	243	Implementation entry; inspect this symbol for details
deadline_detail_out	263	Implementation entry; inspect this symbol for details
validate_deadline_write	279	Implementation entry; inspect this symbol for details
save_deadline_version	287	Implementation entry; inspect this symbol for details
list_time_zones	321	Implementation entry; inspect this symbol for details
preview_academic_deadline	327	Implementation entry; inspect this symbol for details
list_academic_deadlines	338	Implementation entry; inspect this symbol for details
academic_deadline_detail	373	Implementation entry; inspect this symbol for details
create_academic_deadline	385	Implementation entry; inspect this symbol for details
create_academic_deadline_version	415	Implementation entry; inspect this symbol for details
delete_academic_deadline	447	Remove a deadline and its recorded versions when no rule references it.
list_template_references	478	Implementation entry; inspect this symbol for details
template_reference_detail	488	Implementation entry; inspect this symbol for details
create_template_reference	517	Implementation entry; inspect this symbol for details
create_template_version	541	Implementation entry; inspect this symbol for details
create_rule_version	548	Implementation entry; inspect this symbol for details
delete_template_reference_version	557	Implementation entry; inspect this symbol for details
rubric_version_out	575	Implementation entry; inspect this symbol for details
rubric_detail_out	594	Implementation entry; inspect this symbol for details
save_rubric_version	607	Implementation entry; inspect this symbol for details
list_evaluation_rubrics	651	Implementation entry; inspect this symbol for details
evaluation_rubric_detail	686	Implementation entry; inspect this symbol for details
create_evaluation_rubric	698	Implementation entry; inspect this symbol for details
create_evaluation_rubric_version	722	Implementation entry; inspect this symbol for details
create_cycle	759	Implementation entry; inspect this symbol for details
update_cycle	776	Implementation entry; inspect this symbol for details
delete_cycle	796	Implementation entry; inspect this symbol for details
set_group_policy	815	Implementation entry; inspect this symbol for details
delete_group_policy	834	Implementation entry; inspect this symbol for details
health	850	Report application liveness only; this does not check database connectivity.
readiness	856	Report readiness only when the database and vector extension are available.

Symbol	Line	Purpose / source docstring
login	880	Authenticate an active account and issue a new opaque cookie session.
account_response	950	Return an account principal and, for students, its shared group context.
current_user	1004	Return the authenticated active account's safe fields.
admin_current_user	1019	Return the current account only when its persisted role is ADMIN.
admin_registration_mode	1038	Expose demo-mode status only to authenticated administrators.
admin_user_error	1049	Return a safe management failure without leaking persistence details.
add_admin_audit_event	1054	Append only non-sensitive account metadata; never credentials or sessions.
shared_team_account	1068	Implementation entry; inspect this symbol for details
lock_and_protect_last_active_admin	1072	Serialize changes that could remove the final active ADMIN account.
revoke_user_sessions	1089	Implementation entry; inspect this symbol for details
supervisor_capacity_counts	1097	Implementation entry; inspect this symbol for details
supervisor_capacity_out	1119	Implementation entry; inspect this symbol for details
locked_eligible_supervisor_and_cycle	1159	Lock eligibility records so role/status changes cannot race capacity writes.
locked_eligible_supervisor_and_period	1172	Lock an active supervisor before changing annual capacity.
assigned_groups_for_period	1182	Implementation entry; inspect this symbol for details
list_admin_users	1187	Implementation entry; inspect this symbol for details
create_admin_user	1217	Implementation entry; inspect this symbol for details
change_admin_user_status	1250	Implementation entry; inspect this symbol for details
change_admin_user_role	1286	Implementation entry; inspect this symbol for details
delete_admin_user	1328	Permanently remove an unused account; linked records stay protected by FK guards.
list_supervisor_capacities	1358	Implementation entry; inspect this symbol for details
supervisor_capacity_workspace	1368	Return the signed-in supervisor's read-only annual capacity and assignment detail.
read_supervisor_capacity	1410	Implementation entry; inspect this symbol for details
create_supervisor_capacity	1420	Implementation entry; inspect this symbol for details
update_supervisor_capacity	1454	Implementation entry; inspect this symbol for details
delete_supervisor_capacity	1492	Implementation entry; inspect this symbol for details
logout	1523	Revoke the presented session when present and always clear its browser cookie.
register_student	1548	Reject the retired individual public signup flow.
register_team	1568	Atomically create one shared student account, team, and member contacts.
register_student_group	1823	Create one shared STUDENT group account from member contacts only.
student_group_detail	1852	Serialize the shared student principal without exposing credentials.
current_student_team	1893	Implementation entry; inspect this symbol for details
student_group_me	1912	Implementation entry; inspect this symbol for details
student_group_members	1920	Implementation entry; inspect this symbol for details
list_student_groups	1929	Implementation entry; inspect this symbol for details
get_student_group	1946	Implementation entry; inspect this symbol for details
set_student_group_status	1957	Implementation entry; inspect this symbol for details
activate_student_group	1979	Implementation entry; inspect this symbol for details
deactivate_student_group	1984	Implementation entry; inspect this symbol for details
delete_student_group	1989	Implementation entry; inspect this symbol for details
reminder_calculation	2015	Implementation entry; inspect this symbol for details
reminder_version_out	2037	Implementation entry; inspect this symbol for details
reminder_detail	2042	Implementation entry; inspect this symbol for details
save_reminder_version	2048	Implementation entry; inspect this symbol for details
preview_reminder_rule	2058	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
list_reminder_rules	2064	Implementation entry; inspect this symbol for details
read_reminder_rule	2074	Implementation entry; inspect this symbol for details
create_reminder_rule	2082	Implementation entry; inspect this symbol for details
create_reminder_rule_version	2104	Implementation entry; inspect this symbol for details
delete_reminder_rule	2127	Implementation entry; inspect this symbol for details

backend/app/models/__init__.py

Lines: 34. Application SQLAlchemy models.

backend/app/models/academic.py

Lines: 54.

Tables declared: academic_cycles, academic_group_policies, supervisor_capacity_policies.

Symbol	Line	Purpose / source docstring
AcademicCycle	6	Implementation entry; inspect this symbol for details
AcademicGroupPolicy	23	Implementation entry; inspect this symbol for details
SupervisorCapacityPolicy	39	Implementation entry; inspect this symbol for details

backend/app/models/academic_deadline.py

Lines: 39. Non-operational academic deadline configuration models.

Tables declared: academic_deadlines, academic_deadline_versions.

Symbol	Line	Purpose / source docstring
AcademicDeadline	11	Stable deadline identity fixed to one academic cycle.
AcademicDeadlineVersion	22	An immutable recorded deadline configuration.

backend/app/models/agent_supervision.py

Lines: 25. Controlled, append-only supervision-agent run records.

Tables declared: supervision_agent_runs.

Symbol	Line	Purpose / source docstring
SupervisionAgentRun	9	Implementation entry; inspect this symbol for details

backend/app/models/archive_import_batch.py

Lines: 22. Import history for administrator archive workbook uploads.

Tables declared: archive_import_batches.

Symbol	Line	Purpose / source docstring
ArchiveImportBatch	11	Implementation entry; inspect this symbol for details

backend/app/models/archived_project.py

Lines: 102. Logical FYP archive records and immutable document versions.

Tables declared: archived_projects, archived_project_documents.

Symbol	Line	Purpose / source docstring
ArchivedProjectDocumentType	13	Implementation entry; inspect this symbol for details
ArchivedProject	26	Implementation entry; inspect this symbol for details
ArchivedProjectDocument	83	Implementation entry; inspect this symbol for details

backend/app/models/audit_event.py

Lines: 31. Append-only audit records for administrator account mutations.

Tables declared: admin_audit_events.

Symbol	Line	Purpose / source docstring
AdminAuditEvent	13	A safe, immutable record of one successful administrator account mutation.

backend/app/models/auth_session.py

Lines: 27. Persistent, hash-only browser authentication sessions.

Tables declared: auth_sessions.

Symbol	Line	Purpose / source docstring
AuthSession	11	An opaque browser session. The raw bearer token is never stored.

backend/app/models/dashboard_requirement.py

Lines: 29. Role-scoped dashboard deadline and requirement records.

Tables declared: dashboard_requirements.

Symbol	Line	Purpose / source docstring
DashboardRequirement	8	Implementation entry; inspect this symbol for details

backend/app/models/document_requirement.py

Lines: 59.

Tables declared: document_upload_requirements, document_submissions, document_submission_versions.

Symbol	Line	Purpose / source docstring
DocumentUploadRequirement	7	Implementation entry; inspect this symbol for details
DocumentSubmission	29	Implementation entry; inspect this symbol for details
DocumentSubmissionVersion	46	Implementation entry; inspect this symbol for details

backend/app/models/evaluation.py

Lines: 100. Persisted role-owned FYP evaluation records and result audit history.

Tables declared: evaluation_records, evaluation_audit_events, evaluation_corrections, evaluation_aggregation_configs.

Symbol	Line	Purpose / source docstring
EvaluationRecord	12	Implementation entry; inspect this symbol for details
EvaluationAuditEvent	58	Implementation entry; inspect this symbol for details
EvaluationCorrection	73	Implementation entry; inspect this symbol for details
EvaluationAggregationConfig	86	Implementation entry; inspect this symbol for details

backend/app/models/evaluation_rubric.py

Lines: 53. Draft evaluation-rubric configuration models.

Tables declared: evaluation_rubrics, evaluation_rubric_versions, evaluation_rubric_criteria.

Symbol	Line	Purpose / source docstring
EvaluationRubric	11	Stable identity shared by every immutable draft version.
EvaluationRubricVersion	21	A complete immutable snapshot of one draft rubric.
EvaluationRubricCriterion	40	An ordered criterion belonging exclusively to one rubric version.

backend/app/models/group_message.py

Lines: 19.

Tables declared: group_messages.

Symbol	Line	Purpose / source docstring
GroupMessage	6	Implementation entry; inspect this symbol for details

backend/app/models/module2_collaboration.py

Lines: 83. Module 2 collaboration records: notifications, channel reads and moderation.

Tables declared: module2_notifications, module2_channel_reads, module2_chatbot_sessions, collaboration_channels, collaboration_participants, collaboration_messages.

Symbol	Line	Purpose / source docstring
Module2Notification	8	Implementation entry; inspect this symbol for details
Module2ChannelRead	19	Implementation entry; inspect this symbol for details
Module2ChatbotSession	30	Implementation entry; inspect this symbol for details
CollaborationChannel	41	Implementation entry; inspect this symbol for details
CollaborationParticipant	58	Implementation entry; inspect this symbol for details
CollaborationMessage	73	Implementation entry; inspect this symbol for details

backend/app/models/password_reset_token.py

Lines: 26. Hash-only, short-lived password reset tokens.

Tables declared: password_reset_tokens.

Symbol	Line	Purpose / source docstring
PasswordResetToken	11	A single-use reset token; the raw bearer value is never persisted.

backend/app/models/project_workflow.py

Lines: 131. Persisted project-progress, defense scheduling, and panel-evaluation records.

Tables declared: project_milestones, project_tasks, supervision_meetings, defense_schedules, defense_panel_members, panel_evaluations, panel_evaluation_scores.

Symbol	Line	Purpose / source docstring
ProjectMilestone	12	Implementation entry; inspect this symbol for details
ProjectTask	30	Implementation entry; inspect this symbol for details
SupervisionMeeting	50	Implementation entry; inspect this symbol for details
DefenseSchedule	65	Implementation entry; inspect this symbol for details
DefensePanelMember	87	Implementation entry; inspect this symbol for details
PanelEvaluation	98	Implementation entry; inspect this symbol for details
PanelEvaluationScore	119	Implementation entry; inspect this symbol for details

backend/app/models/proposal.py

Lines: 28. Non-operational identity and version foundations for proposals.

Tables declared: proposals, proposal_versions.

Symbol	Line	Purpose / source docstring
Proposal	8	Implementation entry; inspect this symbol for details
ProposalVersion	18	Implementation entry; inspect this symbol for details

backend/app/models/proposal_ai_review.py

Lines: 24. Persisted advisory proposal AI-review runs.

Tables declared: proposal_ai_review_runs.

Symbol	Line	Purpose / source docstring
ProposalAIReviewRun	12	Implementation entry; inspect this symbol for details

backend/app/models/proposal_compliance.py

Lines: 58. Stored Module 3 proposal submissions and compliance report snapshots.

Tables declared: proposal_compliance_submissions, proposal_compliance_reports.

Symbol	Line	Purpose / source docstring
ProposalComplianceSubmission	12	Implementation entry; inspect this symbol for details
ProposalComplianceReport	45	Implementation entry; inspect this symbol for details

backend/app/models/proposal_reference.py

Lines: 43. Authorized archived proposal references and separate PDF documents.

Tables declared: proposal_references, proposal_reference_documents.

Symbol	Line	Purpose / source docstring
ProposalReference	12	Implementation entry; inspect this symbol for details
ProposalReferenceDocument	32	Implementation entry; inspect this symbol for details

backend/app/models/reminder_rule.py

Lines: 26. Non-operational reminder and escalation rule snapshots.

Tables declared: reminder_rules, reminder_rule_versions.

Symbol	Line	Purpose / source docstring
ReminderRule	7	Implementation entry; inspect this symbol for details
ReminderRuleVersion	13	Implementation entry; inspect this symbol for details

backend/app/models/supervisor_assignment.py

Lines: 22.

Tables declared: supervisor_assignments.

Symbol	Line	Purpose / source docstring
SupervisorAssignment	6	Implementation entry; inspect this symbol for details

backend/app/models/supervisor_request.py

Lines: 149. Student supervisor-selection request records and append-only history.

Tables declared: supervisor_requests, supervisor_request_documents, supervisor_request_status_history, supervisor_request_submission_note_versions.

Symbol	Line	Purpose / source docstring
SupervisorRequestStatus	13	Implementation entry; inspect this symbol for details
SupervisorRequestExtractionStatus	22	Implementation entry; inspect this symbol for details
SupervisorRequest	29	Implementation entry; inspect this symbol for details
SupervisorRequestDocument	85	Implementation entry; inspect this symbol for details
SupervisorRequestStatusHistory	114	Implementation entry; inspect this symbol for details
SupervisorRequestSubmissionNoteVersion	133	Immutable snapshots of notes submitted with a supervisor request.

backend/app/models/supervisor_request_conversation.py

Lines: 19. Append-only messages scoped to one supervisor request.

Tables declared: supervisor_request_messages.

Symbol	Line	Purpose / source docstring
SupervisorRequestMessage	11	Implementation entry; inspect this symbol for details

backend/app/models/team.py

Lines: 21.

Tables declared: teams, team_members.

Symbol	Line	Purpose / source docstring
Team	6	Implementation entry; inspect this symbol for details
TeamMember	18	Implementation entry; inspect this symbol for details

backend/app/models/template_reference.py

Lines: 18.

Tables declared: template_references, template_reference_versions, academic_rule_versions.

Symbol	Line	Purpose / source docstring
TemplateType	7	Implementation entry; inspect this symbol for details
TemplateReference	8	Implementation entry; inspect this symbol for details
TemplateReferenceVersion	11	Implementation entry; inspect this symbol for details
AcademicRuleVersion	16	Implementation entry; inspect this symbol for details

backend/app/models/user.py

Lines: 74. User and role database foundation.

Tables declared: users.

Symbol	Line	Purpose / source docstring
UserRole	12	Implementation entry; inspect this symbol for details
User	20	A future-authentication user record; password hashes only.

backend/app/module2_collaboration.py

Lines: 185. Complete Module 2 workspace endpoints and cross-module lifecycle surface.

Symbol	Line	Purpose / source docstring
team_for_actor	16	Implementation entry; inspect this symbol for details
team_payload	37	Implementation entry; inspect this symbol for details
workspace	52	Implementation entry; inspect this symbol for details
TransferLeadership	59	Implementation entry; inspect this symbol for details
LeaveTeam	62	Implementation entry; inspect this symbol for details
transfer_leadership	67	Implementation entry; inspect this symbol for details
dissolve	83	Implementation entry; inspect this symbol for details
leave_team	97	Implementation entry; inspect this symbol for details
notifications	123	Implementation entry; inspect this symbol for details
mark_notification_read	132	Implementation entry; inspect this symbol for details
mark_channel_read	142	Implementation entry; inspect this symbol for details
unread_counts	155	Implementation entry; inspect this symbol for details
proposal_versions	168	Implementation entry; inspect this symbol for details
tasks	175	Implementation entry; inspect this symbol for details
deadlines	182	Implementation entry; inspect this symbol for details

backend/app/modules/__init__.py

Lines: 1. Application feature modules.

backend/app/modules/module_03_proposal/__init__.py

Lines: 1. Module 3: student proposal and university-template compliance.

backend/app/modules/module_03_proposal/compliance_engine.py

Lines: 269. Deterministic comparison between a submission and an active template.

Symbol	Line	Purpose / source docstring
_text	10	Implementation entry; inspect this symbol for details
_same	14	Implementation entry; inspect this symbol for details
_status	22	Implementation entry; inspect this symbol for details
_category_status	30	Implementation entry; inspect this symbol for details
_similar_section	39	Implementation entry; inspect this symbol for details
_field_value	47	Implementation entry; inspect this symbol for details
_tokens	54	Implementation entry; inspect this symbol for details
_consistency_item	59	Implementation entry; inspect this symbol for details
build_change_summary	88	Produce a safe, auditable AI-style summary without sending proposal text externally.
build_compliance_report	112	Implementation entry; inspect this symbol for details
_pill	240	Implementation entry; inspect this symbol for details
_category_detail	244	Implementation entry; inspect this symbol for details
_violation	252	Implementation entry; inspect this symbol for details
_section_number	267	Implementation entry; inspect this symbol for details

backend/app/modules/module_03_proposal/router.py

Lines: 604. Module 3 APIs: template ingestion, proposal submissions, and reports.

Symbol	Line	Purpose / source docstring
ProposalStatusUpdate	35	Implementation entry; inspect this symbol for details
_error	42	Implementation entry; inspect this symbol for details
_student_team	46	Resolve the existing shared project group for its login or an active member.
_same_group	62	Implementation entry; inspect this symbol for details
_owner_scope	66	Implementation entry; inspect this symbol for details
_safe_filename	70	Implementation entry; inspect this symbol for details
_storage_path	75	Implementation entry; inspect this symbol for details
_template_rules	81	Implementation entry; inspect this symbol for details
_template_out	86	Implementation entry; inspect this symbol for details
_active_template	90	Implementation entry; inspect this symbol for details
_allowed_upload	98	Implementation entry; inspect this symbol for details
_allowed_template_upload	103	Implementation entry; inspect this symbol for details
_detected_headings	111	Implementation entry; inspect this symbol for details
_submission_out	135	Implementation entry; inspect this symbol for details
_version_history	141	Implementation entry; inspect this symbol for details
_can_view	164	Implementation entry; inspect this symbol for details
upload_template	185	Implementation entry; inspect this symbol for details
list_templates	226	Implementation entry; inspect this symbol for details
activate_template	232	Implementation entry; inspect this symbol for details
deactivate_template	250	Stop using a baseline without deleting its immutable version history.
delete_template	265	Delete an unused uploaded template version and its extracted rule snapshot.
save_template_rules	284	Save an admin-confirmed rule review as a new immutable rule snapshot.
submit_proposal	320	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
_run_submission	380	Implementation entry; inspect this symbol for details
submit_draft	402	Implementation entry; inspect this symbol for details
withdraw_submission	416	Withdraw a student-owned draft/submission without deleting its history.
update_submission_status	432	Allow an authorized reviewer to record a lifecycle decision.
my_submissions	451	Implementation entry; inspect this symbol for details
get_submission	459	Implementation entry; inspect this symbol for details
report_queue	468	Implementation entry; inspect this symbol for details
get_report	474	Implementation entry; inspect this symbol for details
rerun_report	485	Explicitly re-run a report against the current active baseline.
get_original	498	Implementation entry; inspect this symbol for details
get_annotated_document	509	Return a safe annotated copy; the student's original file is never modified.
download_report_pdf	528	Implementation entry; inspect this symbol for details
_render_pdf	539	Implementation entry; inspect this symbol for details
_render_annotated_document	556	Annotate each affected page with red issue callouts and append full evidence.
display_value	601	Implementation entry; inspect this symbol for details

backend/app/modules/module_03_proposal/template_extractor.py

Lines: 73. Deterministic extraction of measurable template rules.

Symbol	Line	Purpose / source docstring
_sections	15	Implementation entry; inspect this symbol for details
extract_template_rules	41	Return only rules exposed by the uploaded template representation.

backend/app/notifications.py

Lines: 31. Authenticated notifications kept outside the retired team-collaboration module.

Symbol	Line	Purpose / source docstring
notifications	15	Implementation entry; inspect this symbol for details
mark_notification_read	24	Implementation entry; inspect this symbol for details

backend/app/password_reset.py

Lines: 213. Secure, enumeration-resistant password recovery endpoints.

Symbol	Line	Purpose / source docstring
_smtp_from	36	Implementation entry; inspect this symbol for details
_smtp_configured	40	Implementation entry; inspect this symbol for details
_development_mode	46	Implementation entry; inspect this symbol for details
_development_or_demo_mode	50	Implementation entry; inspect this symbol for details
_is_valid_email	54	Implementation entry; inspect this symbol for details
_rate_limit_allows	62	Implementation entry; inspect this symbol for details
_send_reset_email	75	Implementation entry; inspect this symbol for details
_audit	99	Implementation entry; inspect this symbol for details
_generic_response	112	Implementation entry; inspect this symbol for details
forgot_password	117	Create a reset token without revealing whether an account exists.
password_reset_status	167	Expose only the non-sensitive delivery mode needed by the local recovery UI.
development_password_reset_link	176	Return a local reset link only when the backend is explicitly in development mode.
reset_password	192	Consume a valid reset token and revoke every existing browser session.

backend/app/pdf_extraction.py

Lines: 347. Deterministic, local extraction for supervisor-request PDFs and DOCX files.

Symbol	Line	Purpose / source docstring
_docx_formatting	21	Read measurable formatting from the first DOCX section and its runs.
_pdf_formatting	83	Read measurable page, margin, and typography metadata from a PDF.
PdfExtractionError	168	Implementation entry; inspect this symbol for details
extract_proposal_document	175	Extract proposal fields from PDF, DOCX, or UTF-8 TXT documents.
ExtractionResult	229	Implementation entry; inspect this symbol for details
_field	255	Implementation entry; inspect this symbol for details
_normalize_text	261	Normalize pypdf output while retaining line boundaries used by section parsing.
_heading	272	Implementation entry; inspect this symbol for details
_extract_fields	284	Implementation entry; inspect this symbol for details
extract_one_page_pdf	316	Implementation entry; inspect this symbol for details

backend/app/project_workflow.py

Lines: 328. Authorized, persisted project progress, defense scheduling, and evaluation workflows.

Symbol	Line	Purpose / source docstring
failure	34	Implementation entry; inspect this symbol for details
_utc	38	Implementation entry; inspect this symbol for details
_team	42	Implementation entry; inspect this symbol for details
_is_team_member	49	Implementation entry; inspect this symbol for details
_assigned_supervisor	66	Implementation entry; inspect this symbol for details
_can_access_team	70	Implementation entry; inspect this symbol for details
_require_team_access	79	Implementation entry; inspect this symbol for details
_audit	86	Implementation entry; inspect this symbol for details
_milestone_out	92	Implementation entry; inspect this symbol for details
_task_out	96	Implementation entry; inspect this symbol for details
_health	100	Implementation entry; inspect this symbol for details
project_progress	110	Implementation entry; inspect this symbol for details
create_milestone	119	Implementation entry; inspect this symbol for details
update_milestone	128	Implementation entry; inspect this symbol for details
create_task	139	Implementation entry; inspect this symbol for details
update_task	152	Implementation entry; inspect this symbol for details
list_meetings	165	Implementation entry; inspect this symbol for details
create_meeting	172	Implementation entry; inspect this symbol for details
_schedule_out	183	Implementation entry; inspect this symbol for details
_schedule_access	190	Implementation entry; inspect this symbol for details
_check_schedule_conflicts	198	Implementation entry; inspect this symbol for details
_validate_panel_members	221	Implementation entry; inspect this symbol for details
list_schedules	228	Implementation entry; inspect this symbol for details
schedule_options	236	Implementation entry; inspect this symbol for details
schedule_detail	244	Implementation entry; inspect this symbol for details
create_schedule	252	Implementation entry; inspect this symbol for details
reschedule	266	Implementation entry; inspect this symbol for details
_evaluation_out	280	Implementation entry; inspect this symbol for details
_evaluation_access	285	Implementation entry; inspect this symbol for details
list_evaluations	291	Implementation entry; inspect this symbol for details
save_evaluation	299	Implementation entry; inspect this symbol for details
resolve_evaluation	321	Implementation entry; inspect this symbol for details

backend/app/proposal_reference.py

Lines: 90. Authorized, advisory search over archived proposal references.

Symbol	Line	Purpose / source docstring
normalize_title	20	Normalize case, punctuation, and repeated whitespace for advisory matching.
_similarity	25	Implementation entry; inspect this symbol for details
_result	33	Implementation entry; inspect this symbol for details
search_references	50	Implementation entry; inspect this symbol for details
reference_document	78	Implementation entry; inspect this symbol for details

backend/app/registration_policy.py

Lines: 25. Centralized, reversible policy for public registration behavior.

Symbol	Line	Purpose / source docstring
RegistrationMode	7	Implementation entry; inspect this symbol for details
get_registration_mode	16	Return the configured mode, failing closed for missing/invalid values.
registration_mode	22	Read the current process configuration without exposing it publicly.

backend/app/schemas/__init__.py

Lines: 1. API request and response schemas.

backend/app/schemas/academic.py

Lines: 55.

Symbol	Line	Purpose / source docstring
AcademicSession	6	Implementation entry; inspect this symbol for details
CycleOut	10	Implementation entry; inspect this symbol for details
PolicyOut	13	Implementation entry; inspect this symbol for details
CycleCreate	16	Implementation entry; inspect this symbol for details
CycleUpdate	28	Implementation entry; inspect this symbol for details
PolicyUpdate	43	Implementation entry; inspect this symbol for details

API contract: CycleOut

Field	Type	Default / validation declaration
id	int	Required
code	str	Required
display_name	str	Required
academic_year	int / None	None
session	AcademicSession / None	None
registration_open	bool	Required
created_at	datetime	Required
updated_at	datetime	Required

API contract: PolicyOut

Field	Type	Default / validation declaration
id	int	Required
academic_cycle_id	int	Required
max_group_size	int	Required
min_group_size	int	1

Field	Type	Default / validation declaration
formation_deadline	datetime / None	None
matching_weights	dict	{}
created_at	datetime	Required
updated_at	datetime	Required

API contract: CycleCreate

Field	Type	Default / validation declaration
academic_year	int	Required
session	AcademicSession	Required

API contract: CycleUpdate

Field	Type	Default / validation declaration
code	str / None	None
display_name	str / None	None
registration_open	bool / None	None

API contract: PolicyUpdate

Field	Type	Default / validation declaration
max_group_size	int	Required
min_group_size	int	1
formation_deadline	datetime / None	None
matching_weights	dict	{}

backend/app/schemas/academic_deadlines.py

Lines: 100. Strict deadline configuration and timezone-preview schemas.

Symbol	Line	Purpose / source docstring
DeadlinePreviewRequest	8	Implementation entry; inspect this symbol for details
DeadlineWrite	34	Implementation entry; inspect this symbol for details
DeadlineCreate	55	Implementation entry; inspect this symbol for details
DeadlinePreviewOut	67	Implementation entry; inspect this symbol for details
DeadlineVersionOut	74	Implementation entry; inspect this symbol for details
DeadlineDetailOut	82	Implementation entry; inspect this symbol for details
DeadlineListOut	90	Implementation entry; inspect this symbol for details

API contract: DeadlinePreviewRequest

Field	Type	Default / validation declaration
local_due_at	str	Field(min_length=16, max_length=19)
time_zone	str	Field(min_length=1, max_length=100)
utc_offset	str / None	Field(default=None, pattern='^[+-](?:0\\d/1\\d/2[0-3]):[0-5]\\d$')

API contract: DeadlineWrite

Field	Type	Default / validation declaration
name	str	Field(min_length=1, max_length=160)
description	str / None	Field(default=None, max_length=4000)
due_at_utc	datetime	Required

API contract: DeadlineCreate

Field	Type	Default / validation declaration
academic_cycle_id	int	Field(gt=0)

API contract: DeadlinePreviewOut

Field	Type	Default / validation declaration
local_due_at	str	Required
time_zone	str	Required
utc_offset	str	Required
due_at_utc	datetime	Required

API contract: DeadlineVersionOut

Field	Type	Default / validation declaration
id	int	Required
version_number	int	Required
name	str	Required
description	str / None	Required
created_at	datetime	Required

API contract: DeadlineDetailOut

Field	Type	Default / validation declaration
id	int	Required
academic_cycle_id	int	Required
academic_cycle_name	str	Required
created_at	datetime	Required
versions	list[DeadlineVersionOut]	Required

API contract: DeadlineListOut

Field	Type	Default / validation declaration
id	int	Required
academic_cycle_id	int	Required
academic_cycle_name	str	Required
name	str	Required
description	str / None	Required
latest_version_number	int	Required
due_at_utc	datetime	Required
local_due_at	str	Required
time_zone	str	Required
utc_offset	str	Required

backend/app/schemas/admin_users.py

Lines: 83. Strict, safe contracts for ADMIN-only account management.

Symbol	Line	Purpose / source docstring
AdminUserResponse	12	Implementation entry; inspect this symbol for details
AdminUserPage	24	Implementation entry; inspect this symbol for details
AdminUserCreate	31	Implementation entry; inspect this symbol for details
AdminUserRoleChange	76	Implementation entry; inspect this symbol for details
AdminUserStatusChange	81	Implementation entry; inspect this symbol for details

API contract: AdminUserResponse

Field	Type	Default / validation declaration
id	int	Required
full_name	str	Required
email	str	Required
registration_number	str / None	Required
role	UserRole	Required
is_active	bool	Required
created_at	datetime	Required

API contract: AdminUserPage

Field	Type	Default / validation declaration
items	list[AdminUserResponse]	Required
page	int	Required
page_size	int	Required
total	int	Required

API contract: AdminUserCreate

Field	Type	Default / validation declaration
full_name	str	Required
email	EmailStr	Required
password	str	Required
role	UserRole	Required
registration_number	str / None	None

API contract: AdminUserRoleChange

Field	Type	Default / validation declaration
role	UserRole	Required

API contract: AdminUserStatusChange

Field	Type	Default / validation declaration
is_active	bool	Required

backend/app/schemas/auth.py

Lines: 100. Schemas for public authentication-related endpoints.

Symbol	Line	Purpose / source docstring
RegistrationRequest	12	Strict public registration input; privileged fields are forbidden.
RegistrationResponse	42	Safe fields returned after a student account is registered.
LoginRequest	56	Strict credentials accepted by the sign-in endpoint.
CurrentUserResponse	83	Safe account data returned after session authentication.

API contract: RegistrationRequest

Field	Type	Default / validation declaration
full_name	str	Required
email	EmailStr	Required
registration_number	str	Required
password	str	Required

API contract: RegistrationResponse

Field	Type	Default / validation declaration
id	int	Required
full_name	str	Required
email	str	Required
registration_number	str	Required
role	UserRole	Required
is_active	bool	Required
created_at	datetime	Required

API contract: LoginRequest

Field	Type	Default / validation declaration
email	EmailStr / None	None
registration_number	str / None	None
password	str	Required
requested_role	UserRole / None	None

API contract: CurrentUserResponse

Field	Type	Default / validation declaration
id	int	Required
full_name	str	Required
email	str	Required
registration_number	str / None	Required
role	UserRole	Required
is_active	bool	Required
group_id	int / None	None
group_email	str / None	None
principal_type	str	'ACCOUNT'
group_name	str / None	None
academic_cycle	str / None	None
group_status	str / None	None
members	list[dict[str, object]]	Field(default_factory=list)

backend/app/schemas/dashboard_requirements.py

Lines: 43.

Symbol	Line	Purpose / source docstring
DashboardRequirementWrite	9	Implementation entry; inspect this symbol for details
DashboardRequirementOut	37	Implementation entry; inspect this symbol for details

API contract: DashboardRequirementWrite

Field	Type	Default / validation declaration
academic_deadline_id	int / None	Field(default=None, gt=0)
academic_cycle_id	int	Field(gt=0)
title	str	Field(min_length=3, max_length=200)
description	str / None	Field(default=None, max_length=4000)
requirement_type	RequirementType	Required
module	str	Field(min_length=1, max_length=120)
priority	Priority	'MEDIUM'
role_visibility	list[Roles]	Field(min_length=1)

Field	Type	Default / validation declaration
assigned_user_id	int / None	Field(default=None, gt=0)
assigned_group_id	int / None	Field(default=None, gt=0)
due_at	datetime	Required
completion_percent	int	Field(default=0, ge=0, le=100)
is_required	bool	True

API contract: DashboardRequirementOut

Field	Type	Default / validation declaration
id	int	Required
title	str	Required
description	str / None	Required
due_date	datetime	Required
timezone	str	Required
academic_cycle	int	Required
module	str	Required
requirement_type	str	Required
priority	str	Required
status	str	Required
completion_percent	int	Required
is_required	bool	Required
source_deadline_id	int / None	Required
assigned_to_current_user	bool	Required
can_update	bool	Required

backend/app/schemas/document_requirements.py

Lines: 57.

Symbol	Line	Purpose / source docstring
RequirementWrite	5	Implementation entry; inspect this symbol for details
RequirementPatch	39	Implementation entry; inspect this symbol for details
RequirementOut	48	Implementation entry; inspect this symbol for details
VersionOut	52	Implementation entry; inspect this symbol for details
SubmissionOut	56	Implementation entry; inspect this symbol for details

API contract: RequirementWrite

Field	Type	Default / validation declaration
title	str	Field(min_length=1, max_length=200)
due_date	datetime / None	None
assigned_to	str / None	None
description	str / None	None
academic_cycle_id	int / None	Field(default=None, gt=0)
module	str	'General'
visible_to_roles	list[str]	['STUDENT']
assigned_user_id	int / None	Field(default=None, gt=0)
assigned_group_id	int / None	Field(default=None, gt=0)
allowed_file_types	list[str]	['pdf']
max_file_size_mb	int	Field(default=10, ge=1, le=100)
due_at	datetime / None	None
reminder_days_before	int	Field(default=3, ge=0, le=365)
is_required	bool	True
late_upload_allowed	bool	False

API contract: RequirementPatch

Field	Type	Default / validation declaration
title	str / None	Field(default=None, min_length=1, max_length=200)
description	str / None	None
due_at	datetime / None	None
allowed_file_types	list[str] / None	None
max_file_size_mb	int / None	Field(default=None, ge=1, le=100)
reminder_days_before	int / None	Field(default=None, ge=0, le=365)
late_upload_allowed	bool / None	None

API contract: RequirementOut

Field	Type	Default / validation declaration
id	int	Required
title	str	Required
description	str / None	Required
academic_cycle_id	int	Required
module	str	Required
visible_to_roles	list	Required
assigned_user_id	int / None	Required
assigned_group_id	int / None	Required
allowed_file_types	list	Required
max_file_size_mb	int	Required
due_at	datetime	Required
reminder_days_before	int	Required
is_required	bool	Required
status	str	Required
late_upload_allowed	bool	Required
created_at	datetime	Required
updated_at	datetime	Required
submission_count	int	0

API contract: VersionOut

Field	Type	Default / validation declaration
id	int	Required
version_number	int	Required
original_filename	str	Required
mime_type	str	Required
file_size	int	Required
storage_reference	str	Required
uploaded_at	datetime	Required
download_url	str / None	None

API contract: SubmissionOut

Field	Type	Default / validation declaration
id	int	Required
requirement_id	int	Required
status	str	Required
is_late	bool	Required
review_comment	str / None	Required

Field	Type	Default / validation declaration
uploaded_at	datetime	Required
updated_at	datetime	Required
versions	list[VersionOut]	[]

backend/app/schemas/email.py

Lines: 9. Shared normalization for Pydantic email fields.

Symbol	Line	Purpose / source docstring
normalize_email	4	Trim and case-normalize email input before EmailStr validation.

backend/app/schemas/evaluation_rubrics.py

Lines: 88. Strict request and response schemas for draft Evaluation Criteria.

Symbol	Line	Purpose / source docstring
RubricCriterionWrite	8	Implementation entry; inspect this symbol for details
RubricVersionWrite	31	Implementation entry; inspect this symbol for details
RubricCriterionOut	61	Implementation entry; inspect this symbol for details
RubricVersionOut	68	Implementation entry; inspect this symbol for details
RubricDetailOut	77	Implementation entry; inspect this symbol for details
RubricListOut	83	Implementation entry; inspect this symbol for details

API contract: RubricCriterionWrite

Field	Type	Default / validation declaration
name	str	Field(min_length=1, max_length=160)
description	str / None	Field(default=None, max_length=2000)
order	int	Field(ge=1)

API contract: RubricVersionWrite

Field	Type	Default / validation declaration
name	str	Field(min_length=1, max_length=160)
description	str / None	Field(default=None, max_length=4000)
criteria	list[RubricCriterionWrite]	Field(min_length=1, max_length=100)

API contract: RubricCriterionOut

Field	Type	Default / validation declaration
id	int	Required
name	str	Required
description	str / None	Required
order	int	Required

API contract: RubricVersionOut

Field	Type	Default / validation declaration
id	int	Required
version_number	int	Required
name	str	Required
description	str / None	Required
created_at	datetime	Required
criteria	list[RubricCriterionOut]	Required

API contract: RubricDetailOut

Field	Type	Default / validation declaration
id	int	Required
created_at	datetime	Required
versions	list[RubricVersionOut]	Required

API contract: RubricListOut

Field	Type	Default / validation declaration
id	int	Required
name	str	Required
description	str / None	Required
latest_version_number	int	Required
created_at	datetime	Required

backend/app/schemas/evaluations.py

Lines: 160. Request and response contracts for role-based FYP evaluation.

Symbol	Line	Purpose / source docstring
EvaluationCategory	9	Implementation entry; inspect this symbol for details
EvaluationStatus	17	Implementation entry; inspect this symbol for details
EvaluationWrite	34	Implementation entry; inspect this symbol for details
EvaluationOut	64	Implementation entry; inspect this symbol for details
AssignmentMemberOut	85	Implementation entry; inspect this symbol for details
EvaluationAssignmentOut	92	Implementation entry; inspect this symbol for details
EvaluationReturn	99	Implementation entry; inspect this symbol for details
EvaluationCorrectionWrite	112	Implementation entry; inspect this symbol for details
EvaluationAggregationWrite	127	Implementation entry; inspect this symbol for details
EvaluationAggregationOut	133	Implementation entry; inspect this symbol for details
ResultCategoryOut	140	Implementation entry; inspect this symbol for details
ConsolidatedResultOut	149	Implementation entry; inspect this symbol for details

API contract: EvaluationWrite

Field	Type	Default / validation declaration
student_id	int / None	Field(default=None, gt=0)
group_id	int / None	Field(default=None, gt=0)
team_id	int / None	Field(default=None, gt=0)
evaluation_category	EvaluationCategory	Required
obtained_marks	int / None	Field(default=None, ge=0)
feedback	str / None	Field(default=None, max_length=10000)
submit	bool	False

API contract: EvaluationOut

Field	Type	Default / validation declaration
id	int	Required
student_id	int / None	Required
group_id	int / None	Required
evaluator_id	int	Required
evaluator_role	str	Required
evaluation_category	str	Required

Field	Type	Default / validation declaration
maximum_marks	int	Required
obtained_marks	int / None	Required
feedback	str / None	Required
status	str	Required
returned_reason	str / None	Required
submitted_at	datetime / None	Required
approved_at	datetime / None	Required
published_at	datetime / None	Required
created_at	datetime	Required
updated_at	datetime	Required

API contract: AssignmentMemberOut

Field	Type	Default / validation declaration
id	int	Required
full_name	str	Required
registration_number	str	Required
university_email	str	Required

API contract: EvaluationAssignmentOut

Field	Type	Default / validation declaration
group_id	int	Required
academic_cycle_id	int	Required
students	list[AssignmentMemberOut]	Required
categories	list[str]	Required

API contract: EvaluationReturn

Field	Type	Default / validation declaration
reason	str	Field(min_length=3, max_length=5000)

API contract: EvaluationCorrectionWrite

Field	Type	Default / validation declaration
new_marks	int	Field(ge=0)
reason	str	Field(min_length=3, max_length=5000)

API contract: EvaluationAggregationWrite

Field	Type	Default / validation declaration
evaluation_category	EvaluationCategory	Required
method	str	Field(pattern='^(AVERAGE/MAXIMUM/MEDIAN)$')

API contract: EvaluationAggregationOut

Field	Type	Default / validation declaration
id	int	Required
evaluation_category	str	Required
method	str	Required
is_active	bool	Required

API contract: ResultCategoryOut

Field	Type	Default / validation declaration
category	str	Required
maximum_marks	int	Required
obtained_marks	float / None	Required
status	str	Required
evaluator_ids	list[int]	Required
feedback	list[str]	Required

API contract: ConsolidatedResultOut

Field	Type	Default / validation declaration
student_id	int	Required
group_id	int / None	Required
student_name	str	Required
registration_number	str / None	Required
categories	list[ResultCategoryOut]	Required
total_marks	float	Required
maximum_marks	int	Required
percentage	float	Required
grade	str / None	Required
gpa	float / None	Required
published	bool	Required

backend/app/schemas/password_reset.py

Lines: 46. Public schemas for password recovery without exposing account existence.

Symbol	Line	Purpose / source docstring
ForgotPasswordRequest	8	Implementation entry; inspect this symbol for details
ResetPasswordRequest	29	Implementation entry; inspect this symbol for details
PasswordResetMessage	45	Implementation entry; inspect this symbol for details

API contract: ForgotPasswordRequest

Field	Type	Default / validation declaration
identifier	str / None	Field(default=None, min_length=1, max_length=320)
email	str / None	Field(default=None, min_length=1, max_length=320)
registration_number	str / None	Field(default=None, min_length=1, max_length=100)

API contract: ResetPasswordRequest

Field	Type	Default / validation declaration
token	str	Field(min_length=20, max_length=256)
password	str	Field(validation_alias=AliasChoices('password', 'new_password'))

API contract: PasswordResetMessage

Field	Type	Default / validation declaration
message	str	Required

backend/app/schemas/project_workflow.py

Lines: 91. Validated request payloads for project progress, defense, and evaluation APIs.

Symbol	Line	Purpose / source docstring
MilestoneWrite	8	Implementation entry; inspect this symbol for details
TaskWrite	24	Implementation entry; inspect this symbol for details
MeetingWrite	29	Implementation entry; inspect this symbol for details
ScheduleWrite	45	Implementation entry; inspect this symbol for details
ScheduleReschedule	61	Implementation entry; inspect this symbol for details
EvaluationScoreWrite	70	Implementation entry; inspect this symbol for details
EvaluationWrite	78	Implementation entry; inspect this symbol for details
ResolutionWrite	88	Implementation entry; inspect this symbol for details

API contract: MilestoneWrite

Field	Type	Default / validation declaration
title	str	Field(min_length=1, max_length=255)
description	str / None	Field(default=None, max_length=5000)
due_at	datetime / None	None
completion_percent	int	Field(default=0, ge=0, le=100)

API contract: TaskWrite

Field	Type	Default / validation declaration
milestone_id	int / None	Field(default=None, gt=0)
owner_user_id	int / None	Field(default=None, gt=0)

API contract: MeetingWrite

Field	Type	Default / validation declaration
held_at	datetime	Required
minutes	str	Field(min_length=1, max_length=20000)
decisions	str / None	Field(default=None, max_length=10000)
action_items	list[dict[str, str]]	Field(default_factory=list, max_length=100)

API contract: ScheduleWrite

Field	Type	Default / validation declaration
team_id	int	Field(gt=0)
academic_cycle_id	int	Field(gt=0)
start_at	datetime	Required
end_at	datetime	Required
room	str / None	Field(default=None, max_length=255)
online_link	str / None	Field(default=None, max_length=1000)
panel_member_ids	list[int]	Field(default_factory=list, max_length=20)

API contract: ScheduleReschedule

Field	Type	Default / validation declaration
start_at	datetime	Required
end_at	datetime	Required
room	str / None	Field(default=None, max_length=255)
online_link	str / None	Field(default=None, max_length=1000)
panel_member_ids	list[int] / None	Field(default=None, max_length=20)

API contract: EvaluationScoreWrite

Field	Type	Default / validation declaration
criterion_key	str	Field(min_length=1, max_length=160)
score	int	Field(ge=0)
max_score	int	Field(gt=0)
feedback	str / None	Field(default=None, max_length=5000)

API contract: EvaluationWrite

Field	Type	Default / validation declaration
schedule_id	int	Field(gt=0)
scores	list[EvaluationScoreWrite]	Field(min_length=1, max_length=100)
overall_score	int / None	Field(default=None, ge=0, le=100)
overall_feedback	str / None	Field(default=None, max_length=10000)
decision	str / None	Field(default=None, max_length=64)
submit	bool	False

API contract: ResolutionWrite

Field	Type	Default / validation declaration
status	str	Field(pattern='^(RECHECK_REQUESTED/RESOLVED)$')
note	str	Field(min_length=1, max_length=10000)

backend/app/schemas/reminder_rules.py

Lines: 24. Strict configuration-only rule requests.

Symbol	Line	Purpose / source docstring
RulePreview	5	Implementation entry; inspect this symbol for details
RuleWrite	11	Implementation entry; inspect this symbol for details
RuleCreate	23	Implementation entry; inspect this symbol for details

API contract: RulePreview

Field	Type	Default / validation declaration
deadline_version_id	int	Field(gt=0, strict=True)
direction	Literal['BEFORE', 'AFTER']	Required
elapsed_minutes	int	Field(gt=0, le=2147483647, strict=True)

API contract: RuleWrite

Field	Type	Default / validation declaration
name	str	Field(min_length=1, max_length=160)
description	str / None	Field(default=None, max_length=4000)
category	Literal['REMINDER', 'ESCALATION']	Required

API contract: RuleCreate

Field	Type	Default / validation declaration
academic_cycle_id	int	Field(gt=0, strict=True)

backend/app/schemas/supervisor_capacity.py

Lines: 42.

Symbol	Line	Purpose / source docstring
CapacitySession	6	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
SupervisorCapacityWrite	11	Implementation entry; inspect this symbol for details
SupervisorCapacityOut	18	Implementation entry; inspect this symbol for details

API contract: SupervisorCapacityWrite

Field	Type	Default / validation declaration
supervisor_id	StrictInt	Required
academic_year	StrictInt	Field(ge=1900, le=2200)
maximum_groups	StrictInt	Field(gt=0)

API contract: SupervisorCapacityOut

Field	Type	Default / validation declaration
id	int	Required
supervisor_id	int	Required
supervisor_name	str	Required
supervisor_email	str	Required
supervisor_is_eligible	bool	Required
academic_year	int / None	Required
maximum_groups	int	Required
fall_assigned	int	Required
spring_assigned	int	Required
total_assigned	int	Required
remaining_capacity	int	Required
status	str	Required
academic_cycle_id	int / None	None
academic_cycle_name	str / None	None
max_group_capacity	int / None	None
assigned_groups	int / None	None
remaining_groups	int / None	None
session	CapacitySession / None	None
assigned	int / None	None
remaining	int / None	None
full	bool / None	None
created_at	datetime	Required
updated_at	datetime	Required

backend/app/schemas/team_registration.py

Lines: 142. Strict request and safe response schemas for shared-team registration.

Symbol	Line	Purpose / source docstring
TeamMemberInput	12	One group leader or colleague record; contact email is legacy-only.
TeamRegistrationRequest	29	Public input for one shared student login and its team contacts.
SafeAccountOut	47	Implementation entry; inspect this symbol for details
SafeTeamOut	59	Implementation entry; inspect this symbol for details
SafeCycleOut	67	Implementation entry; inspect this symbol for details
SafeTeamMemberOut	75	Implementation entry; inspect this symbol for details
TeamRegistrationResponse	87	Implementation entry; inspect this symbol for details
StudentGroupMemberInput	94	Implementation entry; inspect this symbol for details
StudentGroupRegistrationRequest	120	Implementation entry; inspect this symbol for details

API contract: TeamMemberInput

Field	Type	Default / validation declaration
full_name	str	Required
registration_number	str	Required
university_email	EmailStr / None	None

API contract: TeamRegistrationRequest

Field	Type	Default / validation declaration
academic_cycle_id	int	Required
creator	TeamMemberInput	Required
shared_login_email	EmailStr	Required
password	str	Required
confirm_password	str / None	None
members	list[TeamMemberInput]	Required

API contract: SafeAccountOut

Field	Type	Default / validation declaration
id	int	Required
full_name	str	Required
email	str	Required
registration_number	str	Required
role	UserRole	Required
is_active	bool	Required
created_at	datetime	Required

API contract: SafeTeamOut

Field	Type	Default / validation declaration
id	int	Required
academic_cycle_id	int	Required
created_at	datetime	Required

API contract: SafeCycleOut

Field	Type	Default / validation declaration
id	int	Required
code	str	Required
display_name	str	Required

API contract: SafeTeamMemberOut

Field	Type	Default / validation declaration
id	int	Required
full_name	str	Required
registration_number	str	Required
university_email	str / None	Required
is_creator	bool	Required
is_active	bool	Required
created_at	datetime	Required

API contract: TeamRegistrationResponse

Field	Type	Default / validation declaration
account	SafeAccountOut	Required
team	SafeTeamOut	Required
cycle	SafeCycleOut	Required
members	list[SafeTeamMemberOut]	Required

API contract: StudentGroupMemberInput

Field	Type	Default / validation declaration
email	EmailStr	Required
registration_number	str	Required
full_name	str	Required
designation	str / None	None

API contract: StudentGroupRegistrationRequest

Field	Type	Default / validation declaration
academic_cycle_id	int	Required
group_name	str / None	None
group_email	EmailStr	Required
password	str	Required
confirm_password	str	Required
members	list[StudentGroupMemberInput]	Required

backend/app/schemas/templates.py

Lines: 22.

Symbol	Line	Purpose / source docstring
RuleFields	4	Implementation entry; inspect this symbol for details
TemplateCreate	7	Implementation entry; inspect this symbol for details
TemplateVersionCreate	9	Implementation entry; inspect this symbol for details
RuleCreate	11	Implementation entry; inspect this symbol for details
TemplateOut	13	Implementation entry; inspect this symbol for details
TemplateListOut	15	Implementation entry; inspect this symbol for details
RuleOut	17	Implementation entry; inspect this symbol for details
TemplateHistoryVersion	19	Implementation entry; inspect this symbol for details
TemplateDetail	21	Implementation entry; inspect this symbol for details

API contract: RuleFields

Field	Type	Default / validation declaration
page_size	str / None	None
font_family	str / None	None
font_size_pt	float / None	Field(default=None, gt=0)
margin_mm	float / None	Field(default=None, gt=0)
line_spacing	float / None	Field(default=None, gt=0)
heading_style	str / None	None
required_sections	list[str] / None	None

API contract: TemplateCreate

Field	Type	Default / validation declaration
template_type	TemplateType	Required
title	str	Field(min_length=1, max_length=160)

Field	Type	Default / validation declaration
provenance	str	Field(pattern='^(OBSERVED_REFERENCE/CONFIRMED_REQUIREMENT)$')
source_note	str	Field(min_length=1, max_length=500)
rules	RuleFields	Field(default_factory=RuleFields)

API contract: TemplateVersionCreate

Field	Type	Default / validation declaration
title	str	Field(min_length=1, max_length=160)
provenance	str	Field(pattern='^(OBSERVED_REFERENCE/CONFIRMED_REQUIREMENT)$')
source_note	str	Field(min_length=1, max_length=500)

API contract: RuleCreate

Field	Type	Default / validation declaration
provenance	str	Field(pattern='^(OBSERVED_REFERENCE/CONFIRMED_REQUIREMENT)$')
rules	RuleFields	Field(default_factory=RuleFields)

API contract: TemplateOut

Field	Type	Default / validation declaration
id	int	Required
template_type	TemplateType	Required
version_number	int	Required
title	str	Required
provenance	str	Required
source_note	str	Required
created_at	datetime	Required
is_active	bool	False
activated_at	datetime / None	None
activated_by_id	int / None	None

API contract: TemplateListOut

Field	Type	Default / validation declaration
reference_id	int	Required

API contract: RuleOut

Field	Type	Default / validation declaration
id	int	Required
template_reference_version_id	int	Required
version_number	int	Required
provenance	str	Required
rules	dict	Required
created_at	datetime	Required

API contract: TemplateHistoryVersion

Field	Type	Default / validation declaration
template	TemplateOut	Required
rules	list[RuleOut]	Required

API contract: TemplateDetail

Field	Type	Default / validation declaration
reference_id	int	Required
template_type	TemplateType	Required
versions	list[TemplateHistoryVersion]	Required

backend/app/security.py

Lines: 74. Password and opaque-session security utilities.

Symbol	Line	Purpose / source docstring
normalize_full_name	15	Implementation entry; inspect this symbol for details
validate_full_name	19	Implementation entry; inspect this symbol for details
password_policy_errors	29	Return safe, user-facing password requirements that are not met.
validate_password	47	Validate a newly supplied account password before it is hashed.
hash_password	54	Return an Argon2id hash without modifying the supplied password.
verify_password	59	Verify an unmodified password against an Argon2id hash.
generate_session_token	67	Return an opaque, URL-safe session bearer value; never persist this value.
hash_session_token	72	Return the deterministic database representation of an opaque session token.

backend/app/supervisor_assignments.py

Lines: 89.

Symbol	Line	Purpose / source docstring
AssignmentWrite	13	Implementation entry; inspect this symbol for details
failure	20	Implementation entry; inspect this symbol for details
out	22	Implementation entry; inspect this symbol for details
teams	25	Implementation entry; inspect this symbol for details
listing	29	Implementation entry; inspect this symbol for details
assign	32	Implementation entry; inspect this symbol for details
end	73	Implementation entry; inspect this symbol for details

backend/app/supervisor_request_conversations.py

Lines: 78. Request-scoped conversation endpoints with server-side authorization.

Symbol	Line	Purpose / source docstring
RequestMessageWrite	15	Implementation entry; inspect this symbol for details
_request_for_actor	28	Implementation entry; inspect this symbol for details
_message_out	46	Implementation entry; inspect this symbol for details
conversation_history	52	Implementation entry; inspect this symbol for details
send_conversation_message	59	Implementation entry; inspect this symbol for details

backend/app/supervisor_requests.py

Lines: 1082. Cycle-scoped supervisor selection and proposal-request workflow.

Symbol	Line	Purpose / source docstring
RevisionAction	57	Implementation entry; inspect this symbol for details
DecisionAction	63	Implementation entry; inspect this symbol for details
ProposalAIReviewFinding	79	Implementation entry; inspect this symbol for details
ProposalAIReviewResponse	91	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
_normalize_submission_note	123	Implementation entry; inspect this symbol for details
_reject_text_for_supervisor_selection	128	Keep the legacy supervisor-selection upload contract PDF/DOCX-only.
failure	140	Implementation entry; inspect this symbol for details
_ai_fallback	144	Implementation entry; inspect this symbol for details
_review_prompt	162	Implementation entry; inspect this symbol for details
_ai_out	180	Implementation entry; inspect this symbol for details
_team_for_student	188	Resolve the existing shared group for either its login or an active member.
_actor_has_team_access	202	Implementation entry; inspect this symbol for details
_cycle_for_request	215	Implementation entry; inspect this symbol for details
_lock_capacity	227	Implementation entry; inspect this symbol for details
_capacity_used	247	Implementation entry; inspect this symbol for details
_ensure_capacity	267	Implementation entry; inspect this symbol for details
_owner_request	272	Implementation entry; inspect this symbol for details
_append_history	287	Implementation entry; inspect this symbol for details
_audit	299	Implementation entry; inspect this symbol for details
_history_out	313	Implementation entry; inspect this symbol for details
_document_out	333	Implementation entry; inspect this symbol for details
_missing_required	357	Implementation entry; inspect this symbol for details
_apply_document_snapshot	372	Implementation entry; inspect this symbol for details
_next_document_version	382	Implementation entry; inspect this symbol for details
_next_note_version	387	Implementation entry; inspect this symbol for details
_note_versions_out	392	Implementation entry; inspect this symbol for details
_audit_document	401	Implementation entry; inspect this symbol for details
_out	413	Implementation entry; inspect this symbol for details
_load_for_view	469	Implementation entry; inspect this symbol for details
active_cycles	478	Implementation entry; inspect this symbol for details
eligible_supervisors	487	Implementation entry; inspect this symbol for details
my_requests	523	Implementation entry; inspect this symbol for details
create_request	542	Implementation entry; inspect this symbol for details
update_draft_document	633	Implementation entry; inspect this symbol for details
confirm_extraction	698	Implementation entry; inspect this symbol for details
submit_request	736	Implementation entry; inspect this symbol for details
withdraw_request	801	Implementation entry; inspect this symbol for details
supervisor_queue	832	Implementation entry; inspect this symbol for details
decide_request	850	Implementation entry; inspect this symbol for details
proposal_document	922	Implementation entry; inspect this symbol for details
_monitor_statement	944	Implementation entry; inspect this symbol for details
admin_monitor	973	Implementation entry; inspect this symbol for details
coordinator_monitor	988	Implementation entry; inspect this symbol for details
read_request	1003	Implementation entry; inspect this symbol for details
run_proposal_ai_review	1021	Implementation entry; inspect this symbol for details
latest_proposal_ai_review	1073	Implementation entry; inspect this symbol for details

backend/app/supervisor_workspace.py

Lines: 155. Supervisor-owned group workspace assembled from persisted workflow records.

Symbol	Line	Purpose / source docstring
owned_request	20	Implementation entry; inspect this symbol for details
owned_group	27	Implementation entry; inspect this symbol for details
workspace	35	Implementation entry; inspect this symbol for details
assigned_groups	89	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
group_detail	95	Implementation entry; inspect this symbol for details
group_section	105	Implementation entry; inspect this symbol for details
requests	110	Implementation entry; inspect this symbol for details
request_detail	116	Implementation entry; inspect this symbol for details
DecisionInput	125	Implementation entry; inspect this symbol for details
decision	133	Implementation entry; inspect this symbol for details
group_channel	138	Implementation entry; inspect this symbol for details
conversation	147	Implementation entry; inspect this symbol for details
send_message	153	Implementation entry; inspect this symbol for details

backend/app/team_compatibility.py

Lines: 77. Equal-weight aggregation for team compatibility assessments.

Symbol	Line	Purpose / source docstring
aggregate_compatibility	18	Aggregate comparable categories, leaving missing assessments unscored.
_tokens	30	Implementation entry; inspect this symbol for details
_overlap	38	Implementation entry; inspect this symbol for details
_similar_scale	45	Implementation entry; inspect this symbol for details
_complementary	53	Implementation entry; inspect this symbol for details
hybrid_compatibility	62	Reproducible, explainable score; Ollama is never involved in this function.

backend/app/team_lifecycle.py

Lines: 31. Deterministic Module 2 lifecycle helpers shared by modules 2, 3 and 5.

Symbol	Line	Purpose / source docstring
refresh_team_status	8	Implementation entry; inspect this symbol for details
lock_team	17	Implementation entry; inspect this symbol for details
cancel_open_team_requests	28	Implementation entry; inspect this symbol for details

frontend/src/AcademicCycleDashboard.css

Lines: 1.

Frontend entry import graph: not reached by static imports.

frontend/src/AcademicCycleDashboard.tsx

Lines: 85.

Frontend entry import graph: not reached by static imports.

Imports: react, ./AdminCycleManager, ./AcademicCycleDashboard.css.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/academic-cycles/${detailId}/overview.

Symbol	Line	Purpose / source docstring
AcademicCycleDashboard	15	Implementation entry; inspect this symbol for details

frontend/src/AdminCycleManager.css

Lines: 57.

Frontend entry import graph: reachable.

frontend/src/AdminCycleManager.tsx

Lines: 41.

Frontend entry import graph: reachable.

Imports: react, ./AdminCycleManager.css.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/academic-cycles/${cycle.id}/group-policy, /api/v1/admin/academic-cycles, /api/v1/admin/academic-cycles/${row.id}, /api/v1/admin/academic-cycles/${row.id}/group-policy.

Symbol	Line	Purpose / source docstring
AdminCycleManager	7	Implementation entry; inspect this symbol for details

frontend/src/AdminDeadlines.tsx

Lines: 135.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/admin/academic-deadlines, /api/v1/admin/academic-deadlines${query}, /api/v1/admin/academic-deadlines/${deadlineId}, /api/v1/admin/academic-deadlines/${selected.id}/versions, /api/v1/admin/academic-deadlines/preview, /api/v1/admin/time-zones.

Symbol	Line	Purpose / source docstring
AdminDeadlines	16	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManager.tsx

Lines: 3.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements, /api/v1/document-requirements/${id}/${name}.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManager	3	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerAllFormats.tsx

Lines: 7.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerAllFormats	7	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerClean.tsx

Lines: 35.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/document-requirements, /api/v1/document-requirements/${, /api/v1/document-requirements/${item.id}${name, /api/v1/document-requirements/${item.id}/submissions.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerClean	25	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerComplete.tsx

Lines: 5.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements, /api/v1/document-requirements/${id}/${name}.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerComplete	5	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerCreate.tsx

Lines: 16.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements, /api/v1/document-requirements/${id}/${name}.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerCreate	9	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerFinal.tsx

Lines: 6.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerFinal	6	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerFormats.tsx

Lines: 6.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerFormats	6	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerSimple.tsx

Lines: 5.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements, /api/v1/document-requirements/${id}/${name}.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerSimple	5	Implementation entry; inspect this symbol for details

frontend/src/AdminDocumentRequirementManagerUI.tsx

Lines: 8.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements, /api/v1/document-requirements/${id}/${name}.

Symbol	Line	Purpose / source docstring
AdminDocumentRequirementManagerUI	8	Implementation entry; inspect this symbol for details

frontend/src/AdminOverview.css

Lines: 142.

Frontend entry import graph: reachable.

frontend/src/AdminOverview.tsx

Lines: 21.

Frontend entry import graph: reachable.

Imports: react, ./AdminOverview.css.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/academic-cycles/${cycle.id}/group-policy.

Symbol	Line	Purpose / source docstring
AdminOverview	8	Implementation entry; inspect this symbol for details

frontend/src/AdminPortal.tsx

Lines: 38.

Frontend entry import graph: reachable.

Imports: react, ./AdminCycleManager, ./AdminOverview, ./AdminUsersRoles, ./AdminStudentGroups, ./AdminSupervisorCapacity, ./AdminReminderRules, ./AdminDeadlines, ./AdminDocumentRequirementManagerClean, ./modules/module-3/AdminProposalTemplates, ./portals/admin/modules/archive/ArchiveManagement, ./portals/shared/AdminSidebar, ./AdminSidebar.css.

API references found in source (dynamic expressions may be partial): /api/v1/admin/registration-mode.

Symbol	Line	Purpose / source docstring
AdminPortal	27	Implementation entry; inspect this symbol for details

frontend/src/AdminReminderRules.tsx

Lines: 37.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/admin/academic-deadlines/${row.id}, /api/v1/admin/academic-deadlines?academic_cycle_id, /api/v1/admin/reminder-rules${cycle, /api/v1/admin/reminder-rules${selected, /api/v1/admin/reminder-rules/${id}, /api/v1/admin/reminder-rules/preview.

Symbol	Line	Purpose / source docstring
AdminReminderRules	16	Implementation entry; inspect this symbol for details

frontend/src/AdminSidebar.css

Lines: 119.

Frontend entry import graph: reachable.

frontend/src/AdminStudentGroups.tsx

Lines: 21.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/admin/student-groups, /api/v1/admin/student-groups/${group.id}${action.

Symbol	Line	Purpose / source docstring
AdminStudentGroups	6	Implementation entry; inspect this symbol for details

frontend/src/AdminSupervisorCapacity.tsx

Lines: 21.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/admin/supervisor-capacities, /api/v1/admin/supervisor-capacities${isEdit, /api/v1/admin/supervisor-capacities/${policy.id}, /api/v1/admin/users?role.

Symbol	Line	Purpose / source docstring
AdminSupervisorCapacity	11	Implementation entry; inspect this symbol for details

frontend/src/AdminUsersRoles.tsx

Lines: 489.

Frontend entry import graph: reachable.

Imports: react, ./PasswordField, ./passwordPolicy, ./emailValidation.

API references found in source (dynamic expressions may be partial): /api/v1/admin/users, /api/v1/admin/users/${confirm.user.id}, /api/v1/admin/users/${confirm.user.id}/${confirm.type}, /api/v1/admin/users?${query}.

Symbol	Line	Purpose / source docstring
AdminUsersRoles	36	Implementation entry; inspect this symbol for details

frontend/src/App.tsx

Lines: 1324.

Frontend entry import graph: reachable.

Imports: react, ./AdminPortal, ./PasswordField, ./passwordPolicy, ./emailValidation, ./portals/student/StudentPortal, ./portals/supervisor/SupervisorPortal, ./portals/coordinator/CoordinatorPortal, ./portals/student/modules/StudentEvaluatorPortal, ./assets/sign-in-students.webp.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/academic-cycles/${cycleId}/group-policy, /api/v1/auth/dev/password-reset-link?identifier, /api/v1/auth/forgot-password, /api/v1/auth/group-signup, /api/v1/auth/login, /api/v1/auth/logout, /api/v1/auth/me, /api/v1/auth/password-reset-status, /api/v1/auth/reset-password.

Symbol	Line	Purpose / source docstring
API	73	Implementation entry; inspect this symbol for details
dashboardPageForRole	91	Implementation entry; inspect this symbol for details
App	263	Implementation entry; inspect this symbol for details

frontend/src/auth.css

Lines: 57.

Frontend entry import graph: not reached by static imports.

frontend/src/components/DashboardDeadlinesRequirements.tsx

Lines: 7.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/dashboard/requirements.

Symbol	Line	Purpose / source docstring
DashboardDeadlinesRequirements	3	Implementation entry; inspect this symbol for details

frontend/src/dashboard-theme.css

Lines: 204.

Frontend entry import graph: not reached by static imports.

frontend/src/design-system.css

Lines: 205.

Frontend entry import graph: reachable.

frontend/src/emailValidation.ts

Lines: 40.

Frontend entry import graph: reachable.

Symbol	Line	Purpose / source docstring
EMAIL_DOMAIN_MESSAGE	1	Implementation entry; inspect this symbol for details
EMAIL_FORMAT_MESSAGE	3	Implementation entry; inspect this symbol for details
normalizeEmail	8	Implementation entry; inspect this symbol for details
emailValidationMessage	13	Implementation entry; inspect this symbol for details
isValidEmail	38	Implementation entry; inspect this symbol for details

frontend/src/EvaluationMarksWorkspace.tsx

Lines: 177.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/admin/evaluations, /api/v1/admin/evaluations/aggregation, /api/v1/admin/results/group/${groupId}, /api/v1/evaluations, /api/v1/evaluations/${evaluation.id}, /api/v1/evaluations/${evaluation.id}/report.pdf, /api/v1/evaluations/${row.id}/${action}, /api/v1/evaluations/${row.id}/report.pdf, /api/v1/evaluations/assignments, /api/v1/evaluations/my, /api/v1/student/results.

Symbol	Line	Purpose / source docstring
EvaluationMarksWorkspace	32	Implementation entry; inspect this symbol for details

frontend/src/index.css

Lines: 355.

Frontend entry import graph: reachable.

frontend/src/inputValidation.ts

Lines: 21.

Frontend entry import graph: not reached by static imports.

Symbol	Line	Purpose / source docstring
normalizeFullName	3	Implementation entry; inspect this symbol for details
fullNameValidationMessage	7	Implementation entry; inspect this symbol for details
academicTextValidationMessage	15	Implementation entry; inspect this symbol for details

frontend/src/main.tsx

Lines: 6.

Frontend entry import graph: reachable.

Imports: react, react-dom/client, ./App, ./index.css, ./design-system.css.

frontend/src/modules/module-3/AdminProposalTemplates.tsx

Lines: 101.

Frontend entry import graph: reachable.

Imports: react, ./module-3.css.

API references found in source (dynamic expressions may be partial): /api/v1/proposal-compliance/templates, /api/v1/proposal-compliance/templates/${selected.id}/rules, /api/v1/proposal-compliance/templates/${template.id}/activate, /api/v1/proposal-compliance/templates/${template.id}/deactivate, /api/v1/proposal-compliance/templates/upload.

Symbol	Line	Purpose / source docstring
AdminProposalTemplates	9	Implementation entry; inspect this symbol for details

frontend/src/modules/module-3/module-3.css

Lines: 10.

Frontend entry import graph: reachable.

frontend/src/modules/module-3/ProposalComplianceWorkspace.tsx

Lines: 140.

Frontend entry import graph: reachable.

Imports: react, ./module-3.css.

API references found in source (dynamic expressions may be partial): /api/v1/proposal-compliance/${reviewerMode, /api/v1/proposal-compliance/reports/${id}, /api/v1/proposal-compliance/reports/${selectedId}/annotated, /api/v1/proposal-compliance/reports/${selectedId}/original, /api/v1/proposal-compliance/reports/${selectedId}/pdf, /api/v1/proposal-compliance/reports/${selectedId}/run, /api/v1/proposal-compliance/submissions, /api/v1/proposal-compliance/submissions/${id}/submit, /api/v1/proposal-compliance/submissions/${id}/withdraw, /api/v1/proposal-compliance/submissions/${selectedId}/status.

Symbol	Line	Purpose / source docstring
ProposalComplianceWorkspace	18	Implementation entry; inspect this symbol for details

frontend/src/observability.ts

Lines: 21.

Frontend entry import graph: not reached by static imports.

Symbol	Line	Purpose / source docstring
classifyHttpStatus	7	Implementation entry; inspect this symbol for details
observeFetch	9	Implementation entry; inspect this symbol for details

frontend/src/PanelPortal.tsx

Lines: 2.

Frontend entry import graph: not reached by static imports.

Imports: ./portals/evaluator/EvaluatorPortal.

frontend/src/PasswordField.tsx

Lines: 48.

Frontend entry import graph: reachable.

Imports: react, ./passwordPolicy, ./passwordPolicy, ./passwordPolicy.

Symbol	Line	Purpose / source docstring
PasswordField	27	Implementation entry; inspect this symbol for details

frontend/src/passwordPolicy.ts

Lines: 32.

Frontend entry import graph: reachable.

Symbol	Line	Purpose / source docstring
PASSWORD_MIN_LENGTH	1	Implementation entry; inspect this symbol for details
PASSWORD_MAX_LENGTH	2	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
getPasswordGuidance	12	Implementation entry; inspect this symbol for details
passwordPolicyError	29	Implementation entry; inspect this symbol for details

frontend/src/portals/admin/modules/archive/AdminArchiveProjects.tsx

Lines: 185.

Frontend entry import graph: reachable.

Imports: react, ./OldProjectArchiveImport.

API references found in source (dynamic expressions may be partial): /api/v1/admin/archive/projects/${deleteTarget.id}, /api/v1/admin/archive/projects/${editing.id}, /api/v1/admin/archive/projects/repair-imported, /api/v1/admin/archive/projects?${query}.

Symbol	Line	Purpose / source docstring
AdminArchiveProjects	46	Implementation entry; inspect this symbol for details

frontend/src/portals/admin/modules/archive/ArchiveManagement.tsx

Lines: 5.

Frontend entry import graph: reachable.

Imports: ./AdminArchiveProjects.

Symbol	Line	Purpose / source docstring
ArchiveManagement	3	Implementation entry; inspect this symbol for details

frontend/src/portals/admin/modules/archive/OldProjectArchiveImport.tsx

Lines: 54.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/admin/archive/import/confirm.

Symbol	Line	Purpose / source docstring
OldProjectArchiveImport	5	Implementation entry; inspect this symbol for details

frontend/src/portals/admin/modules/supervisor-requests/AdminSupervisorRequests.tsx

Lines: 18.

Frontend entry import graph: not reached by static imports.

Imports: ./AdminSupervisorRequestsPdf.

Symbol	Line	Purpose / source docstring
AdminSupervisorRequests	16	Implementation entry; inspect this symbol for details

frontend/src/portals/admin/modules/supervisor-requests/AdminSupervisorRequestsPdf.tsx

Lines: 81.

Frontend entry import graph: not reached by static imports.

Imports: react, ./RequestConversation.

API references found in source (dynamic expressions may be partial): /api/v1/admin/audit-events?target_type, /api/v1/supervisor-requests/admin/monitor${suffix}.

Symbol	Line	Purpose / source docstring
AdminSupervisorRequestsPdf	26	Implementation entry; inspect this symbol for details

frontend/src/portals/admin/modules/supervisor-requests/ProposalReferenceSearch.tsx

Lines: 1.

Frontend entry import graph: not reached by static imports.

Imports: ../../../student/modules/supervisor-selection/ProposalReferenceSearch.

frontend/src/portals/admin/modules/supervisor-requests/RequestConversation.tsx

Lines: 1.

Frontend entry import graph: not reached by static imports.

Imports: ../../../student/modules/supervisor-selection/RequestConversation.

frontend/src/portals/coordinator/CoordinatorPortal.tsx

Lines: 23.

Frontend entry import graph: reachable.

Imports: react, ./modules/supervisor-requests/CoordinatorRequestsPdf, ./CoordinatorSidebar, ./../modules/module-3/ProposalComplianceWorkspace, ../student/student.css, ./supervisor/modules/supervisor-requests/supervisor-requests.css, ./modules/supervisor-requests/coordinator-requests.css.

Symbol	Line	Purpose / source docstring
CoordinatorPortal	11	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/CoordinatorSidebar.tsx

Lines: 9.

Frontend entry import graph: reachable.

Imports: ../shared/RoleSidebar.

Symbol	Line	Purpose / source docstring
coordinatorNav	3	Implementation entry; inspect this symbol for details
CoordinatorSidebar	7	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/CoordinatorArchive.tsx

Lines: 24.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/archived-projects/search${suffix}.

Symbol	Line	Purpose / source docstring
CoordinatorArchive	5	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/CoordinatorGroupPolicies.tsx

Lines: 11.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/academic-cycles/${cycle.id}/group-policy.

Symbol	Line	Purpose / source docstring
CoordinatorGroupPolicies	6	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/CoordinatorLiveList.tsx

Lines: 35.

Frontend entry import graph: not reached by static imports.

Imports: react.

Symbol	Line	Purpose / source docstring
CoordinatorLiveList	18	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/CoordinatorOverview.tsx

Lines: 43.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/academic-cycles, /api/v1/dashboard/deadlines, /api/v1/document-requirements, /api/v1/supervisor-requests/coordinator/monitor.

Symbol	Line	Purpose / source docstring
CoordinatorOverview	18	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/supervisor-requests/coordinator-requests.css

Lines: 1.

Frontend entry import graph: reachable.

frontend/src/portals/coordinator/modules/supervisor-requests/CoordinatorRequests.tsx

Lines: 17.

Frontend entry import graph: not reached by static imports.

Imports: ./CoordinatorRequestsPdf.

Symbol	Line	Purpose / source docstring
CoordinatorRequests	15	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/supervisor-requests/CoordinatorRequestsPdf.tsx

Lines: 90.

Frontend entry import graph: reachable.

Imports: react, ./RequestConversation, ./PanelScheduling.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor-requests/coordinator/monitor${suffix}.

Symbol	Line	Purpose / source docstring
CoordinatorRequestsPdf	33	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/supervisor-requests/PanelScheduling.tsx

Lines: 17.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/panel-schedules, /api/v1/panel-schedules/${row.id}, /api/v1/panel-schedules/options.

Symbol	Line	Purpose / source docstring
PanelScheduling	7	Implementation entry; inspect this symbol for details

frontend/src/portals/coordinator/modules/supervisor-requests/ProposalReferenceSearch.tsx

Lines: 1.

Frontend entry import graph: not reached by static imports.

Imports: ../../../student/modules/supervisor-selection/ProposalReferenceSearch.

frontend/src/portals/coordinator/modules/supervisor-requests/RequestConversation.tsx

Lines: 1.

Frontend entry import graph: reachable.

Imports: ../../../student/modules/supervisor-selection/RequestConversation.

frontend/src/portals/evaluator/evaluator.css

Lines: 1.

Frontend entry import graph: reachable.

frontend/src/portals/evaluator/EvaluatorOverview.tsx

Lines: 39.

Frontend entry import graph: reachable.

Symbol	Line	Purpose / source docstring
EvaluatorOverview	13	Implementation entry; inspect this symbol for details

frontend/src/portals/evaluator/EvaluatorPortal.tsx

Lines: 48.

Frontend entry import graph: reachable.

Imports: react, ./EvaluatorOverview, ./EvaluatorSidebar, ../student/student.css, ./evaluator.css.

Symbol	Line	Purpose / source docstring
EvaluatorPortal	20	Implementation entry; inspect this symbol for details

frontend/src/portals/evaluator/EvaluatorSidebar.tsx

Lines: 28.

Frontend entry import graph: reachable.

Imports: ./EvaluatorPortal, ../shared/RoleSidebar.

Symbol	Line	Purpose / source docstring
EvaluatorSidebar	16	Implementation entry; inspect this symbol for details

frontend/src/portals/shared/AdminSidebar.tsx

Lines: 22.

Frontend entry import graph: reachable.

Imports: ./RoleSidebar.

Symbol	Line	Purpose / source docstring
AdminSidebar	18	Implementation entry; inspect this symbol for details

frontend/src/portals/shared/CollaborationChannels.tsx

Lines: 16.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/${role}/collaboration.

Symbol	Line	Purpose / source docstring
CollaborationChannels	6	Implementation entry; inspect this symbol for details

frontend/src/portals/shared/dashboard-nav.css

Lines: 375.

Frontend entry import graph: not reached by static imports.

frontend/src/portals/shared/portal-shell.css

Lines: 216.

Frontend entry import graph: not reached by static imports.

frontend/src/portals/shared/PortalShell.tsx

Lines: 41.

Frontend entry import graph: not reached by static imports.

Imports: react, ./portal-shell.css.

Symbol	Line	Purpose / source docstring
PortalShell	17	Implementation entry; inspect this symbol for details

frontend/src/portals/shared/role-sidebar.css

Lines: 217.

Frontend entry import graph: reachable.

frontend/src/portals/shared/RoleSidebar.tsx

Lines: 110.

Frontend entry import graph: reachable.

Imports: react, ./role-sidebar.css.

Symbol	Line	Purpose / source docstring
SidebarIcon	66	Implementation entry; inspect this symbol for details
RoleSidebar	70	Implementation entry; inspect this symbol for details

frontend/src/portals/shared/sidebar-state.ts

Lines: 15.

Frontend entry import graph: not reached by static imports.

Symbol	Line	Purpose / source docstring
readSidebarCollapsed	1	Implementation entry; inspect this symbol for details
writeSidebarCollapsed	9	Implementation entry; inspect this symbol for details

frontend/src/portals/shared/SidebarProfile.tsx

Lines: 9.

Frontend entry import graph: not reached by static imports.

Symbol	Line	Purpose / source docstring
SidebarProfile	3	Implementation entry; inspect this symbol for details

frontend/src/portals/student/components/GroupMembers.tsx

Lines: 40.

Frontend entry import graph: reachable.

Symbol	Line	Purpose / source docstring
GroupMembers	10	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/AgentSupervisionPanel.tsx

Lines: 17.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/agent-supervision/${teamId}/runs.

Symbol	Line	Purpose / source docstring
AgentSupervisionPanel	5	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/archive-knowledge.css

Lines: 1.

Frontend entry import graph: reachable.

frontend/src/portals/student/modules/ArchivedProjectDetail.tsx

Lines: 3.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/archived-projects/${projectId}.

Symbol	Line	Purpose / source docstring
ArchivedProjectDetail	3	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/EvaluatorWorkflow.tsx

Lines: 16.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/panel-evaluations, /api/v1/panel-evaluations/schedule/${selected.id}, /api/v1/panel-schedules.

Symbol	Line	Purpose / source docstring
EvaluatorWorkflow	8	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/knowledge-search/ArchivedProjectDetail.tsx

Lines: 1.

Frontend entry import graph: not reached by static imports.

Imports: ../ArchivedProjectDetail.

frontend/src/portals/student/modules/knowledge-search/KnowledgeSearch.tsx

Lines: 1.

Frontend entry import graph: reachable.

Imports: ../KnowledgeSearch.

frontend/src/portals/student/modules/KnowledgeSearch.tsx

Lines: 26.

Frontend entry import graph: reachable.

Imports: react, ./ArchivedProjectDetail.

API references found in source (dynamic expressions may be partial): /api/v1/archived-projects/search${params.toString.

Symbol	Line	Purpose / source docstring
KnowledgeSearch	7	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentEvaluatorPortal.tsx

Lines: 18.

Frontend entry import graph: reachable.

Imports: react, ../../evaluator/EvaluatorOverview, ../../evaluator/EvaluatorSidebar, ./EvaluatorWorkflow, ../../../modules/module-3/ProposalComplianceWorkspace, ../student.css, ../../evaluator/evaluator.css.

Symbol	Line	Purpose / source docstring
StudentEvaluatorPortal	11	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentPhasePlaceholder.tsx

Lines: 12.

Frontend entry import graph: reachable.

Imports: ../StudentSidebar, ./StudentProgressDashboard, ../../../App, ./StudentRequiredDocumentsComplete.

Symbol	Line	Purpose / source docstring
StudentPhasePlaceholder	5	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentProgressDashboard.tsx

Lines: 115.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/group-channels/teams, /api/v1/project-progress/${kind}/${item.id}, /api/v1/project-progress/${teamId}, /api/v1/project-progress/${teamId}/${path}.

Symbol	Line	Purpose / source docstring
StudentProgressDashboard	34	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentRequiredDocuments.tsx

Lines: 3.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements/${item.id}/upload, /api/v1/document-requirements/student.

Symbol	Line	Purpose / source docstring
StudentRequiredDocuments	3	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentRequiredDocumentsComplete.tsx

Lines: 6.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/document-requirements/${id}/upload, /api/v1/document-requirements/${x.id}/submission, /api/v1/document-requirements/submissions/${sub.id}/versions/${v.id}/download, /api/v1/student/document-requirements.

Symbol	Line	Purpose / source docstring
StudentRequiredDocumentsComplete	6	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentRequiredDocumentsDashboard.tsx

Lines: 27.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/student/group/requirements, /api/v1/student/group/requirements/${item.id}/${endpoint}, /api/v1/student/group/requirements/${item.id}/submission.

Symbol	Line	Purpose / source docstring
StudentRequiredDocumentsDashboard	16	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/StudentSupervisorInfo.tsx

Lines: 36.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/student/group/me.

Symbol	Line	Purpose / source docstring
StudentSupervisorInfo	17	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/supervisor-selection/ProposalReferenceSearch.tsx

Lines: 23.

Frontend entry import graph: not reached by static imports.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/proposal-references/search?q.

Symbol	Line	Purpose / source docstring
ProposalReferenceSearch	5	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/supervisor-selection/RequestConversation.tsx

Lines: 52.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor-requests/${requestId}/conversation.

Symbol	Line	Purpose / source docstring
RequestConversation	5	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/supervisor-selection/StudentSupervisorSelection.tsx

Lines: 32.

Frontend entry import graph: reachable.

Imports: ./StudentSupervisorSelectionPdf.

Symbol	Line	Purpose / source docstring
StudentSupervisorSelection	30	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/supervisor-selection/StudentSupervisorSelectionPdf.tsx

Lines: 41.

Frontend entry import graph: reachable.

Imports: react, ./supervisor-selection.css, ./RequestConversation.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor-requests, /api/v1/supervisor-requests/${preview.id}/confirm, /api/v1/supervisor-requests/${preview.id}/submit, /api/v1/supervisor-requests/${record.id}/withdraw, /api/v1/supervisor-requests/cycles, /api/v1/supervisor-requests/draft/${editingId}/document, /api/v1/supervisor-requests/mine, /api/v1/supervisor-requests/supervisors?academic_cycle_id.

Symbol	Line	Purpose / source docstring
StudentSupervisorSelectionPdf	27	Implementation entry; inspect this symbol for details

frontend/src/portals/student/modules/supervisor-selection/supervisor-selection.css

Lines: 2.

Frontend entry import graph: reachable.

frontend/src/portals/student/student-admin-theme.css

Lines: 134.

Frontend entry import graph: reachable.

frontend/src/portals/student/student-content.css

Lines: 197.

Frontend entry import graph: reachable.

frontend/src/portals/student/student-visibility.css

Lines: 2.

Frontend entry import graph: reachable.

frontend/src/portals/student/student.css

Lines: 3.

Frontend entry import graph: reachable.

frontend/src/portals/student/StudentOverview.tsx

Lines: 56.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/notifications, /api/v1/student/group/me.

Symbol	Line	Purpose / source docstring
StudentOverview	20	Implementation entry; inspect this symbol for details
StudentOverviewWithDocuments	54	Implementation entry; inspect this symbol for details

frontend/src/portals/student/StudentPortal.tsx

Lines: 56.

Frontend entry import graph: reachable.

Imports: react, ./StudentOverview, ./StudentSidebar, ./modules/StudentPhasePlaceholder, ./modules/StudentRequiredDocumentsDashboard, ./modules/supervisor-selection/StudentSupervisorSelection, ./modules/knowledge-search/KnowledgeSearch, ./../modules/module-3/ProposalComplianceWorkspace, ./components/GroupMembers, ./modules/StudentSupervisorInfo, ./student.css, ./student-content.css, ./student-admin-theme.css, ./modules/archive-knowledge.css, ./student-visibility.css.

Symbol	Line	Purpose / source docstring
STUDENT_PHASES	35	Implementation entry; inspect this symbol for details
StudentPortal	46	Implementation entry; inspect this symbol for details

frontend/src/portals/student/StudentSidebar.tsx

Lines: 43.

Frontend entry import graph: reachable.

Imports: ../shared/RoleSidebar, react.

Symbol	Line	Purpose / source docstring
StudentSidebar	19	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/modules/supervisor-requests/ProposalReferenceSearch.tsx

Lines: 1.

Frontend entry import graph: not reached by static imports.

Imports: ../../../student/modules/supervisor-selection/ProposalReferenceSearch.

frontend/src/portals/supervisor/modules/supervisor-requests/RequestConversation.tsx

Lines: 1.

Frontend entry import graph: reachable.

Imports: ../../../student/modules/supervisor-selection/RequestConversation.

frontend/src/portals/supervisor/modules/supervisor-requests/supervisor-requests.css

Lines: 2.

Frontend entry import graph: reachable.

frontend/src/portals/supervisor/modules/supervisor-requests/SupervisorRequests.tsx

Lines: 21.

Frontend entry import graph: not reached by static imports.

Imports: ./SupervisorRequestsPdf.

Symbol	Line	Purpose / source docstring
SupervisorRequests	19	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/modules/supervisor-requests/SupervisorRequestsPdf.tsx

Lines: 29.

Frontend entry import graph: reachable.

Imports: react, ./RequestConversation, ../../SupervisorGroupDetails.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor/requests, /api/v1/supervisor/requests/${selected.id}/${action}.

Symbol	Line	Purpose / source docstring
SupervisorRequestsPdf	6	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/supervisor-capacity.css

Lines: 1.

Frontend entry import graph: reachable.

frontend/src/portals/supervisor/SupervisorAssignedTeams.tsx

Lines: 32.

Frontend entry import graph: reachable.

Imports: react, ./SupervisorGroupDetails.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor/assigned-groups.

Symbol	Line	Purpose / source docstring
SupervisorAssignedTeams	4	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorCapacity.tsx

Lines: 18.

Frontend entry import graph: reachable.

Imports: react, ./supervisor-capacity.css.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor/capacity.

Symbol	Line	Purpose / source docstring
SupervisorCapacity	10	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorDecisionHistory.tsx

Lines: 9.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor-requests/supervisor/queue?status.

Symbol	Line	Purpose / source docstring
SupervisorDecisionHistory	5	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorGroupDetails.tsx

Lines: 52.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor/groups/${data.group.id}/conversation, /api/v1/supervisor/groups/${groupId}.

Symbol	Line	Purpose / source docstring
date	18	Implementation entry; inspect this symbol for details
GroupDetailsContent	34	Implementation entry; inspect this symbol for details
SupervisorGroupDetails	47	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorOverview.tsx

Lines: 18.

Frontend entry import graph: reachable.

Imports: react.

API references found in source (dynamic expressions may be partial): /api/v1/supervisor-requests/supervisor/queue?status.

Symbol	Line	Purpose / source docstring
SupervisorOverview	7	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorPortal.tsx

Lines: 20.

Frontend entry import graph: reachable.

Imports: react, ./modules/supervisor-requests/SupervisorRequestsPdf, ./SupervisorSidebar, ./SupervisorOverview, ./SupervisorAssignedTeams, ./SupervisorDecisionHistory, ./SupervisorUnavailable, ./SupervisorCapacity, ../student/student.css, ./modules/supervisor-requests/supervisor-requests.css.

Symbol	Line	Purpose / source docstring
SupervisorPortal	14	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorSidebar.tsx

Lines: 23.

Frontend entry import graph: reachable.

Imports: ../shared/RoleSidebar.

Symbol	Line	Purpose / source docstring
supervisorNav	3	Implementation entry; inspect this symbol for details
SupervisorSidebar	11	Implementation entry; inspect this symbol for details

frontend/src/portals/supervisor/SupervisorUnavailable.tsx

Lines: 3.

Frontend entry import graph: reachable.

Symbol	Line	Purpose / source docstring
SupervisorUnavailable	1	Implementation entry; inspect this symbol for details

backend/alembic/env.py

Lines: 45. Alembic environment using the FYPHub application database configuration.

Symbol	Line	Purpose / source docstring
run_migrations_offline	20	Run migrations without a live database connection.
run_migrations_online	33	Run migrations with the shared synchronous application engine.

backend/alembic/versions/3a1f53f17b1c_add_academic_cycles_and_group_policies.py

Lines: 56. add academic cycles and group policies

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	50	Revert this revision.

backend/alembic/versions/9415b228314f_join_requests_and_creator_proof_binding.py

Lines: 75. join requests and creator proof binding

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	64	Revert this revision.

backend/alembic/versions/9c227dc337bf_create_users_foundation.py

Lines: 49. create users foundation

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	44	Revert this revision.

backend/alembic/versions/a2b3c4d5e6f7_add_academic_deadline_versions.py

Lines: 40. add academic deadline configuration versions

Symbol	Line	Purpose / source docstring
upgrade	16	Implementation entry; inspect this symbol for details
downgrade	38	Implementation entry; inspect this symbol for details

backend/alembic/versions/a3b4c5d6e7f8_bind_anonymous_join_proof_target.py

Lines: 48. bind anonymous join proof target

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	33	Revert this revision.

backend/alembic/versions/a7b8c9d0e1f2_add_group_messages.py

Lines: 14. Add retained team channel messages.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	13	Implementation entry; inspect this symbol for details

backend/alembic/versions/a8467cad28da_invitation_transactional_outbox.py

Lines: 37. invitation transactional outbox

Symbol	Line	Purpose / source docstring
upgrade	21	Implementation entry; inspect this symbol for details
downgrade	34	Implementation entry; inspect this symbol for details

backend/alembic/versions/b2c3d4e5f6a7_student_matching_proof.py

Lines: 17. Allow a distinct anonymous student matching proof.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	14	Implementation entry; inspect this symbol for details

backend/alembic/versions/b3c4d5e6f7a8_add_reminder_rules.py

Lines: 15. Add non-operational reminder rule versions.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	13	Implementation entry; inspect this symbol for details

backend/alembic/versions/b8c9d0e1f2a3_add_team_match_preferences.py

Lines: 13. Add team compatibility preferences.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	12	Implementation entry; inspect this symbol for details

backend/alembic/versions/b8d2e4f6a9c1_add_admin_audit_events.py

Lines: 55. add append-only administrator audit events

Symbol	Line	Purpose / source docstring
upgrade	21	Implementation entry; inspect this symbol for details
downgrade	50	Implementation entry; inspect this symbol for details

backend/alembic/versions/c3d4e5f6a7b8_proposal_foundation.py

Lines: 37. Add non-operational proposal identity and version foundations.

Symbol	Line	Purpose / source docstring
upgrade	15	Implementation entry; inspect this symbol for details
downgrade	35	Implementation entry; inspect this symbol for details

backend/alembic/versions/c4d5e6f7a8b9_add_student_roster.py

Lines: 17. Add approved student roster and import provenance.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	15	Implementation entry; inspect this symbol for details

backend/alembic/versions/c4e7f1a2b9d3_add_audit_targets.py

Lines: 25. add explicit audit targets

Symbol	Line	Purpose / source docstring
upgrade	15	Implementation entry; inspect this symbol for details
downgrade	22	Implementation entry; inspect this symbol for details

backend/alembic/versions/c5d6e7f8a9b0_add_supervisor_requests.py

Lines: 67. add student supervisor-selection requests and immutable history

Symbol	Line	Purpose / source docstring
upgrade	16	Implementation entry; inspect this symbol for details
downgrade	59	Implementation entry; inspect this symbol for details

backend/alembic/versions/c9d0e1f2a3b4_add_match_chat.py

Lines: 16. Add temporary match chats and retained messages.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	14	Implementation entry; inspect this symbol for details

backend/alembic/versions/ce964120f177_add_teams_and_member_duplicate_.py

Lines: 68. add teams and member duplicate constraints

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	60	Revert this revision.

backend/alembic/versions/d0e1f2a3b4c5_match_chat_mutual_reconsent.py

Lines: 17. Allow new match conversations after mutual fresh consent.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	14	Implementation entry; inspect this symbol for details

backend/alembic/versions/d5e6f7a8b9c0_add_template_reference_versions.py

Lines: 10. add internal template reference and rule versions

Symbol	Line	Purpose / source docstring
upgrade	6	Implementation entry; inspect this symbol for details
downgrade	10	Implementation entry; inspect this symbol for details

backend/alembic/versions/d6e7f8a9b0c1_add_supervisor_request_documents.py

Lines: 81. add immutable one-page proposal documents and extraction snapshots

Symbol	Line	Purpose / source docstring
upgrade	18	Implementation entry; inspect this symbol for details
downgrade	63	Implementation entry; inspect this symbol for details

backend/alembic/versions/e1f2a3b4c5d6_individual_verification.py

Lines: 19. Store roster-bound verification challenges and one-use proofs.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
downgrade	15	Implementation entry; inspect this symbol for details

backend/alembic/versions/e5f6a7b8c9d0_add_workflow_policy_drafts.py

Lines: 15. Add non-operational workflow policy drafts.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	13	Implementation entry; inspect this symbol for details

backend/alembic/versions/e7f8a9b0c1d2_add_proposal_references_and_request_conversations.py

Lines: 65. add authorized proposal references and request conversations

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	56	Implementation entry; inspect this symbol for details

backend/alembic/versions/e8f9a0b1c2d3_add_supervisor_capacity_policies.py

Lines: 27. add supervisor capacity policies

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	26	Implementation entry; inspect this symbol for details

backend/alembic/versions/eb100aa929ab_team_invitations.py

Lines: 71. team invitations

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	62	Revert this revision.

backend/alembic/versions/f1a2b3c4d5e6_add_evaluation_rubric_drafts.py

Lines: 47. add evaluation rubric draft versions

Symbol	Line	Purpose / source docstring
upgrade	16	Implementation entry; inspect this symbol for details
downgrade	44	Implementation entry; inspect this symbol for details

backend/alembic/versions/f2a3b4c5d6e7_anonymous_verification.py

Lines: 82. anonymous verification

Symbol	Line	Purpose / source docstring
upgrade	21	Apply this revision.
downgrade	74	Revert this revision.

backend/alembic/versions/f6a7b8c9d0e1_add_supervisor_assignments.py

Lines: 14. Add supervisor allocation history.

Symbol	Line	Purpose / source docstring
upgrade	9	Implementation entry; inspect this symbol for details
downgrade	13	Implementation entry; inspect this symbol for details

backend/alembic/versions/f7a1c3d9e2b4_add_auth_sessions.py

Lines: 40. add hash-only browser authentication sessions

Symbol	Line	Purpose / source docstring
upgrade	20	Implementation entry; inspect this symbol for details
downgrade	37	Implementation entry; inspect this symbol for details

backend/alembic/versions/f9a0b1c2d3e4_add_archived_projects.py

Lines: 69. add logical archived projects and immutable document versions

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	62	Implementation entry; inspect this symbol for details

backend/alembic/versions/fa0b1c2d3e4_allow_archived_title_reuse_after_deactivation.py

Lines: 20. allow archived title reuse after deactivation

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	18	Implementation entry; inspect this symbol for details

backend/alembic/versions/fb1c2d3e4f5_add_historical_archive_fields.py

Lines: 39. add structured historical project archive fields

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	36	Implementation entry; inspect this symbol for details

backend/alembic/versions/fb1c2d3e4f5a_add_module03_proposal_compliance.py

Lines: 53. add Module 03 dynamic proposal compliance tables

Symbol	Line	Purpose / source docstring
_jsonb	14	Implementation entry; inspect this symbol for details
upgrade	18	Implementation entry; inspect this symbol for details
downgrade	47	Implementation entry; inspect this symbol for details

backend/alembic/versions/fc2d3e4f5a6b_add_supervisor_request_submission_notes.py

Lines: 38. add optional immutable supervisor-request submission note versions

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	32	Implementation entry; inspect this symbol for details

backend/alembic/versions/fc6b7c8d9e0f_add_private_collaboration_channels.py

Lines: 60. add assignment-scoped private Module 2 collaboration channels

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	53	Implementation entry; inspect this symbol for details

backend/alembic/versions/fd3e4f5a6b7c_add_fyp_registry_metadata.py

Lines: 32. add safe administrative metadata for the FYP registry

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	30	Implementation entry; inspect this symbol for details

backend/alembic/versions/fe4f5a6b7c8d_add_password_reset_tokens.py

Lines: 32. add hash-only password reset tokens

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	29	Implementation entry; inspect this symbol for details

backend/alembic/versions/ff5a6b7c8d9e_add_progress_scheduling_evaluation.py

Lines: 126. add persisted progress, defense scheduling, and panel evaluation workflows

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	115	Implementation entry; inspect this symbol for details

backend/alembic/versions/fg6b7c8d9e0f_add_module03_template_versions.py

Lines: 37. add immutable Module 03 template metadata and rule version provenance

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	27	Implementation entry; inspect this symbol for details

backend/alembic/versions/fh7c8d9e0f1a_add_module03_activation_mode.py

Lines: 18. Persist complete or partial Module 03 baseline activation mode.

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	17	Implementation entry; inspect this symbol for details

backend/alembic/versions/fi8d9e0f1a2b_continue_proposal_chain.py

Lines: 14. Continue the proposal foundation migration chain.

Symbol	Line	Purpose / source docstring
upgrade	9	Converge the already-applied migration branches without changing data.

Symbol	Line	Purpose / source docstring
downgrade	13	Re-expose the proposal foundation parent for historical downgrades.

backend/alembic/versions/fj9e0f1a2b3_add_proposal_type.py

Lines: 19. Store the selected SHORT/LONG type on supervisor-request proposals.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	17	Implementation entry; inspect this symbol for details

backend/alembic/versions/fk0f1a2b3c4_add_proposal_metadata_fields.py

Lines: 19. Add editable proposal metadata snapshots to the existing lifecycle records.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	17	Implementation entry; inspect this symbol for details

backend/alembic/versions/fl1a2b3c4d5_add_agent_supervision_runs.py

Lines: 33. Add controlled supervision agent run records.

Symbol	Line	Purpose / source docstring
upgrade	11	Implementation entry; inspect this symbol for details
downgrade	31	Implementation entry; inspect this symbol for details

backend/alembic/versions/gm2b3c4d5e6f_add_dashboard_requirements.py

Lines: 42. Add role-scoped dashboard requirements.

Symbol	Line	Purpose / source docstring
upgrade	11	Implementation entry; inspect this symbol for details
downgrade	38	Implementation entry; inspect this symbol for details

backend/alembic/versions/hn3c4d5e6f7a_add_document_upload_requirements.py

Lines: 35. add admin-managed document upload requirements

Symbol	Line	Purpose / source docstring
upgrade	15	Implementation entry; inspect this symbol for details
downgrade	34	Implementation entry; inspect this symbol for details

backend/alembic/versions/jk4d5e6f7a8b_document_storage_reference.py

Lines: 15. make document version content optional for persistent file storage

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	14	Implementation entry; inspect this symbol for details

backend/alembic/versions/kl5e6f7a8b9c_add_role_based_fyp_evaluations.py

Lines: 78. add independent role-based FYP evaluations and result audit history

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	69	Implementation entry; inspect this symbol for details

backend/alembic/versions/lm6f7a8b9c0d_remove_evaluation_marks_module.py

Lines: 37. Remove the retired Evaluation & Marks module.

Symbol	Line	Purpose / source docstring
upgrade	16	Implementation entry; inspect this symbol for details
downgrade	36	Implementation entry; inspect this symbol for details

backend/alembic/versions/m123a001_modules_1_3_lifecycle.py

Lines: 43. Join existing histories; add proposal lifecycle and membership consent only.

Symbol	Line	Purpose / source docstring
upgrade	15	Implementation entry; inspect this symbol for details
downgrade	34	Implementation entry; inspect this symbol for details

backend/alembic/versions/mn7c8d9e0f1a_restore_role_based_evaluations.py

Lines: 97. restore the role-based FYP evaluation and results module

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	87	Implementation entry; inspect this symbol for details

backend/alembic/versions/mo8r9e0f1a2b_add_proposal_ai_review_runs.py

Lines: 32. Persist advisory AI review runs for supervisor proposal requests.

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	30	Implementation entry; inspect this symbol for details

backend/alembic/versions/n456a001_add_academic_cycle_year_session.py

Lines: 40. Persist the year and session selected when creating an academic cycle.

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	35	Implementation entry; inspect this symbol for details

backend/alembic/versions/oa1b2c3d4e5f_repair_module1_evaluation_schema.py

Lines: 89. Repair Module 1 evaluation tables after removing the retired compliance branch.

Symbol	Line	Purpose / source docstring
upgrade	18	Implementation entry; inspect this symbol for details
downgrade	86	Implementation entry; inspect this symbol for details

backend/alembic/versions/ob1c2d3e4f5a_repair_module1_identity_columns.py

Lines: 38. Add identity generation to tables created by the Module 1 repair migration.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	37	Implementation entry; inspect this symbol for details

backend/alembic/versions/oc1d2e3f4a5b_add_template_activation.py

Lines: 24. Compatibility migration for the existing template activation state.

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	20	Implementation entry; inspect this symbol for details

backend/alembic/versions/p567a001_supervisor_capacity_year_session.py

Lines: 75. Add year/session identity to supervisor capacity policies.

Symbol	Line	Purpose / source docstring
upgrade	17	Implementation entry; inspect this symbol for details
downgrade	70	Implementation entry; inspect this symbol for details

backend/alembic/versions/pa1b2c3d4e5f_add_module03_compliance_workspace.py

Lines: 62. Add the Module 3 proposal compliance workspace.

Symbol	Line	Purpose / source docstring
upgrade	14	Implementation entry; inspect this symbol for details
downgrade	57	Implementation entry; inspect this symbol for details

backend/alembic/versions/qa1b2c3d4e5f_complete_module02_collaboration.py

Lines: 122. Complete Module 2 team lifecycle, matching and collaboration foundations.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	88	Implementation entry; inspect this symbol for details

backend/alembic/versions/r678a001_remove_registry_roster_identity_modules.py

Lines: 55. Remove registry, roster, workflow-draft, invitation and identity modules.

Symbol	Line	Purpose / source docstring
upgrade	17	Implementation entry; inspect this symbol for details
downgrade	54	Implementation entry; inspect this symbol for details

backend/alembic/versions/r678a002_remove_invitation_outbox.py

Lines: 17. Remove the transactional outbox belonging to retired invitations.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
downgrade	16	Implementation entry; inspect this symbol for details

backend/alembic/versions/r678a003_annual_supervisor_capacity.py

Lines: 101. Make supervisor capacity annual and shared across Fall and Spring.

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	93	Implementation entry; inspect this symbol for details

backend/alembic/versions/rd7e8f9a0b1c_repair_archive_import_columns.py

Lines: 44. Repair archive storage columns on the active collaboration migration branch.

Symbol	Line	Purpose / source docstring
upgrade	13	Implementation entry; inspect this symbol for details
downgrade	42	Implementation entry; inspect this symbol for details

backend/alembic/versions/re8f9a0b1c2d_archive_import_batches.py

Lines: 52. Persist archive import batches and append-safe duplicate keys.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	46	Implementation entry; inspect this symbol for details

backend/alembic/versions/rf9a0b1c2d3e_document_requirement_lifecycle.py

Lines: 33. Normalize document requirement lifecycle and bind submissions to groups.

Symbol	Line	Purpose / source docstring
upgrade	12	Implementation entry; inspect this symbol for details
downgrade	29	Implementation entry; inspect this symbol for details

backend/tests/__init__.py

Lines: 1. Backend test package.

backend/tests/test_academic_cycle_overview.py

Lines: 146. Authorized academic-cycle overview coverage for Admin and Coordinator portals.

Symbol	Line	Purpose / source docstring
request	22	Implementation entry; inspect this symbol for details
AcademicCycleOverviewTests	66	Implementation entry; inspect this symbol for details

Test cases: test_admin_and_coordinator_receive_empty_structured_overview, test_overview_rejects_missing_session_student_and_unknown_cycle, test_overview_route_is_registered_in_openapi, test_fall_2027_list_id_resolves_for_admin_and_coordinator.

backend/tests/test_academic_deadlines.py

Lines: 227. Rollback-isolated coverage for non-operational academic deadlines.

Symbol	Line	Purpose / source docstring
AcademicDeadlineTests	23	Implementation entry; inspect this symbol for details

Test cases: test_authorization_and_origin_rejection, test_unknown_cycle_and_strict_validation, test_naive_writes_leave_versions_and_audit_unchanged, test_missing_timezone_data_returns_service_error,

test_timezone_conversion_mismatch_gap_and_overlap, test_initial_subsequent_and_ordered_history,test_audit_failure_rolls_back_subsequent_version, test_controlled_concurrent_version_allocation.

backend/tests/test_admin_users.py

Lines: 238. Rollback-isolated coverage for M1-B administrator account management.

Symbol	Line	Purpose / source docstring
request	21	Implementation entry; inspect this symbol for details
AdminUserTests	49	Implementation entry; inspect this symbol for details

Test cases: test_listing_authorization_search_and_safe_values, test_pagination_uses_the_requested_page, test_last_admin_lock_is_exclusive_across_database_connections, test_creation_duplicate_and_atomic_audit, test_creation_rejects_malformed_email, test_admin_can_create_each_fixed_non_public_role, test_role_status_session_revocation_and_last_admin_protection, test_shared_team_protection_and_append_only_database_trigger, test_cycle_policy_management_and_audit_target, test_academic_cycle_creation_requires_year_and_session, test_academic_cycle_creation_persists_year_and_session.

backend/tests/test_ai_proposal_review.py

Lines: 45.

Symbol	Line	Purpose / source docstring
AiProposalReviewTests	6	Implementation entry; inspect this symbol for details

Test cases: test_deterministic_fallback_returns_structured_findings, test_missing_document_fallback_is_unavailable_and_structured, test_fallback_preserves_deterministic_evidence.

backend/tests/test_archive_import.py

Lines: 91. Excel imports preserve source columns without an archive schema.

Symbol	Line	Purpose / source docstring
ArchiveImportValidationTests	9	Implementation entry; inspect this symbol for details

Test cases: test_arbitrary_columns_and_values_are_preserved, test_legacy_xls_keeps_columns_values_and_dates, test_no_required_fields_or_cycle_matching, test_duplicate_and_blank_headers_and_blank_cells, test_all_worksheets_and_repeated_rows_are_kept, test_headerless_and_explicit_heading_row, test_formulas_and_excel_errors_are_not_imported_as_content, test_stops_after_last_email_and_ignores_trailing_formula_or_spacing_rows, test_whitespace_only_cells_are_blank.

backend/tests/test_archive_import_persistence.py

Lines: 73. Confirm and edit flexible imports inside a rolled-back database transaction.

Symbol	Line	Purpose / source docstring
ArchiveImportPersistenceTests	15	Implementation entry; inspect this symbol for details

Test cases: test_confirm_and_edit_only_original_columns, test_http_preview_and_confirm_use_flexible_columns.

backend/tests/test_archived_projects.py

Lines: 142. Focused logical archive, versioning, ranking, and role-isolation tests.

Symbol	Line	Purpose / source docstring
ArchivedProjectTests	23	Implementation entry; inspect this symbol for details

Test cases: test_admin_create_search_detail_and_panel_denial, test_pdf_versions_are_separate_and_student_read_only, test_metadata_refresh_duplicate_protection_and_inactive_filtering, test_metadata_audit_snapshot_can_be_rolled_back.

backend/tests/test_audit_history.py

Lines: 59. Rollback-isolated read-only audit viewer tests.

Symbol	Line	Purpose / source docstring
AuditHistoryTests	11	Implementation entry; inspect this symbol for details

Test cases: test_authorization_and_missing, test_filter_pagination_order_and_read_only, test_legacy_and_sensitive_data_exclusion, test_strict_filters.

backend/tests/test_auth_sessions.py

Lines: 479. Rollback-isolated integration tests for cookie-backed authentication.

Symbol	Line	Purpose / source docstring
request	26	Invoke ASGI without adding a test client dependency.
AuthSessionTests	65	Implementation entry; inspect this symbol for details

Test cases: test_login_safe_cookie_and_me, test_login_session_is_usable_by_proposal_endpoints, test_bad_unknown_and_inactive_credentials_are_generic, test_malformed_email_login_is_rejected_before_authentication, test_creator_registration_login_and_selected_role_cannot_escalate, test_admin_endpoint_requires_persisted_admin_role, test_missing_tampered_expired_revoked_and_deactivated_sessions, test_session_rotation_logout_and_idempotent_logout, test_same_valid_session_supports_concurrent_requests, test_deactivation_revokes_every_active_session_for_the_user, test_origin_cors_and_role_helpers, test_authenticated_admin_can_create_document_requirement, test_admin_published_document_requirement_is_visible_to_student, test_forgot_password_is_generic_and_stores_only_a_hash, test_reset_password_rejects_weak_and_consumes_token_revoking_sessions, test_new_request_invalidates_old_token_and_expired_token_is_rejected, test_development_without_smtp_exposes_only_loopback_reset_link, test_development_with_smtp_never_exposes_local_link, test_production_without_smtp_keeps_safe_unavailable_status, test_demo_mode_unknown_email_uses_development_message_without_smtp.

backend/tests/test_collaboration_channels.py

Lines: 27. Focused contract tests for the private Module 2 collaboration channel API.

Symbol	Line	Purpose / source docstring
CollaborationChannelContractTests	11	Implementation entry; inspect this symbol for details

Test cases: test_blank_messages_are_rejected, test_message_text_is_trimmed, test_private_channel_routes_exist.

backend/tests/test_coordinator_operations.py

Lines: 18.

Symbol	Line	Purpose / source docstring
test_coordinator_read_only_routes_are_registered	5	Implementation entry; inspect this symbol for details
test_coordinator_operations_are_read_only_get_routes	17	Implementation entry; inspect this symbol for details

backend/tests/test_cycle_concurrency.py

Lines: 282. True multi-connection coverage for cycle configuration and team registration.

Symbol	Line	Purpose / source docstring
_RequestWithoutOrigin	42	The direct endpoint call has the same permitted no-Origin boundary as tests.
CycleConcurrencyTests	48	Exercise the production cycle-then-policy lock order with real sessions.

Test cases: test_registration_finishes_before_a_waiting_registration_closure, test_policy_reduction_rejects_after_waiting_registration_commits.

backend/tests/test_document_requirements_contract.py

Lines: 46.

Symbol	Line	Purpose / source docstring
test_document_requirement_models_and_routes_exist	7	Implementation entry; inspect this symbol for details
test_supported_file_mimes_and_states	14	Implementation entry; inspect this symbol for details
test_exact_document_requirement_paths_are_registered_in_openapi	20	Implementation entry; inspect this symbol for details
test_storage_reference_is_relative_and_path_traversal_is_rejected	34	Implementation entry; inspect this symbol for details
test_file_format_adapter_preserves_single_and_combined_formats	40	Implementation entry; inspect this symbol for details

backend/tests/test_document_requirements_visibility.py

Lines: 62.

Symbol	Line	Purpose / source docstring
_Rows	8	Implementation entry; inspect this symbol for details
_Db	16	Implementation entry; inspect this symbol for details
_student	24	Implementation entry; inspect this symbol for details
_requirement	28	Implementation entry; inspect this symbol for details
test_published_all_students_requirement_appears	37	Implementation entry; inspect this symbol for details
test_student_specific_requirement_appears_only_for_that_student	44	Implementation entry; inspect this symbol for details
test_draft_and_archived_requirements_stay_hidden	51	Implementation entry; inspect this symbol for details
test_requirement_for_another_student_stays_hidden	58	Implementation entry; inspect this symbol for details

backend/tests/test_evaluation_rubrics.py

Lines: 184. Rollback-isolated coverage for internal evaluation-rubric drafts.

Symbol	Line	Purpose / source docstring
EvaluationRubricTests	21	Implementation entry; inspect this symbol for details

Test cases: test_authorization_origin_and_missing_record, test_validation_rejects_blank_duplicate_order_and_extra_fields, test_ordered_history_preserves_prior_version, test_audit_failure_rolls_back_complete_version, test_concurrent_version_numbers_are_distinct.

backend/tests/test_evaluations.py

Lines: 135. Integration coverage for role-owned marks and published result aggregation.

Symbol	Line	Purpose / source docstring
EvaluationWorkflowTests	18	Implementation entry; inspect this symbol for details

Test cases: test_role_ownership_and_submission_lock, test_panel_records_are_independent_and_admin_can_monitor, test_approve_publish_and_student_results, test_report_access_and_unauthorized_student, test_admin_correction_and_panel_aggregation_are_audited.

backend/tests/test_groq_configuration.py

Lines: 13.

Symbol	Line	Purpose / source docstring
GroqConfigurationTests	6	Implementation entry; inspect this symbol for details

Test cases: test_key_is_optional_for_ollama.

backend/tests/test_groq_provider.py

Lines: 63.

Symbol	Line	Purpose / source docstring
_Response	9	Implementation entry; inspect this symbol for details
_StatusClient	16	Implementation entry; inspect this symbol for details
_Client	25	Implementation entry; inspect this symbol for details
GroqProviderTests	33	Implementation entry; inspect this symbol for details

Test cases: test_success_uses_exact_endpoint_headers_and_model, test_missing_key_falls_back_without_request, test_unauthorized_and_invalid_model_are_classified.

backend/tests/test_group_access.py

Lines: 96.

Symbol	Line	Purpose / source docstring
request_scope	16	Implementation entry; inspect this symbol for details

Symbol	Line	Purpose / source docstring
GroupAccessTests	32	Implementation entry; inspect this symbol for details

Test cases: test_group_login_returns_saved_role_and_group_context, test_registration_number_login_is_rejected, test_dissolved_group_cannot_login.

backend/tests/test_group_ai.py

Lines: 53.

Symbol	Line	Purpose / source docstring
GroupAITests	13	Implementation entry; inspect this symbol for details

Test cases: test_authorization_and_team_isolation, test_configured_model_and_history, test_missing_configuration_timeout_and_runtime_error, test_agent_model_falls_back_to_configured_local_model.

backend/tests/test_group_channels.py

Lines: 51.

Symbol	Line	Purpose / source docstring
GroupChannelTests	12	Implementation entry; inspect this symbol for details

Test cases: test_permissions_all_channels, test_history_reassignment_and_safe_rollback.

backend/tests/test_module3_compliance_engine.py

Lines: 107.

Symbol	Line	Purpose / source docstring
test_template_upload_contract_only_accepts_pdf_and_docx	7	Implementation entry; inspect this symbol for details
test_compliance_report_reconciles_findings_and_locations	16	Implementation entry; inspect this symbol for details
test_compliance_report_checks_structure_and_academic_consistency	52	Implementation entry; inspect this symbol for details
test_version_change_summary_is_structured_and_advisory	88	Implementation entry; inspect this symbol for details
test_module3_lifecycle_and_review_routes_are_exposed	101	Implementation entry; inspect this symbol for details

backend/tests/test_observability.py

Lines: 15.

Symbol	Line	Purpose / source docstring
ObservabilityTests	6	Implementation entry; inspect this symbol for details

Test cases: test_workflow_names_are_safe_and_specific.

backend/tests/test_ollama_provider.py

Lines: 66.

Symbol	Line	Purpose / source docstring
_Response	9	Implementation entry; inspect this symbol for details
_Client	19	Implementation entry; inspect this symbol for details
OllamaProviderTests	29	Implementation entry; inspect this symbol for details

Test cases: test_chat_request_is_local_non_streaming_and_schema_validated, test_invalid_response_returns_fallback, test_timeout_returns_fallback.

backend/tests/test_password_policy.py

Lines: 62.

Symbol	Line	Purpose / source docstring
PasswordPolicyTests	10	Implementation entry; inspect this symbol for details

Test cases: test_shared_policy_accepts_compliant_password, test_shared_policy_reports_each_missing_requirement, test_shared_policy_rejects_overlong_values, test_student_and_admin_creation_schemas_use_shared_policy, test_admin_registration_number_is_role_dependent.

backend/tests/test_pdf_extraction.py

Lines: 71.

Symbol	Line	Purpose / source docstring
PdfExtractionFieldTests	9	Implementation entry; inspect this symbol for details

Test cases: test_demo_headings_extract_all_sections_until_next_heading, test_labels_are_case_insensitive_and_aliases_are_supported, test_missing_field_remains_not_detected_without_fabrication, test_normalization_preserves_sections_and_colon_spacing, test_docx_upload_extracts_fields_with_immutable_file_metadata, test_unsupported_upload_is_rejected.

backend/tests/test_project_workflow.py

Lines: 93. Focused coverage for persisted progress, scheduling, and evaluation workflows.

Symbol	Line	Purpose / source docstring
ProjectWorkflowTests	18	Implementation entry; inspect this symbol for details

Test cases: test_progress_milestones_tasks_meetings_and_health, test_progress_cross_team_denial, test_schedule_assignment_conflict_and_reschedule, test_panel_evaluation_submission_disagreement_and_resolution.

backend/tests/test_proposal_foundation.py

Lines: 48. Isolated checks for the non-operational proposal schema.

Symbol	Line	Purpose / source docstring
ProposalFoundationTests	7	Implementation entry; inspect this symbol for details

Test cases: test_short_long_and_independent_version_sequences, test_invalid_type_and_missing_team_are_rejected, test_invalid_version_duplicate_and_missing_parent_are_rejected, test_proposal_lifecycle_routes_are_exposed.

backend/tests/test_proposal_reference_and_conversation.py

Lines: 123. Reference-search and request-conversation authorization coverage.

Symbol	Line	Purpose / source docstring
ProposalReferenceAndConversationTests	21	Implementation entry; inspect this symbol for details

Test cases: test_title_search_exact_similar_and_no_close, test_reference_pdf_and_panel_denial, test_request_conversation_scope_and_panel_denial.

backend/tests/test_registration.py

Lines: 228. Integration tests for public student registration.

Symbol	Line	Purpose / source docstring
request_json	19	Make an in-process ASGI request without requiring a test HTTP client package.
RegistrationIntegrationTests	59	Registration tests whose database writes are rolled back as one outer transaction.

Test cases: test_password_hashing_and_verification, test_individual_public_registration_is_rejected, test_successful_student_registration_stores_only_hash, test_demo_mode_allows_unlisted_student_and_records_mode, test_duplicate_normalized_email_returns_conflict, test_common_consumer_and_institutional_domains_are_accepted, test_duplicate_normalized_registration_number_returns_conflict, test_invalid_email_missing_fields_and_password_lengths_are_rejected, test_privileged_request_fields_are_rejected.

backend/tests/test_reminder_rules.py

Lines: 97. Isolated non-operational reminder configuration coverage.

Symbol	Line	Purpose / source docstring
ReminderTests	16	Implementation entry; inspect this symbol for details

Test cases: test_auth_origin_and_validation, test_dst_exact_link_history_and_cross_cycle, test_overflow_and_atomic_audit, test_concurrent_allocation.

backend/tests/test_retired_modules.py

Lines: 23. Focused assertions for the retired registry/roster/identity surface.

Symbol	Line	Purpose / source docstring
test_retired_api_routes_are_not_registered	6	Implementation entry; inspect this symbol for details

backend/tests/test_supervisor_assignments.py

Lines: 84.

Symbol	Line	Purpose / source docstring
AssignmentTests	13	Implementation entry; inspect this symbol for details

Test cases: test_authorization_validation_history_and_capacity, test_wrong_cycle_ineligible_and_audit_rollback, test_fall_and_spring_share_annual_capacity_and_counts, test_00_concurrent_capacity.

backend/tests/test_supervisor_capacity.py

Lines: 145. Rollback-isolated coverage for M1-E.1 supervisor capacity configuration.

Symbol	Line	Purpose / source docstring
request	21	Implementation entry; inspect this symbol for details
SupervisorCapacityTests	44	Implementation entry; inspect this symbol for details

Test cases: test_00_capacity_advisory_lock_is_exclusive_across_connections, test_authorization_validation_eligibility_and_duplicate, test_update_historical_listing_and_capacity_lock, test_audit_failure_rolls_back_capacity, test_annual_policy_is_shared_by_fall_and_spring, test_capacity_status_and_update_cannot_drop_below_assignments.

backend/tests/test_supervisor_requests.py

Lines: 496. Focused authorization, lifecycle, capacity, and audit tests for supervisor requests.

Symbol	Line	Purpose / source docstring
SupervisorRequestTests	24	Implementation entry; inspect this symbol for details

Test cases: test_eligible_supervisor_list_and_role_isolation, test_create_submit_and_student_list, test_duplicate_cross_cycle_and_stale_request_protection, test_supervisor_only_decision_comment_and_accept_assignment, test_accept_requires_registered_team_and_keeps_request_submitted, test_duplicate_assignment_is_rejected_without_accepting_request, test_accept_capacity_conflict_rolls_back_assignment_and_status, test_accept_audit_failure_rolls_back_assignment_and_status, test_reject_changes_resubmit_withdraw_and_history, test_capacity_full_inactive_supervisor_monitoring_and_audit_rollback, test_pdf_validation_and_missing_required_fields, test_pdf_preview_confirmation_and_authorized_original_document, test_owner_confirm_and_submit_only, test_submit_requires_confirmation_and_current_revision, test_cross_team_owner_and_missing_required_fields_are_denied, test_concurrent_submission_revision_safety, test_submission_note_is_optional_visible_and_immutable_by_version, test_fyp_registry_role_scopes_filters_and_supervisor_assignment, test_fyp_registry_detail_and_admin_metadata_are_audited_without_mutating_proposal, test_fyp_registry_student_ownership_and_cross_scope_are_enforced.

backend/tests/test_supervisor_workspace.py

Lines: 163. Transactional integration checks for the supervisor group workspace.

Symbol	Line	Purpose / source docstring
SupervisorWorkspaceTests	19	Implementation entry; inspect this symbol for details

Test cases: test_workspace_request_and_acceptance, test_workspace_ownership_and_download_denial, test_workspace_decision_comments, test_workspace_capacity_enforced, test_workspace_conversation_and_empty_states, test_workspace_empty_group, test_workspace_real_upload_download_and_refresh, test_student_submission_to_supervisor_assignment_flow.

backend/tests/test_team_compatibility.py

Lines: 12.

Symbol	Line	Purpose / source docstring
TeamCompatibilityTests	4	Implementation entry; inspect this symbol for details

Test cases: test_equal_weight_and_unanswered_excluded, test_no_answers_is_not_zero_score, test_unknown_or_non_boolean_assessment_is_rejected.

backend/tests/test_team_constraints.py

Lines: 340. Database integration tests for the applied team/member constraints.

Symbol	Line	Purpose / source docstring
TeamConstraintTests	20	Exercise constraints supplied by Alembic revision ce964120f177.

Test cases: test_duplicate_registration_within_one_team_is_rejected, test_duplicate_active_registration_across_teams_in_one_cycle_is_rejected, test_duplicate_active_contact_email_in_one_cycle_is_rejected, test_noncanonical_registration_and_email_values_are_rejected, test_same_name_students_with_distinct_identifiers_are_allowed, test_member_team_cycle_mismatch_is_rejected, test_second_creator_is_rejected, test_shared_user_cannot_link_to_two_teams, test_member_identifiers_can_recur_in_different_academic_cycles.

backend/tests/test_team_registration.py

Lines: 598. Rollback-isolated API tests for shared-team registration.

Symbol	Line	Purpose / source docstring
request_json	24	Call the endpoint in process, without starting a server or adding a client library.
cors_preflight	67	Exercise CORS without a running HTTP server.
TeamRegistrationApiTests	108	API writes share an outer transaction and never retain fixture data.
TeamEligibilityTests	485	Roster enforcement uses the selected cycle and each member's contact email.

Test cases: test_successful_team_registration_creates_one_user_and_one_creator, test_student_group_registration_creates_one_shared_student_account, test_student_group_registration_rejects_privileged_role_fields, test_group_registration_accepts_only_names_and_registration_numbers, test_group_registration_enforces_configured_minimum, test_cors_allows_only_the_configured_frontend_origin, test_unknown_closed_and_missing_policy_cycles_return_safe_errors, test_configured_maximum_boundary_and_excess_members, test_request_and_existing_record_duplicate_conflicts, test_creator_registration_used_by_individual_account_is_rejected, test_invalid_and_privileged_fields_are_rejected, test_late_member_insert_failure_rolls_back_all_new_records, test_safe_error_for_existing_shared_login_email, test_password_confirmation_mismatch_is_rejected_before_creation, test_missing_creator_and_member_are_field_specific_and_atomic, test_wrong_cycle_and_mismatched_pair_are_rejected, test_normalization_and_shared_login_is_not_contact_email, test_invalid_semester_is_rejected_by_roster_constraint, test_demo_mode_allows_unlisted_one_two_and_three_member_teams, test_restricted_mode_and_invalid_mode_keep_roster_validation, test_demo_mode_records_audit_metadata_without_changing_security_validation, test_demo_mode_still_enforces_group_limit.

backend/tests/test_template_references.py

Lines: 156. Rollback-isolated regression coverage for internal template references.

Symbol	Line	Purpose / source docstring
TemplateReferenceTests	17	Implementation entry; inspect this symbol for details
TemplateReferenceConcurrencyTests	128	Implementation entry; inspect this symbol for details

Test cases: test_access_validation_and_history, test_origin_and_numeric_rule_validation, test_audit_failure_rolls_back_initial_reference, test_audit_failure_rolls_back_metadata_version, test_audit_failure_rolls_back_rule_version, test_concurrent_metadata_version_allocation.

frontend/tests/academic-cycle-dashboard.test.mjs

Lines: 69.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/account-creation.test.mjs

Lines: 50.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/admin-theme.test.mjs

Lines: 28.

Imports: node:assert/strict, node:fs, node:path, node:url.

frontend/tests/archive-import.test.mjs

Lines: 13.

Imports: node:assert/strict, node:fs.

frontend/tests/archive-search.test.mjs

Lines: 16.

Imports: node:assert/strict, node:fs, node:path, node:url.

frontend/tests/auth-persistence.test.mjs

Lines: 45.

Imports: node:assert/strict, node:fs, node:test.

Symbol	Line	Purpose / source docstring
API	9	Implementation entry; inspect this symbol for details

frontend/tests/collaboration-channel.test.mjs

Lines: 16.

Imports: node:assert/strict, node:fs.

frontend/tests/coordinator-admin-monitoring.test.mjs

Lines: 36.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/coordinator-workspace.test.mjs

Lines: 18.

Imports: node:assert/strict, node:fs.

frontend/tests/dashboard-navigation.test.mjs

Lines: 49.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/dashboard-presentation.test.mjs

Lines: 34.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/dashboard-theme.test.mjs

Lines: 29.

Imports: node:assert/strict, node:fs, node:test, \.\/dashboard-theme\.css.

frontend/tests/design-system.test.mjs

Lines: 30.

Imports: node:assert/strict, node:fs, node:test, \.\/index\.css, \.\/auth\.css, \.\/design-system\.css, ./index.css, ./design-system.css.

frontend/tests/document-requirements.test.mjs

Lines: 50.

Imports: node:assert/strict, node:fs.

frontend/tests/email-validation.test.mjs

Lines: 39.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/encoding.test.mjs

Lines: 32.

Imports: node:assert/strict, node:fs, node:path, node:url.

frontend/tests/evaluation-marks.test.mjs

Lines: 36.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/evaluator-portal.test.mjs

Lines: 29.

Imports: node:assert/strict, node:fs.

frontend/tests/group-account-access.test.mjs

Lines: 25.

Imports: node:assert/strict, node:fs, node:test.

API references found in source (dynamic expressions may be partial): /api/v1/team-registration.

frontend/tests/manual-delete-controls.test.mjs

Lines: 19.

Imports: node:test, node:assert/strict, node:fs.

frontend/tests/module3-proposal.test.mjs

Lines: 27.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/password-policy.test.mjs

Lines: 47.

Imports: node:assert/strict, node:fs, node:path, node:url, \.\/passwordPolicy.

frontend/tests/password-reset.test.mjs

Lines: 52.

Imports: node:assert/strict, node:fs, node:path, node:url.

frontend/tests/portal-contrast.test.mjs

Lines: 28.

Imports: node:assert/strict, node:fs.

frontend/tests/project-workflow.test.mjs

Lines: 28.

Imports: node:assert/strict, node:fs.

frontend/tests/registration-mode.test.mjs

Lines: 22.

Imports: node:assert/strict, node:fs/promises, node:test.

frontend/tests/retired-modules.test.mjs

Lines: 33.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/role-sidebar.test.mjs

Lines: 44.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/student-entry-flow.test.mjs

Lines: 34.

Imports: node:assert/strict, node:fs, node:path, node:url.

frontend/tests/supervisor-capacity.test.mjs

Lines: 33.

Imports: node:assert/strict, node:fs, node:test.

frontend/tests/supervisor-portal.test.mjs

Lines: 40.

Imports: node:assert/strict, node:fs, node:path, node:url.

frontend/tests/supervisor-request-workflow.test.mjs

Lines: 73.

Imports: node:assert/strict, node:fs, node:path, node:url.

scripts/bootstrap_first_admin.py

Lines: 69. Interactively create the first FYPHub administrator, once only.

Symbol	Line	Purpose / source docstring
main	27	Prompt locally for a new first-admin account without exposing credentials.

scripts/build_defence_guide.py

Lines: 351. Build a source-backed defence guide. Database inspection is read-only.

Symbol	Line	Purpose / source docstring
command	70	Implementation entry; inspect this symbol for details
source_index	77	Implementation entry; inspect this symbol for details
snapshot	157	Implementation entry; inspect this symbol for details
cell	172	Implementation entry; inspect this symbol for details
build	176	Implementation entry; inspect this symbol for details
render	247	Implementation entry; inspect this symbol for details
main	342	Implementation entry; inspect this symbol for details

scripts/configure_p1_cycle.py

Lines: 107. Configure the approved local Fall 2026 academic-cycle bootstrap.

Symbol	Line	Purpose / source docstring
main	31	Create the approved local setting, or verify it already matches exactly.

scripts/seed_archived_projects.py

Lines: 69. Development-only idempotent archive fixtures; never run at application startup.

Symbol	Line	Purpose / source docstring
sample_pdf	28	Implementation entry; inspect this symbol for details
main	46	Implementation entry; inspect this symbol for details

scripts/seed_proposal_references.py

Lines: 79. Explicit development-only seed for proposal reference search.

Symbol	Line	Purpose / source docstring
sample_pdf	35	Implementation entry; inspect this symbol for details
main	58	Implementation entry; inspect this symbol for details

database/init/001_extensions.sql

Lines: 1.

docker-compose.yml

Lines: 155. Service topology, initialization jobs, ports, volumes, networks and health checks.

backend/Dockerfile

Lines: 28. Python runtime image, dependency installation and Uvicorn launch command.

frontend/Dockerfile

Lines: 23. Frontend build stage and Nginx static runtime image.

frontend/nginx.conf

Lines: 27. SPA fallback and reverse proxy routes for API and health endpoints.

frontend/vite.config.ts

Lines: 15. Vite React/Tailwind plugins, development host/port and API proxy.

frontend/package.json

Lines: 1. Frontend dependency declarations and dev/build/lint commands.

frontend/package-lock.json

Lines: 3383. Exact resolved npm dependency graph for reproducible installs.

frontend/tsconfig.json

Lines: 1. TypeScript project-reference configuration.

frontend/tsconfig.app.json

Lines: 1. Browser TypeScript compiler rules.

frontend/tsconfig.node.json

Lines: 1. Node/toolchain TypeScript compiler rules.

frontend/eslint.config.js

Lines: 5. Lint rules and React hooks checks.

backend/requirements.txt

Lines: 16. Pinned Python runtime dependency versions.

backend/alembic.ini

Lines: 4. Migration tool configuration; environment builds the effective connection.

frontend/index.html

Lines: 1. HTML shell containing the React root and module entry script.

.gitignore

Lines: 36. Excludes environment files, dependencies, caches and runtime uploads from Git.

.env.example

Lines: 35. Environment-variable template; real private values are intentionally not inspected or exported.

Appendix E - Migration revision graph

Revision	Parent(s)	File
3a1f53f17b1c	9c227dc337bf	backend/alembic/versions/3a1f53f17b1c_add_academic_cycles_and_group_policies.py
9415b228314f	a3b4c5d6e7f8	backend/alembic/versions/9415b228314f_join_requests_and_creator_proof_binding.py
9c227dc337bf	None	backend/alembic/versions/9c227dc337bf_create_users_foundation.py
a2b3c4d5e6f7	f1a2b3c4d5e6	backend/alembic/versions/a2b3c4d5e6f7_add_academic_deadline_versions.py
a3b4c5d6e7f8	f2a3b4c5d6e7	backend/alembic/versions/a3b4c5d6e7f8_bind_anonymous_join_proof_target.py
a7b8c9d0e1f2	f6a7b8c9d0e1	backend/alembic/versions/a7b8c9d0e1f2_add_group_messages.py
a8467cad28da	eb100aa929ab	backend/alembic/versions/a8467cad28da_invitation_transactional_outbox.py
b2c3d4e5f6a7	a8467cad28da	backend/alembic/versions/b2c3d4e5f6a7_student_matching_proof.py
b3c4d5e6f7a8	a2b3c4d5e6f7	backend/alembic/versions/b3c4d5e6f7a8_add_reminder_rules.py
b8c9d0e1f2a3	a7b8c9d0e1f2	backend/alembic/versions/b8c9d0e1f2a3_add_team_match_preferences.py
b8d2e4f6a9c1	f7a1c3d9e2b4	backend/alembic/versions/b8d2e4f6a9c1_add_admin_audit_events.py
c3d4e5f6a7b8	b2c3d4e5f6a7	backend/alembic/versions/c3d4e5f6a7b8_proposal_foundation.py
c4d5e6f7a8b9	b3c4d5e6f7a8	backend/alembic/versions/c4d5e6f7a8b9_add_student_roster.py
c4e7f1a2b9d3	b8d2e4f6a9c1	backend/alembic/versions/c4e7f1a2b9d3_add_audit_targets.py
c5d6e7f8a9b0	b2c3d4e5f6a7	backend/alembic/versions/c5d6e7f8a9b0_add_supervisor_requests.py
c9d0e1f2a3b4	b8c9d0e1f2a3	backend/alembic/versions/c9d0e1f2a3b4_add_match_chat.py
ce964120f177	3a1f53f17b1c	backend/alembic/versions/ce964120f177_add_teams_and_member_duplicate_.py
d0e1f2a3b4c5	c9d0e1f2a3b4	backend/alembic/versions/d0e1f2a3b4c5_match_chat_mutual_reconsent.py
d5e6f7a8b9c0	c4e7f1a2b9d3	backend/alembic/versions/d5e6f7a8b9c0_add_template_reference_versions.py
d6e7f8a9b0c1	c5d6e7f8a9b0	backend/alembic/versions/d6e7f8a9b0c1_add_supervisor_request_documents.py
e1f2a3b4c5d6	d0e1f2a3b4c5	backend/alembic/versions/e1f2a3b4c5d6_individual_verification.py
e5f6a7b8c9d0	c4d5e6f7a8b9	backend/alembic/versions/e5f6a7b8c9d0_add_workflow_policy_drafts.py
e7f8a9b0c1d2	d6e7f8a9b0c1	backend/alembic/versions/e7f8a9b0c1d2_add_proposal_references_and_request_conversations.py
e8f9a0b1c2d3	d5e6f7a8b9c0	backend/alembic/versions/e8f9a0b1c2d3_add_supervisor_capacity_policies.py
eb100aa929ab	9415b228314f	backend/alembic/versions/eb100aa929ab_team_invitations.py
f1a2b3c4d5e6	e8f9a0b1c2d3	backend/alembic/versions/f1a2b3c4d5e6_add_evaluation_rubric_drafts.py
f2a3b4c5d6e7	e1f2a3b4c5d6	backend/alembic/versions/f2a3b4c5d6e7_anonymous_verification.py
f6a7b8c9d0e1	e5f6a7b8c9d0	backend/alembic/versions/f6a7b8c9d0e1_add_supervisor_assignments.py
f7a1c3d9e2b4	ce964120f177	backend/alembic/versions/f7a1c3d9e2b4_add_auth_sessions.py
f9a0b1c2d3e4	e7f8a9b0c1d2	backend/alembic/versions/f9a0b1c2d3e4_add_archived_projects.py
fa0b1c2d3e4	f9a0b1c2d3e4	backend/alembic/versions/fa0b1c2d3e4_allow_archived_title_reuse_after_deactivation.py
fb1c2d3e4f5	fb1c2d3e4f5a	backend/alembic/versions/fb1c2d3e4f5_add_historical_archive_fields.py
fb1c2d3e4f5a	fa0b1c2d3e4	backend/alembic/versions/fb1c2d3e4f5a_add_module03_proposal_compliance.py

Revision	Parent(s)	File
fc2d3e4f5a6b	fa0b1c2d3e4	backend/alembic/versions/fc2d3e4f5a6b_add_supervisor_request_submission_notes.py
fc6b7c8d9e0f	r678a003	backend/alembic/versions/fc6b7c8d9e0f_add_private_collaboration_channels.py
fd3e4f5a6b7c	fc2d3e4f5a6b	backend/alembic/versions/fd3e4f5a6b7c_add_fyp_registry_metadata.py
fe4f5a6b7c8d	fd3e4f5a6b7c	backend/alembic/versions/fe4f5a6b7c8d_add_password_reset_tokens.py
ff5a6b7c8d9e	fe4f5a6b7c8d	backend/alembic/versions/ff5a6b7c8d9e_add_progress_scheduling_evaluation.py
fg6b7c8d9e0f	ff5a6b7c8d9e	backend/alembic/versions/fg6b7c8d9e0f_add_module03_template_versions.py
fh7c8d9e0f1a	fg6b7c8d9e0f	backend/alembic/versions/fh7c8d9e0f1a_add_module03_activation_mode.py
fi8d9e0f1a2b	c3d4e5f6a7b8	backend/alembic/versions/fi8d9e0f1a2b_continue_proposal_chain.py
fj9e0f1a2b3	fi8d9e0f1a2b	backend/alembic/versions/fj9e0f1a2b3_add_proposal_type.py
fk0f1a2b3c4	fj9e0f1a2b3	backend/alembic/versions/fk0f1a2b3c4_add_proposal_metadata_fields.py
fl1a2b3c4d5	fk0f1a2b3c4	backend/alembic/versions/fl1a2b3c4d5_add_agent_supervision_runs.py
gm2b3c4d5e6f	fl1a2b3c4d5	backend/alembic/versions/gm2b3c4d5e6f_add_dashboard_requirements.py
hn3c4d5e6f7a	gm2b3c4d5e6f	backend/alembic/versions/hn3c4d5e6f7a_add_document_upload_requirements.py
jk4d5e6f7a8b	hn3c4d5e6f7a	backend/alembic/versions/jk4d5e6f7a8b_document_storage_reference.py
kl5e6f7a8b9c	jk4d5e6f7a8b	backend/alembic/versions/kl5e6f7a8b9c_add_role_based_fyp_evaluations.py
lm6f7a8b9c0d	kl5e6f7a8b9c	backend/alembic/versions/lm6f7a8b9c0d_remove_evaluation_marks_module.py
m123a001	('c3d4e5f6a7b8', 'fh7c8d9e0f1a')	backend/alembic/versions/m123a001_modules_1_3_lifecycle.py
mn7c8d9e0f1a	lm6f7a8b9c0d	backend/alembic/versions/mn7c8d9e0f1a_restore_role_based_evaluations.py
mo8r9e0f1a2b	('mn7c8d9e0f1a', 'ff5a6b7c8d9e')	backend/alembic/versions/mo8r9e0f1a2b_add_proposal_ai_review_runs.py
n456a001	m123a001	backend/alembic/versions/n456a001_add_academic_cycle_year_session.py
oa1b2c3d4e5f	mo8r9e0f1a2b	backend/alembic/versions/oa1b2c3d4e5f_repair_module1_evaluation_schema.py
ob1c2d3e4f5a	oa1b2c3d4e5f	backend/alembic/versions/ob1c2d3e4f5a_repair_module1_identity_columns.py
oc1d2e3f4a5b	ob1c2d3e4f5a	backend/alembic/versions/oc1d2e3f4a5b_add_template_activation.py
p567a001	n456a001	backend/alembic/versions/p567a001_supervisor_capacity_year_session.py
pa1b2c3d4e5f	('ob1c2d3e4f5a', 'oc1d2e3f4a5b')	backend/alembic/versions/pa1b2c3d4e5f_add_module03_compliance_workspace.py
qa1b2c3d4e5f	pa1b2c3d4e5f	backend/alembic/versions/qa1b2c3d4e5f_complete_module02_collaboration.py
r678a001	p567a001	backend/alembic/versions/r678a001_remove_registry_roster_identity_modules.py
r678a002	r678a001	backend/alembic/versions/r678a002_remove_invitation_outbox.py
r678a003	r678a002	backend/alembic/versions/r678a003_annual_supervisor_capacity.py
rd7e8f9a0b1c	fc6b7c8d9e0f	backend/alembic/versions/rd7e8f9a0b1c_repair_archive_import_columns.py
re8f9a0b1c2d	rd7e8f9a0b1c	backend/alembic/versions/re8f9a0b1c2d_archive_import_batches.py
rf9a0b1c2d3e	re8f9a0b1c2d	backend/alembic/versions/rf9a0b1c2d3e_document_requirement_lifecycle.py

Appendix F - Reproducibility and evidence files

The machine-readable companion technical-inventory.json contains the full inspected schema, ORM metadata, endpoint ownership, source symbols, frontend imports and test names. scripts/build_defence_guide.py rebuilds it using read-only database catalog queries. FYPHub_Defence_Guide_Source.md is the authored explanatory part. Generated documents are snapshots and should be regenerated after code/schema changes.